Living the Change: Designerly modes of real-life experimentation

MARTIN SJÖMAN



Academic dissertation which, with due permission from KTH Royal Institute of Technology, is submitted for public defence for the Degree of Doctor of Philisophy on Thursday the 21:st of September 2023, in Kollegiesalen at KTH, Brinellvägen 6, Stockholm.

Doctoral Dissertation 2023
TRITA-ITM-AVL 2023:20
ISBN: 978-91-8040-679-6

KTH Royal Institute of Technology
School of Industrial Engineering and Management
Department of Machine Design; Product and Service Design
Brinellvägen 83
100 44 Stockholm
Sweden

© Martin Sjöman 2023
Photographs without specific attribution are either taken as part of the research projects or taken by the author.
Some quotes have been translated from Swedish to English by the author.
Graphic design and production: Martin Sjöman
Printed in Sweden by Universitetsservice AB

Abstract

The sustainability transitions required to address the climate crisis cannot be achieved by technology alone; radical lifestyle changes are needed. To contribute to meeting this critical challenge, design must move its focus from individual needs, desires, and behaviors to the level of the complex socio-technical systems that shape our society. There have been several calls for research that is action-, future- and learning-oriented, to accelerate sustainability transitions. In a broad sense, my research concerns how design practice can be used and further developed to this end.

There is growing consensus that real-life experimentation is required to understand and realize the potentials of sustainability innovations, and an emerging experimental turn can be seen in the proliferation of approaches such as living labs, city labs and transition labs, as well as in policy experiments, pilots, demonstrations, and field trials. There is a broad movement in society to involve users or citizens in learning and experimentation in the complexity of real-life contexts, but as will be discussed in this dissertation, most approaches do not realize these ambitions in practice.

This dissertation presents an approach for design-driven, or designerly living labs for the real-life exploration and demonstration of possible sustainable concepts and futures. Living labs are often described as having their roots in design, and this dissertation represents a move to reclaim that term for more open-ended modes of experimentation. By living the change, these designerly labs have provided rich insights into the entangled social-technical nature of sustainable futures, and identified barriers and pathways towards them.

The dissertation is based on detailed and operative accounts of seven such designerly living labs carried out by design researchers at KTH Royal Institute of Technology in Sweden from 2014 to 2023. A cross-case analysis is presented in order to identify and validate the key characteristics of this emerging approach, and how they connect to design practice. In the
analysis I also investigate how these labs relate to other research approaches in fields such as transition studies, user innovation, participatory design, and action research. I position designerly living labs as an alternative and complement to more mainstream approaches to real-life experimentation—specifically for the early-stage framing of sustainability issues and in exploring sustainable future concepts and lifestyles.

Main findings include a number of factors that were found to demarcate different modes and understandings of real-life experimentation research. These factors include the involvement of users as co-researchers in exploration rather than as testers or co-creators in innovation, and how this more open-ended aim for learning may conflict with notions of developing, evaluating and scaling up. There is also a discussion on how different understandings of these factors can lead to confusion and conflict in transdisciplinary research and offer recommendations for organizing new research projects of this sort.

Sammanfattning

Den hållbarhetsomställning som behövs för att hantera klimatkrisen kan inte begränsas till vad som kan lösas med ny teknologi, utan kräver att vi radikalt förändrar även våra livsstilar. För att vara en del av att möta den här utmaningen så behöver designfältet flytta sitt fokus från individuella behov, önskningar och beteenden, mot de komplexa socio-tekniska system som formar vårt samhälle. För att driva hållbarhetsomställningen snabbare framåt efterfrågas nu mer aktions- och framtids-orienterad forskning med mer fokus på lärande. I ett större perspektiv så handlar min forskning om att besvara hur designpraktik kan utvecklas och användas i detta syfte.

Det finns en ny samstämmighet om behovet av metoder som bygger på mer öppet experimenterande ute i människors verklighet för att innovationer på hållbarhetsområdet ska kunna nå sin fulla potential. Detta är synligt i framväxten av angreppssätt såsom living labs, city-labs och transition labs, policyexperiment, piloter, demonstrationsprojekt och fältförsök. Det finns också en bred rörelse i samhället mot att involvera användare och medborgare i dessa experiment och lärandeprocesser, men som kommer att beskrivas i denna avhandling så är det få av de nämnda angreppssätten som verkligen uppfyller detta.

I denna avhandling presenteras en approach för designdrivna living labs i syfte att utforska och demonstrera möjliga och hållbara framtidskoncept i vanliga människors vardagsliv. Metoderna i living labs beskrivs ofta som framvuxna ur deltagande designarbete, och denna avhandling kan ses som ett steg i att återta begreppet living labs för mer öppet experimenterande forskning. Genom att låta människor leva med och i den förändring som eftersöks, så har dessa ”levande” labb givit rika insikter om socioteknisk komplexa hållbara framtider, och visat på både hinder och möjliga vägar framåt.

Avhandlingen baseras på detaljerade och konkreta framställningar av sju designdrivna living labs, som genomförts av designforskare vid Kungliga Tekniska Högskolan i Stockholm från 2014–2023. Angreppssättets särskiljande egenskaper har identifierats och validerats genom en tvärfallsanalys. Dessa egenskaper har sedan positionerats i relation till designpraktik och metodologier inom andra forskningsfält, såsom omställningsstudier, användardriven innovation, deltagande design och aktionsforskning. Designdrivna living labs befinns vara ett alternativ till

mer vanligt förekommande experimenterande angreppssätt, för att angripa hållbarhetsrelaterade forskningsfrågor i tidiga forskningsskeden. Hållbara framtidskoncept och livsstilar utforskas genom designinterventioner, mitt i människors vardagsliv.

Viktiga forskningsbidrag är de faktorer och strategiska ställningstaganden som skiljer olika ansatser för experimenterande forskning ute i samhället. Dessa inkluderar en beskrivning av hur användare kan engageras som reflektiva medforskare i stället för som testare eller medskapare i innovationsarbete. I avhandlingen diskuteras också hur denna typ av mer öppet utforskande skiljer sig från vanligare förekommande arbetsformer för att utveckla, utvärdera och skala upp teknikinnovationer i samhället, samt hur välfungerande projekt av denna typ kan organiseras.

Acknowledgements

In this dissertation, I attempt to name and describe a research approach that has emerged over many years, through a line of research projects of varying sizes. These projects reflect the joint efforts of many people and they have been participatory in many ways. I have not been part of these developments from the beginning. Instead, I have been concerned with naming, framing, trying out, and further developing a design research phenomenon that was already in motion. My first thanks must therefore go to Sara Ilstedt and Teo Enlund, who I believe played the most important part in initiating the designerly living labs and who pushed to make all this happen. Sara has also been my main supervisor and has inspired me by providing a broader and more grounded perspective on how design research has developed over time, in Sweden and elsewhere. This has included real-world accounts of how research funders and institutions often both seek out and frequently misunderstand design. I have also enjoyed discussing different interpretations of apparent research views and purposes. Teo was my only design partner in the first part of the Mistra SAMS research program and has ever been a champion of design and for action. In a transdisciplinary research context with many conflicting perspectives, Teo was also a direct connection to the core of design practice, embodying designerly ways.

Another central source of support has been Susanne Nilsson, my second supervisor. Thank you for your curiosity and dedication, and for all the long, openly explorative and energizing discussions! Many thanks also to my first readers, Pelle Ehn and Lisbeth Svengren Holm. Their close readings provided perspective and supported reflection, and have improved the text in many ways.

Mia Hesselgren has played an important part in shaping this emerging designerly research approach and has also been my closest colleague. Thank you for the support! I have really enjoyed our collaborations, and especially the important learning experiences from my first research undertaking at KTH in the SMSS project. Working closely together, ideas and understandings about the labs have naturally been passed back and forth between us. Still, we have different perspectives when describing these rich design research projects and have focused on different aspects of this phenomenon, and I believe our two accounts complement each other. Mistra SAMS was the main part of my research and made it possible to secure funding for my PhD program. I send thanks to Anna Kramers, Karolina Isaksson, Mattias Höjer, and many others

for their courageous attempt to integrate a radically new design research approach into the heart of this large research program. Tina Ringensson was my closest colleague in the first part of Mistra SAMS, and I learnt a lot from our collaboration on interviews and analysis, and together with Liridona Sopjani we had many interesting discussions about the nature of 'labs'.

I also want to express my gratitude to my colleagues at the Integrated Product Development and Design group, which has been a very friendly and supportive workplace, as well as to all my research colleagues in CoKitchen, at KTH Live-in-Lab, the Integrated Transport Research Lab and the Swedish National Road and Transport Research Institute. This also includes all the supportive colleagues from public and private research partners, such as Botkyrka municipality, Ericsson, Academic Housing, and many others.

Over these many research projects, I have been continually surprised and impressed by our co-researchers' interest and engagement, their willingness to experiment, and diligence in reporting. There could be no real-life experimentation without people willing to actively engage and to support research. To all of you: I am grateful for being allowed to peek into your lives and thoughts, and I know this cannot be taken for granted.

Finally, thank you Jonas for putting up with all my long workdays, and Lotta for your loving support and contrasting perspective on research. To my friends and family who have stood by me and lured me out into the sun every now and then: you know who you are.

Table of Contents

1. Introduction

1.1. Designerly ways of knowing 1

1.2. Design methods and “labs” to support sustainability transitions 3

Design for sustainable futures 3

Real-life experimentation 4

Design and designerly living labs 5

1.3. Research scope, questions and aims 6

The Designerly Living Labs 6

Research aim 7

Calls for operative accounts of participatory research projects 10

1.4. Thesis outline 10

1.5. List of publications 12

Publication A 12

Publication B 13

Publication C 13

Publication D 14

Publication E 15

Publication F 15

How the publications relate to the research questions 16

2. The research context

2.1. Designerly thinking to frame complex issues 17

2.2. Schools of Design and Design Research 18

Human-Centered Design 19

Co-design and Participatory Design 20

‘Infrastructuring’ and design for social innovation 21

Design for Sustainability 22

Practice-Oriented Design 24

2.3. Modes and views of user-participation 27

2.4. Real-life experimentation 31

Transitions, and why real-life experimentation is needed 31

Modes of experimentation 32

2.5. Living Labs 35

3. Methodology

3.1. Research perspectives 42

Practice-Based Design Research 42
Design and action-oriented research 42
Design research as a reflexive research practice 43

3.2. The cases and the data collection 45
3.3. How the cases were analyzed 49

Cross-case analysis 50

3.4. Limitations 51

4. Empirical projects

4.1. The proposed DLL methodology 53

How the designerly living labs started 53
Explorative studies in early research stages 53
Learning devices as a doing-first strategy 54
Real-life Interventions designed as learning devices 55
Longitudinal experimentation 55
Provoke, disturb or enable experimentation with new practices 56
Engaging users as co-researchers 57
Summary 58

4.2. Earlier projects - antecedents 59

A Car-Free Year (henceforth ***Car-Free***) 59
KTH LEV-Pool 62

4.3. The Restaurant Lab 65
4.4. CoKitchen 68
4.5. Mistra SAMS part 1, 2017-2020 76

Living lab #0, Future playing-rules for mobility. 80
Living lab #1, Work closer, travel smarter 86

4.6. Mistra SAMS part 2, 2021–2024 94

LL#3, Riksten 95

5. Analysis

5.1. Earlier steps in the analysis 106
5.2. The cross-case analysis 109

Method Characteristics / Research Principles 109
User Activity / Engagement 114
Stakeholder Activity / Engagement 116

Innovation outcomes and impact 118

Results and discussion 120

6. Understanding DLLs

6.1. Prototyping futures in early research stages 121

6.2. Provisional tools for learning 125

The research purposes of learning vs. solving 125

The DLLs function as backcasting set in real life 128

6.3. Experimentation in real-life contexts 131

Conceptualizing a living lab as an experiment or an infrastructure 132

Interventions aiming to enable or provoke experimentation 135

6.4. Positioning the DLLs with respect to modes of user participation 139

The DLL-participants' role as co-researchers 139

High user involvement as power or as level of activity? 140

Using Practice-Oriented Design, DLLs enable new practices but do not design them 147

6.5. Types of research outputs and outcomes from DLLs 150

Open inquiry to frame or reframe an issue 150

How holistic design research may both connect and challenge other research fields 151

Actionable knowledge of pathways and barriers 152

DLLs are suited for modelling, not evaluation 153

7. Implementing DLLs

7.1. Challenges and strategies for research collaboration 156

Confusion and conflicts in the DLLs 156

Interdisciplinary and transdisciplinary research collaborations 158

A guiding vision may provide conditions for design 159

7.2. Design strategies for bridging, orchestration and 'making' 164

Design for mediating or 'bridging' conflicts and misunderstandings 164

Staging a DLL requires orchestration and a sensitivity for experiences 166

Designer as operative 'doer' and maker 167

8. The design methodology PhD

Challenges of having both methodological and empirical aims 170

Lack of mandate to make methodological choices 171

The 'operative trap' of a design PhD 172

9. Conclusions

9.1. Research contributions 173
9.2. Implications for practice 176
9.3. Future research 178
9.4. Outro 179

References 180
Appended publications 198

1. Introduction

1.1. Designerly ways of knowing

At its core, this dissertation concerns the nature of design from a designer-practitioner-researcher's point of view and how design can support society in navigating the unimaginably vast sustainability transitions ahead. The research literature strongly connects design to innovation and change, but why or how design has this capacity seems to be a more complex matter. Decades of studies into "designerly" ways of thinking, knowing, and doing have built a rich and diverse body of research, but there are also many diverse, and sometimes contradictory, definitions of what design is. Moreover, the target of these studies is not only in constant flux, but also a field that seems to being growing broader by the day. As design has continually taken on new challenges and entered new fields, it has incorporated new perspectives that stretch and blur its borders. Attempts to define what design is quickly become overly narrow when newly added design richness simply does not fit. Design used to concern products developed in isolation, until participatory design was introduced. Later on, the concept of service design was invented, followed by design for sustainability, as well as critical and speculative design that aims to stimulate, provoke, and engage. These approaches have been followed by policy design, design for social innovation, transformation and transition design—approaches that seek to tackle the great societal challenges of our times. Design has been described as human-centered and experiential, future-oriented and speculative, action-oriented and pragmatist, holistic, reflective and critical, experimental and interventionist, and as being fit for handling open-ended, messy, uncertain and ambiguous situations. Design is concerned with the artificial world and world-making, with problem-solving, problem-finding and "problem-making," with framing and reframing issues, with the situated and with synthesis. It uses abductive reasoning, creativity, intuition, and empathy, as well as interdisci-

plinary teamwork and many tools to facilitate co-creation. This dissertation will provide examples of nearly all of these aspects.

Design is a science, but also a craft that one learns by doing. It is not just a process and a set of elaborate tools; rather, it has been described as a mindset, a mode of thinking, and even as a distinct culture. Sometimes bundled together with engineering, design has been described by writers including Herbert Simon, Nigel Cross, Horst Rittel, Bruce Archer, and Terry Irwin as a "third knowledge culture," alongside or balancing between science and the humanities—a practice-based epistemology all its own. What complicates matters even further is that many of the same authors also view design as an "essential and universal aspect of human culture" (Archer, Baynes, & Roberts, 2005, p 3; see also Manzini, 2015; Sanders & Stappers, 2014), in the sense that the act of designing and the capacity to model future scenarios in the mind is not restricted to a profession but rather is something all humans can do to remedy an unsatisfying situation. But when everything can be called design and everybody a designer, are we still able to distinguish this from the culture, mindset, practices, craft, and skills of trained designers?

Descriptions of design range from step-by-step Design Thinking recipes, to deeper meanings and intricate philosophical meanderings. Reading these texts as a design practitioner, having internalized the tacit knowledge of the craft, some descriptions resonate and others do not. In a way, this dissertation presents a bit of "reverse engineering." Here I investigate a designerly research approach that has been created and carried out by designer-practitioner-researchers, in order to understand why it emerged and how it works. By analyzing and picking apart a series of such research cases, and by testing and experimenting with these research methodologies myself, I have not just produced knowledge of and furthered the development of this research approach as such, but also learned something about what is specifically designerly about it, and about the nature of design.

1.2. Design methods and "labs" to support sustainability transitions

Design for sustainable futures

The sustainability transitions required to address the climate crisis cannot be achieved by technology alone; radical lifestyle changes are needed (Scoones et al., 2015). This means that we need to reevaluate what we value, what we consume, and the very organization of our societies. To contribute to meeting this critical challenge, design must move its focus from individual needs, desires and behaviors to the level of the complex socio-technical systems that shape our society. The currently dominant design approaches—such as Human Centered Design and most accounts of Design Thinking—build upon a product legacy, where the aim has always been to understand what people want and to create attractive products, ultimately aiming to "sell more" (Shove et al., 2008). To support a radical shift towards more sustainable lifestyles, designing new solutions requires a broader scope that also considers social practices and societal institutions. More importantly, design must also contribute to changing these structures. As Robinson (2004) points out, widespread hopes that sustainable development will allow sustainable economic growth must be countered by the understanding that we must also constrain consumption. Especially in the context of housing, mobility, and food, more sustainable patterns of consumption are needed (EEA, 2013).

Ceschin & Gaziulusoy (2016) provide an overview of how design for sustainability has evolved to meet these requirements. Design has moved from an insular product focus to be increasingly viewed as a tool to imagine and explore sustainable futures by creating compelling or provoking future scenarios and prototypes (Sanders & Stappers, 2014). At the same time, design is also increasingly used as a policy instrument to tackle complex socio-technical issues and to initiate change (see, e.g., Kimbell and Bailey, 2017) by engaging citizens in processes of exploring societal problems and designing solutions.

Designing for sustainability transitions challenges traditional and even very recent design approaches. User needs cannot be viewed as static but as something that design can, and often must, seek to change (Shove et al., 2007, Shove et al., 2008). This is very difficult to do in practice. As Henderson and Kyng (1991) have pointed out, user needs and practices cannot be observed or

anticipated before a product or service is designed, but must be investigated during its use. Hence, products or services that aim for sustainable outcomes cannot be viewed in isolation or be designed outside of the complex socio-technical systems that they may affect (Shove et al., 2007). Socio-technical systems change through the constant co-evolution of technical systems and social practices (Rotmans & Loorbach, 2010; Welch & Warde, 2015). As one part of the equation changes, so does the other. In a very broad sense, the research I present here concerns how design practice can be used and further developed to meet the challenges that will arise from the unavoidable shift to more-sustainable lifestyles.

Real-life experimentation

Because transitions require simultaneous social and technological developments, building or changing any complex socio-technical systems requires continuous experimentation and learning (Loorbach, 2007). We cannot possibly plan and design for all aspects of a highly complex system such as the provision of housing or transportation in society, nor can we foresee all the consequences of the changes that we make. For these reasons, real-life experimentation and learning are centerpieces in transitions research (Loorbach et al., 2017; Sengers et al., 2019; Caniglia et al., 2020). Indeed, there is a growing consensus that real-life experimentation is required to explore the potentials of sustainability innovations (Loorbach, 2007; Evans & Karvonen, 2010; Ceschin, 2014; Edwards & Bulkeley, 2017). To understand these innovations, research must be conducted in the context of everyday social practices and lifestyles (Liedtke, 2015; Spaargaren, 2011). This dissertation investigates design-driven and practice-based, designerly modes of research, and more specifically how design can shape and support real-life experimentation research setups.

One common approach to real-life experimentation has been the concept of living labs, an in-situ research approach that leverages active user involvement in innovation processes. Today, the European Network of Living Labs (ENoLL) congregates a main stream of European living labs, which have been defined as “user-centred open innovation ecosystems based on a systematic user co-creation approach, integrating research and innovation processes in real-life communities and settings” (ENoLL, n.d.). The same organization also describes living labs as “research infrastructures” and “open innovation environments attracting inwards investment” (ibid.), implying that the lab is an enabling

structure more than an approach or methodology. These research infrastructures are predominantly large-scale, often emphasizing the later stages of innovation processes, using terms such as exploitation and upscaling. They have also been described as structures for development and real-life testing of novel products, services, systems and solutions (Almirall & Wareham, 2011; Leminen et al., 2015). Even though living labs use a rhetoric that often emphasizes experimentation, design, and user co-creation, in reality many living labs are merely technological testbeds for the development and testing of ICT-solutions. Several studies have found that living lab activities have been limited to evaluative testing in order to improve product and service reliability (Salminen et al., 2011). Furthermore, living labs and similar approaches are mostly carried out in the late implementation or commercialization stages (see, e.g., Steen & van Beuren, 2017), and in practice user roles are often limited to "provid[ing] information and feedback upon request" (Leminen et al., 2014, p 41; Chilvers et al., 2018).

Design and designerly living labs

Still, the notion of labs as a tool for addressing societal issues such as sustainable development is spreading globally, as seen in the rapid proliferation of approaches such as challenge labs, future labs, policy experiments and fablabs, as well as pilots, demonstrations, and field-trials. There is a widespread drive to integrate experimentation and user involvement in society and in real-life contexts, but as outlined above, most labs do not fulfill these ambitions in practice. In this light, there seems to be room for more explorative and constructive approaches that focus more on framing and understanding complex socio-technical issues in the search for pathways to change.

This dissertation looks to design practice to provide an alternative. Design has been suggested as suitable for approaching complex and open-ended questions (Coyne, 2005; Buchanan, 1992; Stolterman, 2008), or "wicked problems" (Rittel & Webber, 1973) such as sustainability issues or societal challenges. Design practitioners often approach complex issues using a "doing-first" mode of thinking (Mintzberg and Westley, 2001): by visualizing, prototyping, making design interventions, or otherwise attempting to make these issues tangible (Sanders and Stappers, 2014). Co-design or participatory strategies are often used as ways to involve users and stakeholders in the process (Sanders, 2008). Design is also concerned with future visions, and designers are known for both

proposing alternative futures and pathways, and for enlisting stakeholders to realize them (Krippendorff, 2012).

> *This dissertation investigates and contributes to an emerging line of real-life experimentation studies aiming at employing design to catalyze sustainable lifestyle change. These studies, or "labs", have largely been envisaged, led and carried out by design researchers and design practitioners at KTH Royal Institute of Technology in Stockholm, Sweden. The lab structures presented here have been clearly design-driven, focused on lived experiences, and differ markedly from other types of labs in the literature. I therefore propose calling them Designerly Living Labs (DLL).*

1.3. Research scope, questions and aims

The Designerly Living Labs

As described above, the term living lab has come to be used mainly for large research infrastructures that in practice mostly focus on testing and evaluating technological solutions in the later development stages, with little active user participation or real-life experimentation. This dissertation aims to reclaim the term living labs for an emerging line of design-driven real-life experimentation setups that are—as was once suggested by Kakee Scott and colleagues—concerned with learning about change "by living change."

Since 2014, KTH's Design for Sustainability group Green Leap has initiated a series of more exploratory and open-ended living labs that have been planned and managed by designer-practitioner-researchers. My PhD program was created in 2017 to support the introduction of design research methods into the transdisciplinary Mistra SAMS research consortium, which included researchers from other KTH units as well as from several other research institutes. My research was also partly funded by the CoKitchen study, which was launched in 2018 by Green Leap's founder, design researcher professor Sara Ilstedt. At the time of writing, six design-driven longitudinal experiments have been carried out, which I have used to identify the core characteristics of DLLs. A seventh lab was launched in the fall of 2022, during the writing of this dissertation.

The DLLs to date:

2014 - 2016[1]	A Car-Free Year (Henceforth, "*Car-Free*")
2015 - 2016	KTH LEV Pool (*LEV-Pool*)
2018	Future Playing-Rules for Mobility (*Playing-Rules*)
2019 - 2021	The Restaurant Lab
2019 - 2022	Work Closer, Travel Smarter (*Work Closer*)
2019 - 2022	CoKitchen (*CoKitchen*)
2022 - 2023	Samsas in Riksten (*Riksten*)[2]

In this dissertation, it has been important to map out the earliest developments of this emerging research approach and the research context where this unfolded. I therefore include key insights from Car-Free and LEV-Pool that took place before I began my PhD research. To capture some important developments concerning stakeholder integration, I also include the Restaurant Lab, even though I took an active part only in the planning stages. Playing-Rules, Work Closer, and Riksten were carried out within the framework of Mistra SAMS. There have been no scientific publications from the Riksten lab thus far, and it is not yet clear whether this living lab will fall within the definition of DLLs in this dissertation. Still, the experiences from its planning stages led to several important insights about the challenges of organizing this type of research collaboration and are therefore included.

Research aim

In 2003, Gert Spaargaren made a call for the identification and analysis of potential routes for the innovation of systems of provision for sustainable lifestyles, including the identification of "lock-in factors" that prevent a broad uptake of such lifestyles, and with a specific focus on the social and societal dynamics of consumption. In an educational framework that proposed "Transi-

1 These timespans concern the length of the research projects. The length of the periods of real-life experimentation can be found in table 4 in section 3.1.

2 "Samsas" is a Swedish expression both for sharing and for co-existing side by side, and Riksten is a suburb to Stockholm, Sweden. In this dissertation, the Riksten lab is not referred to as a DLL since it had only recently been launched and it was unclear to which extent it would be "designerly".

tion Design" as an emerging design discipline, Irwin et al. (2015) called for new ways of designing for transitions in complex societal systems, stressing that everyday life and lifestyles should be the primary context within which to design for sustainable futures. (ibid) Similarly and more broadly, calls have been made for research that enables the exploration of desirable and sustainable futures (Loorbach et al., 2017; Hildén et al., 2017). More recently, 156 researchers collectively issued a strong call for a transformation of our knowledge systems in order to accelerate sustainability transitions (Fazey et al., 2020). This call emphasizes the need to develop more collaborative forms of research that are action-oriented, future-looking, and focused on learning, using methods that are both experimental and experiential.

By exploring and developing the emerging designerly approach for living labs, I aim to support the diffusion of design-driven methods for approaching sustainable lifestyle change. By investigating the seven cases, I aim to understand and position this new approach for real-life experimentation, and to present it as a complementary alternative to the more-common notions of pilots and living labs, as well as to design approaches such as co-design and design for social innovation. There is also an overarching aim to connect DLLs to their roots in design research practice, and thereby foster reflection on the nature of design. These aims have led to the formulation of three broad research questions or *themes* regarding DLLs, along with a fourth, more reflective, theme concerning the notion of a design PhD.

Understanding DLLs

In response to the above-mentioned calls for new forms of research, a first broad research question or theme concerns understanding DLLs. This includes an inquiry into:

- Research situations and aims for which DLLs are suitable.
- How DLLs function and what are their defining characteristics.
- How DLLs can complement other real-life experimentation methods to support sustainability transitions.
- The types of research results and impacts that are generally produced in a DLL.

This research theme is mainly addressed and discussed in chapter six.

Implementing DLLs

When operatively planning and implementing DLLs, and especially with larger and more diverse research collaborations, a second research theme has emerged that concerns:

- Research conditions and forms and types of research collaboration that support effective implementation of DLLs.
- Competences, strategies and 'doings' of designer-practitioner-researchers.

This research theme is mainly answered and discussed in chapter seven.

Designerly ways

A third research question or theme cuts across the others and concerns how the DLLs have emerged from design practice and how their key characteristics and their implementation relate to "designerly ways of knowing." Here, the investigation aims to further our understanding of the function of DLLs and the roles of the designer-practitioner-researchers, in relation to the design literature. Insights and reflections regarding this overarching theme connect to and are interspersed throughout all parts of the discussion.

The design methodology PhD

Later in the process, meta-level reflections from carrying out a design methodology PhD developed to form an additional, shorter discussion in chapter eight.

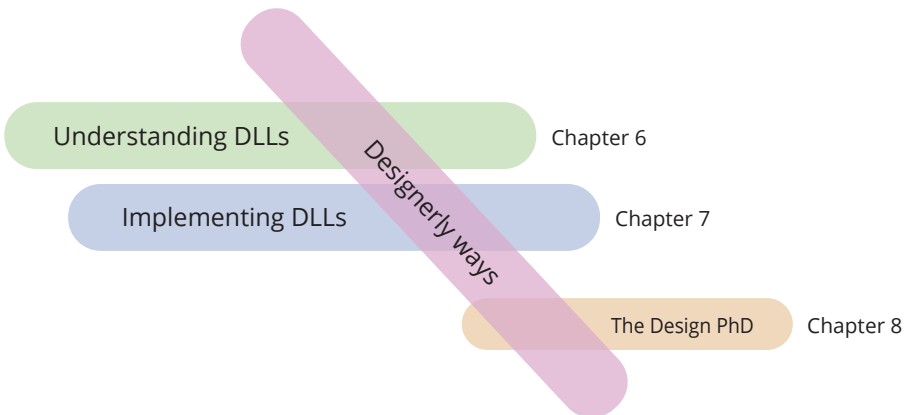


Figure 1. A visual overview of the research questions or themes in this thesis.

Calls for operative accounts of participatory research projects

There have been several calls for research case-studies to include not only methodological descriptions but also "the nitty gritty of operationalizing methods" (Botero et al., 2020, p 3). Woolrych et al. (2011) suggest that instead of comparing method "recipes," we should more broadly investigate how research practitioners mix, improve and make additions to match specific project circumstances. Rich case studies may also include "intermediate designing" and materialization processes, which have been argued to influence the outcomes of participatory and real-life design approaches (Hyysalo & Hyysalo, 2018).

For these reasons, the in-depth empirical case-descriptions presented in chapter four are described on an operational level and discuss not only my key methodological choices but also the situated research context, including pre-planning and funding, practical challenges and trade-offs, as well as organizational issues. This includes important considerations such as matching methods to research aims, and exactly how to involve participating users. The cases where I have personally taken part quite naturally include more operative accounts of these matters.

From my experience, there seems to be a lack of awareness and taken-for-grantedness regarding these research design choices and their practical implications. By discussing these choices, using example from the seven cases, I also aim to support the implementation of real-life experimentation projects, studies, and research spaces. I hope to both encourage and support practitioners and researchers to use more exploratory methods and to engage people and stakeholders more actively in the early stages of research on complex sustainability issues.

1.4. Thesis outline

This is a compilation dissertation, and the included publications can be found in the second part. However, the first part is not merely a summary of the publications but also a provides a synthesis, a deepening of the analysis, and new contributions.

In the first chapter I have briefly outlined the larger perspective on why this research is needed. This by connecting emerging developments in society and research with the research aim and research questions in this dissertation. This chapter also includes a list of the included publications.

Chapter two provides an overview of relevant bodies of literature, which include design and design research, transition studies, and approaches for real-life experimentation. I also offer a broad review of modes for, and perspectives on, user participation.

Although chapter three can be rightly considered a results chapter, the proposed DLL approach is presented here, before the empirical projects and the analysis to support the reader. It is important to point out that both the proposed approach and the theoretical understanding of it have emerged in several stages, through sequences of reflection in and on action (Schön, 1995) during the planning, implementation, and analysis of the six DLLs—a process that started before I began my research for this dissertation.

In chapter four rich, operational accounts of seven design research projects, or "labs" are presented. As outlined in the previous section, these case descriptions are important, as they allow insight into how and why these projects were planned, and how they were implemented in practice.

Chapter five presents a cross-case analysis of the projects that identifies and validates key characteristics and research principles.

In chapter six, the designerly research approach is further discussed in relation to the design literature, as well as in relation to other real-life experimentation methodologies. This discussion provides an additional layer of reflection that adds to and further highlights certain core themes, as well as more recent methodological developments. This chapter focuses on the first broad research question concerning the nature of DLLs.

In chapter seven, I turn to what we can learn about the practical implementation of DLLs. Here I discuss research collaborations and conditions for design research, as well as findings regarding the role of design and design practitioners' skills. This chapter mainly addresses the second research question concerning how DLLs are implemented in practice.

Chapter eight contains a brief discussion of three additional findings regarding

the concept of design methodology PhD.

Finally, in the conclusion in chapter nine I reflect on the main contributions of my research and discuss implications for practice, recommendations for future research, and a few general insights about design.

1.5. List of publications

Publication A

Sjöman, M., Ringenson, T., & Kramers, A. (2020) Exploring everyday mobility in a living lab based on economic interventions. *European Transport Research Review*.

This paper presents the first living lab in Mistra SAMS, *Future playing-rules for mobility*, and was written shortly after its closing. This DLL was conceived to learn about people's images and understandings about the costs of travel and especially car-ownership. Nine people's travels were followed in detail for six months while they were subjected to economic incentives designed to promote sustainable mobility. The paper mainly analyzes and describes the research outcomes of this lab, and in this thesis, it serves as an example of the breadth of the often unexpected findings that commonly emerge from a DLL, and how these findings may challenge or reframe current understandings. Writing this paper, we initially tried to present the findings as one cohesive but multi-faceted story of the many ways that car-use is integrated in society and how people internally legitimize their car-use. The reviewers however demanded that we connect all the separate findings to the literature in their respective fields, and as none of the living lab researchers possessed this breadth of domain-related knowledge, we had to carry out a very broad and time-consuming literature review. In this paper, the DLL as a research method is only very briefly presented as "a living lab based on design methodology" with a few notes on how it differs from the dominant understanding of living labs.

This paper was written in close collaboration with then PhD student Tina Ringensson. I had the main responsibility for framing the paper. Tina wrote most of the introduction and background, and provided a first overall structure for the paper. We carried out the qualitative analysis together, and I and carried

out the quantitative analysis. We discussed and developed the results section and the discussion together. I wrote the main parts of the method section and conclusion, and carried out most of the literature review required for positioning our contributions in the literature.

Publication B

Sjöman, M., and Hesselgren, M. (2020) Designerly Living Labs: Early-stage exploration of future sustainable concepts, in Boess, S., Cheung, M. and Cain, R. (Eds.), *Synergy - DRS International Conference 2020*.

This paper was the first attempt to analyze and describe the designerly living labs methodology more in depth. It was written in parallel with my work in the DLL *Work-Closer*, and frames this work as a step in further exploring and developing the methodology from the KTH living lab antedescents, *Car-Free* and *LEV-Pool*. The analysis also included many examples from the DLL *Playing-Rules*.

In this paper, designerly living labs are described as an explorative 'fuzzy-front end' of research, and as sharing eight characteristics, being early-stage, explorative, interventionist, holistic, participatory, open-ended, and with close researcher-participant relations. An earlier description of interventions as being either soft or strict (Hesselgren et al., 2017) is discussed more in depth, and developed to include participant recruitment strategies. In *Work-Closer*, I added a new way to encourage more active user participation by use of 'Challenges' which is presented here, as an experimental method to balance and handle the issue of strictness. These characteristics and concepts are all further developed in this thesis.

I wrote this conference paper with support from Mia Hesselgren, who read my script, discussed the contents and conclusions, and helped structure it more clearly.

Publication C

Vaddadi, B., Ringenson, T., Sjöman, M., Hesselgren, M., & Kramers, A. (2022). Do they work? Exploring possible potentials of Neighbourhood Telecommuting Centres in supporting sustainable travel. *Travel Behaviour and Society, 29, 34-41*

This paper presents the main findings from the second Mistra SAMS DLL, Work Closer. Like publication A, this paper is provided as an example of the outcomes of a DLL and discusses the difficulties in analyzing mixed data from many sources. Work Closer aimed to explore travel reduction but wound up focusing on working life and related aspects, which serves as an illustration of how research into sustainability transitions has to include interdisciplinary perspectives and societal dimensions. It also illustrates how DLLs tend to uncover such unanticipated issues and may have to flexibly shift their focus over time. The paper also introduces a discussion of how participants' different living conditions may allow some to be "forerunners," making more changes to their lifestyles during the experiment, while other participants may find that there are structures that limit their participation.

This paper was mainly written by then–PhD students Bhavana Vaddadi and Tina Ringensson, with Vaddadi assuming primary responsibility for the paper and conducting the quantitative analysis. Ringensson assumed primary responsibility for the qualitative analysis in MaxQDA. I took part in planning the article and in the qualitative analysis, and wrote parts of the introduction, discussion, and most of the conclusion.

Publication D

Sjöman, M., Hesselgren, M. (2022) Designerly Living Labs: Design-driven experimentation. In: Oldbury, K., Isaksson, K. & Marsden, G. (2022) Experimentation for sustainable transport: Risks, strengths, and governance implications. *Linnefors förlag, Boxholm.*

This anthology chapter is included since it presents another step in my analysis of the DLL-methodology. This text is mainly aimed at practitioners working with mobility experiments, and highlights how DLLs' more exploratory research design and the participants' role as "co-researchers" are two strategic research choices that are suitable for early research stages. DLLs are concerned with framing issues rather than testing and evaluating solutions, and their purpose emphasizes learning rather than immediate problem-solving. This publication also discussed how DLLs are participatory even though the interventions are designed and controlled by the researchers.

I was responsible for the idea, the background literature review, and the analysis of the methodology. Mia Hesselgren rewrote and condensed the text

to clarify and emphasize the two strategic choices, and to tailor the text to the anthology format.

Publication E

Ilstedt, S., Sjöman, M. (2022) The value of being close: Social and ecological sustainability in co-living for students, in Lockton, D., Lenzi, S., Hekkert, P., Oak, A., Sádaba, J., Lloyd, P. (Eds.), *DRS2022*.

This conference paper presents the initial results from the CoKitchen DLL. This DLL aimed to explore co-living as a possible solution for more environmentally and socially sustainable housing. In this project, a testbed area was remodelled twice in two years, from four single studio apartments to one co-living apartment, with more shared and less private space in each step. The paper gives a brief background to the project's methods and interventions, with a focus on a line of service concepts that sought to support the students in developing social and sustainable co-living practices. The paper also gives background to Sweden's current student housing situation and barriers for the development of co-living. In this dissertation, this paper mainly serves as an example of the types of research results produced in a DLL.

Both co-authors were involved in the analysis for this paper. Ilstedt was responsible for the paper and wrote the main structure and most of the introduction and background sections. I wrote most of the sections about the interventions and results, as well as the final parts of the discussion and the conclusion.

Publication F

Sjöman, M. (2022) Designerly Living Labs: Real-life experimentation in early research stages. *Continuous Innovation Network. Proceedings of CINet 2022*.

This conference paper presents DLLs to another audience, in the field of innovation management, and specifically delves into how DLLs differ from other real-life experimentation methodologies. In this paper I use a literature review of critiques of the main European stream of living labs, with the aim of showing how most other living lab formats mostly disregard earlier research stages, and involve user participants only in late-stage testing and evaluation. I then present a process overview that highlights these temporal aspects and the differences in when and how users are engaged. This paper was the runner-up

for the Continuous Innovation Network's "Best Implications for Practice" award.

How the publications relate to the research questions

The publications serve different purposes in this dissertation. In publications B, D, and F I discuss the DLL approach, its rationale, and its characteristics. Publications A, C and E are included mainly to illustrate the types of research outputs that DLLs produce. Answers to research question two, about implementation, are mainly discussed in publication D, while publications B and E provide background for discussing the need for interdisciplinary and transdisciplinary research collaborations. Publications A, C, and E include brief discussions of how the approach relates to design practice, but this theme is mainly developed in the body of this dissertation.

Table 1. The publications' relations to the research questions

	RQ1 Understanding DLLs			**RQ2 Implementing DLLs**		**RQ3 Designerly ways**
	Situations and aims	Character-istics	Outcomes	Collaboration and research conditions	Design competencies and capacities	Relation to design practice
A			x	(x)		
B	x	x	x		(x)	x
C			x	(x)		
D	x	x		(x)	(x)	x
E	x	x	x			(x)
F	x	x	(x)			x

The seven publications and the extent to which they discuss the research questions and related aspects. A publication marked x directly relates to the research question or aspect, while (x) denotes a smaller or more indirect connection.

2. The research context

This dissertation positions the Designerly Living Labs approach in relation to theory and to other research approaches in several fields of research. The broad literature review in this chapter therefore provides a background for a discussion of these relations.

2.1. Designerly thinking to frame complex issues

One important theme in this dissertation is how DLLs have been employed to address the need to find pathways for change in complex socio-technical systems. Design has been suggested as suitable for approaching complex and open-ended questions (Buchanan, 1992; Coyne 2005), often referred to as "wicked problems" (Rittel & Weber, 1973) or planning problems. These problems can be confusing, and clients and decision-makers may have conflicting needs and opinions. They are inherently ill-defined or indeterminate, since the questions can't be fully defined without developing a solution and since there is never one correct solution—only apparently better or worse ones. Rittel and Weber hold that wicked problems such as sustainability cannot be solved directly but instead require further exploration and reframing of the issue. These activities have been found to be typical of design. Carlgren et al.'s (2016) study of design at six large corporations identified this process of further exploring and even expanding a problem—rather than narrowing it down too quickly—as one of five central themes of Design Thinking. They describe this problem-framing process as unconstrained and futuristic in its thinking, and with an openness to the unexpected.

Design is a suitable framework for addressing wicked societal problems, because design creates knowledge by doing and making, using a "doing-first mode of thinking" (Mintzberg and Westley, 2001). Zimmerman et al. (2010) point to Research through Design (RtD) practices of "doing" or "making" as

especially well-suited to address wicked problems such as societal change. As will be further discussed in this dissertation, “doing first” and taking the leap to making is at the core of the DLL approach. Design is interventionist, and making or changing something to start learning is a natural way for designers to approach complex issues. In Buchanan’s description (1992), the designer makes a preliminary invention or a working hypothesis that is then used for further exploration and development. In this way, design frames complex or wicked problems, in parallel with attempts to solve them. This means that “[d]esign activity includes not only solving the problem but also re-defining the problem in the light of emerging solutions” (Archer, Baynes, and Roberts, 2005, p 4). Complexity can also be viewed as richness, and designers do not shy away from the complexity of cultures, emotions, or opposing perspectives. Instead, design is a holistic approach that is known for integrating knowledge and theories from across many disciplines (Zimmerman et al. 2010). Design research methods typically produce rich experiential data, which is needed for inspiration. In contrast to research in other fields, design also has proven capabilities for dealing with mess and ambiguity (Stolterman, 2008) and uses this to create a space for action (Linse, 2017).

Design is also concerned with the future and may “conceive and plan what does not yet exist” and may do so by creating a design as a “working hypothesis” while still firmly within the indeterminacy of a wicked problem (Buchanan, 1992 p 18). Designerly acts of speculative making are useful to explore, frame, and reframe complex challenges. In the words of Anthony Dunne and Fiona Raby (2013), there is a divide between “affirmative” and “critical” design; in the first, one designs for consumption and “for how the world is,” while in the second one designs “for how the world could be” by inventing parallel worlds to provoke and incite debate. These designerly capacities and predispositions for framing, reframing, and approaching complex issues by making are all central to the DLLs and will be further discussed in chapters 6 and 7.

2.2. Schools of Design and Design Research

This dissertation builds on a number of partly overlapping theoretical and methodological frameworks for design research, or schools of design, reviewed in this section. The last two decades have seen a rapid development of new fields and applications for design, addressing the societal challenges of our

time. Human Centered Design (HCD) is the dominant approach in the industry but has come under scrutiny lately as failing to support sustainability. Transformation Design, Practice-Oriented Design, and Participatory Design are three active and developing fields of design research that have inspired the DLL approach, and that this dissertation both builds upon and contributes to.

Human-Centered Design

Human-Centered Design (HCD) is the dominant approach to design today, but it is also a several-decades-long movement in the design field towards *designing for people*. HCD is concerned with the relationship between people and designed artifacts; van der Bijl-Brouwer and Dorst (2017, p 2) describe it as "a group of methods and principles aimed at supporting the design of useful, usable, pleasurable and meaningful products or services for people." One early step in this movement were the Scandinavian experiments with collaborative design in the 1970s and 1980s. These early examples of participatory design inspired a deeper interest in understanding user needs and perceptions. Another important contribution to the development of HCD came from the American design consultancy IDEO, with the concept of *Design Thinking*, which established a strong connection between user insight and design-driven innovation. IDEO described their process as empathic and insights-driven, based on ethnographic research approaches and multidisciplinary teamwork (Kelley, 2009). Today, HCD is widely accepted as a gold standard that is included in most international standards for product development. These documents commonly describe HCD as a detailed process with several steps. According to ISO 9241-210, the design process should be iterative, based upon and evaluated against an explicit understanding of users, tasks and environments (ISO, 2010). The British Standard BS7000 includes examples of common HCD methods: "Research into customer and user attitudes and aspirations, work and life-style is essential. Particularly instructive is observation of how users interact with products" (BSI, 2008). To reliably inform the design process, this research is carried out by cross-functional teams that include not only designers but also anthropologists or behavioral scientists. As Elisabeth Sanders (2005) describes it, HCD today has also firmly expanded into the "fuzzy front end" (Koen et al., 2002) of the design process. In this early and generative phase, inspirational research is increasingly carried out by designers using methods such as cultural probes (Gaver et al., 1999) or the making of rough prototypes or mock-ups—

tools and methods previously reserved mainly for the later phases of development (Sanders & Stappers, 2014).

Using the term *human-centered* rather than *user-centered* can be seen as a further step from the focus on ergonomics, cognition, and usability, towards greater emphasis on human experiences, values, dreams, and aspirations. In contrast to participatory design, even though user involvement is a core principle of HCD the active participation of users is not (see, e.g., ISO, 1999, 2010). In Sanders and Stappers' (2008) terms, HCD is mainly understood as designing for rather than with the user. More broadly, designers' understanding of users has developed to view them as capable of adapting or even co-creating designs, but HCD is still mainly *expert-led*. This means that user involvement is mostly used to inform and inspire professional designers (Sanders, 2005).

Because it reflects the general understanding of how design is used and taught today, HCD forms a backdrop against which other design approaches and methods are often contrasted.

Co-design and Participatory Design

Participation is a general term, but Participatory Design (PD) is also a diverse methodological field that has its origin in efforts to democratize the design of systems in the workplace by engaging skilled workers in the design process (Bødker, 1996; Ehn & Kyng, 1987). In the early development of PD, its methodologies were generally termed Collaborative Design (Bødker et al., 2000). PD often engages in normative issues of power and control (Ehn, 2017), aiming for legitimate participation to support resource-weak groups, opening up for and dealing with controversy (Björgvinsson et al., 2012). The power-related aspects of PD will be further investigated in the next sub-section.

Co-design is a more general term that mainly lacks this association with concepts of power and is more widely used in commercial contexts. Co-creation with users is understood to be particularly helpful in early, generative project stages and when developing more complex product-service systems (Sanders & Stappers, 2014). Co-design is often referred to as participatory, in that the designer is designing *with* the user as a partner rather than *for* the user as a subject (Sanders & Stappers, 2008; Steen et al., 2007), but it is not overtly concerned with democratic processes or user empowerment. In practice, co-design aims to actively involve users to inform the design process through

their knowledge and experiences. Co-design has been described as a deliberate effort to enable co-creation in design processes (Sanders & Stappers, 2008). Co-design workshops can facilitate co-creation using methods such as "experience prototyping" (Buchenau & Suri, 2000) or "generative tool-kits" (Sanders & William, 2001) to elicit user reflection and uncover tacit knowledge. By making physical artefacts such as prototypes or visualizations, designers can help users make sense of a situation and create ideas (Sanders & Stappers, 2014; Vaajakallio & Mattelmäki, 2014).

Co-design activities predominantly take place during the concept design phase (Voss et al., 2009), and in Sanders and Stappers' (2008) widely used co-design process model, the design activities end when a product or service is taken into use. Both participatory design and co-design do, however, include strands of longitudinal real-life research setups, often in the form of in-situ testing and development, or projects aiming to solve local societal challenges. Some of these can be framed as extended co-design approaches that include design during the use-phase (Botero & Hyysalo, 2013). In this view, a designed artefact will keep evolving in-use, and be further designed by users and other stakeholders. Design must therefore support and leave room for heterogenous groups of users to interpret, appropriate, or adapt the design (Henderson & Kyng, 1991). Several such methodologies, such as "co-realization" and "metadesign" (Fischer & Giaccardi, 2004) position the designer as continually present during the use-phase, but are mostly used in commercial workplaces for developing high tech projects in collaboration with expert users (Botero & Hyysalo, 2013). Botero & Hyysalo (2013) describe a rare extended co-design project outside the commercial realm conducting ICT development for a community of senior citizens. The authors note that this example demonstrates the need for strategies for longer and more "evolutionary design collaborations" to support "communities of practice."

This dissertation includes an extensive discussion of how user participants have been engaged in the DLLs. This long-term or "extended" participation differs from how users are commonly engaged in PD and Co-design, a topic that is the focus of subsection 6.4.

'Infrastructuring' and design for social innovation

Similar to extended co-design, the PD concept of *Infrastructuring* emerged

to support user communities in longer processes of building "information infrastructures." This approach may include not only the everyday adaptions and modifications that users make during use but also further implementation, maintenance, and redesign (Bjögvinsson, Ehn & Hillgren, 2010). In the last decade, PD—and especially Infrastructuring—have expanded towards more open-ended processes and more complex socio-technical assemblies. This line of PD research has sought to build, empower, and sustain communities of participants. Infrastructuring, in this sense is not "project-based design", but provides an open-ended and very long-term structure without predefined objectives or projects. This means that the structure is open to support the possibilities that emerge (Hillgren et al., 2011). The designers then provide user communities with socio-technical resources to deal with "matters of concern" (Le Dantec & DiSalvo, 2013). One prominent example is the work of Pelle Ehn and colleagues in the Medea Lab, which promoted grassroots initiatives in a multiethnic, lower-income neighborhood in Malmo, Sweden. In this way, Infrastructuring has been likened to designing a platform for social innovation (Manzini & Rizzo, 2011; Karasti, 2014). Design for Social Innovation (DSI) involves actively engaging stakeholders and user-communities as insiders, and facilitating for them to assume co-ownership of a design process (Bjögvinsson, Ehn, & Hillgren, 2010). In the book *Design When Everybody Designs*, Ezio Manzini (2015) describes an emerging DSI movement that is developing in parallel with a culture of "designing and making." Manzini describes this work as creating conditions that make people willingly engage in participatory 'life projects' to tackle local social issues. Of special importance are places, services, and platforms for social experiments, with design experts promoting and supporting grassroot organizations and creative communities that act as "place makers."

In this dissertation, I discuss the nature of these bottom-up community activation strategies and how these approaches are both similar to and partly conflict with the methods used in the DLLs.

Design for Sustainability

The DLL approach has been applied to address sustainability issues. In an excellent review of how design for sustainability has evolved, Ceschin & Gaziulusoy (2018) distinguish between four levels, from insular to more systemic aims. Early approaches, such as eco-design (Tukker et al., 2000), aimed to

reduce the harmful environmental effects of products. Such approaches include e.g., steps to reduce the use of toxic substances, or to design products that are more energy-efficient during the use-phase—efforts that have not always resulted in significant reductions, however, as efficiency gains can lead to increasing consumption.

On the second level, Design for Sustainable Behavior (DfSB) (Lockton et al. 2008; Lilley, 2009) instead suggests designing products to intentionally change unsustainable behaviors, using "design scripts" (Akrich, 1992) that nudge or inform users so they might make more sustainable choices. Behavior-based design approaches have largely been based on a rational choice-ontology. Examples include ICT solutions such as smart homes and in-home devices that track and manage personal energy consumption. This approach has, however, proven ineffective for the most part, and behavior-based designs tend to lose effectiveness over time (See, e.g., Oudshoorn & Pinch, 2003). Furthermore, as Strengers (2014) has pointed out, this vision of "empowering consumers" to actively manage their energy consumption by making informed choices, will not only be of interest to a small audience but also risks drawing our attention away from the practices that guide consumption.

On the third level, the design of eco-efficient Product-Service Systems (Ceschin, 2013) is a more systemic approach that is obviously relevant for the DLLs but is more closely connected to Niche Management. The design initiatives described in the previous subsection, such as DSI, are also examples of more systemic approaches with similarities to the DLLs but that are often limited to facilitating community-based local or possibly regional solutions.

Transition Design and Transformation Design

Ceschin and Gaziulosoy's fourth level is concerned with addressing more systemic issues and societal challenges. The term *transformation design* (Burns et al., 2006) was first proposed by the British Design Council's RED unit as it aimed to tackle social and economic issues through design led innovation. Later, Irwin et al. (2015) suggested the term *transition design* to refer to design that acts as an agent for change of complex systems. Overall, Transition Design has more strongly emphasized design as a way to "amplify grassroots solutions" (ibid.). An editorial presented at the 2018 Design Research Society conference strongly suggested that transition designers should tackle more-political

concerns, such as decolonialization and community activism (Boehnert et al., 2018). As will be attended to in the discussion, these research ideals have not been central to the DLLs. However, transition design also more strongly emphasizes the importance of local lifestyles as a relevant context for design, and in the same editorial Boehnert et al. also suggest more attention to everyday transition experiences through the use of prototypes or living labs (ibid.). The term transition design also better aligns with the understanding of transitions in the political sciences as referring to more "purposive" change, while transformations are understood as more emergent and less coordinated (Smith et al., 2005).

In summary, the earlier term transformation design placed more focus on design as a practice and a process and lacked the connotations linked to political perspectives. The term transition design has become more established and is more closely connected to relevant research fields, and as such it will be the term used in this dissertation. In transition design discourse, practice-oriented design is emerging as a potentially important approach in design for sustainability, and is outlined in the following subsection.

Practice-Oriented Design

In an influential presentation and manifesto from 2006, Elisabeth Shove introduced the concept of *practice-oriented product design*, suggesting that product designers move from focusing on single products and individual users to engage with social practices as the unit of inquiry, analysis, and intervention. Designers, she argued, should seek to understand and influence the dynamics of people's practices (Shove, 2006). Through an in-depth analysis of the elements of a practice and how it has evolved over time, designers may identify "points for intervention." Practice-oriented design (POD) has been an important inspiration for the DLLs from the start, and some of its concepts are required to understand the methodological choices involved.

Socio-technical regimes, by definition, are both technological regimes and "regimes of consumption," sustained through daily practices that millions of people consider normal (Shove, 2003, 2004). These consumption practices are *routinized behaviors* shaped and reproduced by several interconnected elements, including social norms and values, needs and expectations, habits and temporal routines, and physical and cognitive skills, but also by material

products and infrastructures (Reckwitz, 2002). Any socio-technical transition requires transitions in practice (Geels & Schot, 2007; Shove et al., 2007; Warde, 2005). Social Practice Theory (SPT) describes our daily practices as largely socially constructed, and has been increasingly applied for analyzing the social nature of everyday consumption (Warde, 2005). In direct opposition to rational choice theory, SPT views individual choice as limited, as our behaviors are shaped by contextual parameters, social norms, and material contexts. Changing a practice means forming a new working complex of meanings, materials, and competences (Shove et al. 2007). A sustainability transition, such as moving away from car usage, entails changing people's practices, which in turn necessitates changing an interlocking complex of values and understandings, material resources, and technologies, as well as societal institutions such as norms and regulations.

Practice-oriented design as a shift away from HCD

The last few years have seen a rising interest in more broadly practice-oriented design approaches, suggesting that design be oriented towards enabling change in practices (Spaargaren, 2003; Shove et al. 2007; Julier, 2007; Björgvinsson, 2008; Kuijer, 2014). POD suggests using practices rather than products as the unit of analysis: "not washing machines but washing; not cars, but commuting; not microwaves, but cooking" (Scott et al, 2012, p 283). This is also very close to Don Norman's (2005) concept of "activity-based design." Understanding the mechanisms of how practices change can inform the development of socio-technical innovations for more sustainable living. In viewing our behaviors as dynamic social constructs and our needs as malleable, POD seems to offer a better understanding of the challenges in designing for sustainability and provide ways to counter the limitations of other approaches (Shove et al., 2007). This move from identifying needs to understanding the dynamics of *how needs are formed* may turn out to be a paradigm-shift for design.

From a social-practice perspective, both user-centered and behavior-based design have been described as inadequate approaches to designing for sustainable outcomes (Scott et al., 2011). Human-Centered Design (HCD) aims to observe and deeply understand users and their needs but also tends to view these needs as static. Consequently, HCD often aims to fulfill identified user needs without considering how and why observed behaviors or expressed desires have formed. Using practices as the unit of analysis may reveal other ways to provide

valuable experiences or to replace unsustainable behaviors. In the same line of thinking, Norman (2005) has criticized HCD for failing to understand how technologies take part in shaping behavior and social structures, as well as for failing to address the sequential requirements of higher-level human activities.

The practical implementation of Practice-Oriented Design

Practice theory is not a method or approach that is directly applicable to design but rather a theoretical lens used to better understand social patterns of behaviors. Ida Nilstad Pettersen (2015) notes that integrating practice theory and design comes with some tension, as HCD emphasizes the experiences of individual users while a practice orientation implies that needs and emotions are part of, and result from, practices. On the other hand, common design practices such as the introduction of prototypes or new environments may help users deviate from what is considered normal or de-routinize practices (ibid.). We may also need to question the legitimacy of practices and why they are carried out. As Spaargaren (1997) has pointed out, we rarely question given levels of service. The impact of driving, for example, might be reduced by replacing fossil-fuel cars with electric vehicles, but POD can lead to more difficult questions such as why and when people should drive private cars in the first place.

In a review of early POD studies, Nilstad Pettersen (2015) finds that users have been involved to generate insight into what is normal and taken for granted and to design, explore and experiment with new practices. User participation in practice-oriented design is common and can be fruitful, but is not without challenges. As Spaargaren (2011) has noted, user-driven design approaches are likely limiting, as sustainable lifestyles are often counterintuitive. Sustainable lifestyles are often initially seen as reducing, limiting, or refraining from something, and in the DLLs we have found that the value gained is generally understood only later.

One early influential example of POD methodology was a study set up by Kakee Scott, Jaco Quist, and Conny Bakker in 2008 to find alternative, more sustainable bathing practices to reduce domestic energy consumption (Scott et al., 2011). The authors devised a process of alternating between stages of reflection and stages of experimentation by deviating from current practices. Twelve women were supported in the use of practice theory to first deconstruct their current bathing practices and then to make deviations from them by designing

new “proto-practices.” Overall, the researchers found that the outcomes gave reason for optimism about the use of POD, as the participants clearly found ways to challenge their current routines and understandings. They also noted that the pilot study included only a very short intervention, while “practice-oriented design research should ideally be conducted using an iterative, dynamic, longitudinal form to support the ongoing development of practices in real life” (p 286). In many ways, the DLLs presented in this dissertation answer this call, although certain specific differences are discussed.

2.3. Modes and views of user-participation

In this section follows a broad overview of user participation in several fields. In very general terms, there are substantive, instrumental, and normative arguments for participation (Bidwell, 2016). Substantive arguments view participation as a way to access users’ diverse knowledge and preferences, or their creativity. Instrumental reasons concern its usefulness for achieving set objectives by fostering understanding, trust, or engagement. Normative arguments are connected to democratic values, in the sense that the people who are affected by a change have a right to influence that change. Here, empowerment and active citizenship are also viewed as ends in themselves.

The notion of user is a generic term for those who use products and services to benefit from them (Whalen, 2020). In what is called the “participatory turn” in technoscience (Felt & Fochler, 2010), new forms of user participation are expected to improve the connection between technologies and their users. In practice, however, this participation is often limited to consulting or consuming (Chilvers et al., 2018). Weiland et al. (2017) found that pilot and demonstration projects often formulated technological “target knowledge” quite clearly, while knowledge production goals in the social realm tended to be formulated weakly. Also, because most pilots and demonstrations limit their aims to individual behavior change, user participation rarely goes beyond the testing of new technologies (Ryghaug & Skjölsvold, 2018).

The question of *how* users may participate is not straightforward, and the terminology used to discuss this question is diverse and conflicting. Terms like user involvement, user-centric and user focused are used with diverse meanings that range from vaguely user-oriented approaches and the passive sourcing

of user data, to consulting with users, actively involving users in testing, or working together with users in co-design processes (see, e.g., Kujala, 2003; Steen & van Beuren, 2017). In most cases, the term *user involvement* implies that users have an influence on a research or development process through active partaking or by being consulted. However, in a series of oft-cited papers, Hartwick and Barki (1989, 1994) suggest the term should be used in a sense more in line with psychology, to signify users' level of motivation or personal interest in an issue. In the following, I use the term involvement in the former sense, as a very broad term for the inclusion of user perspectives. In line with Hermans (2015), I distinguish between involvement, which may be passive, and engagement, as including personal motivation.

The term user participation generally implies more *actively* partaking. Damodaran (1996) ranks levels of user involvement from informative through consultative to participative. In product development, Kaulio (1998) has suggested categorizing user involvement into three levels, where designing "for" the customer denotes the use of data on customer preferences, needs, and requirements, while designing "with" the customer implies a greater customer focus that might include displaying different solutions or concepts to elicit customer reactions. The last level, design "by" the customer, means that the customer actively partakes in the design work, which is described as "a qualitative leap in the relationship to the customer where the distinction between customers and designers is blurred" (p. 147). More well-known in the design field, the work of Sanders and Stappers (2008) instead distinguishes only between designing "for" or "with" the user, with the latter denoting active participation in the design work. Using this distinction, designing *with* the user mainly refers to co-design (see section 2.2) and denotes focusing on "how professionals can capacitate lay people to design" (Whalen, 2020, p. 31). Overall, most taxonomies of user participation in co-design concern users' level of ability or their activity in designing products or concepts. In contrast, in user innovation and participatory design, user activity may include a broader range of activities, such as taking part in exploratory activities or in facilitating or mediating collaborative processes. In participatory design and many other research fields, user participation is also often related to power. Building on definitions from the field of community engagement, Steen and van Beuren (2017) characterize user participation in living labs as "having *decision-making power* in the development process" (p. 28, emphasis added). Similarly, participatory design emphasizes democratic values such as users' rights to inclusion in matters that

affect them (see, e.g., Ehn, 2017).

In this dissertation, I discuss user participation in the DLLs as an active engagement in exploratory activities, where a focus on power-related perspectives is mostly not relevant. I will mainly use the term user participants for individuals who take part in a DLL in the role of potential users. I will also use the term co-researchers when highlighting reflexivity as a central research contribution (see further discussion in 6.4).

Ideals of empowerment and insider-led modes of research

When involving users to participate in research, the normative issue of power is frequently discussed. This power dimension and instrumental reasons for user participation are both more relevant in situations when changes will be made that directly affect a group of people. In real-life experimentation studies that aim to develop or implement a societal change in the here and now, the involvement of affected communities is a central concern. It can be argued that people should be allowed to have an influence on changes to their lives and neighborhoods. Therefore, when experimentation is viewed as a form of urban governance, Evans (2016) asks us to critically view the disruptive potential of "improvements" that are sought through experimentation. In many studies, the "disruptive potential" of experimentation can however be viewed as limited. Brown et al. (2008) discuss how urban experimentation is often "bounded," both spatially and temporally, as well as by users recognizing "the effort to be an experiment" (p. 112). Still, no real-life experiment can be completely bounded, and a study will always have some sort of influence on future developments.

In action research, there is always an aim to change participants' current situation or practices (Brown & Dowling, 2001; Elliott, 1991) and to make a difference in the here and now. In line with these aims, Herr and Andersson (2004) suggest that in participatory action research (PAR), user participants are always either in control of the research or participate in shaping the design and methodology of the research. Since the users are affected by the proposed change, their voices, power, and representation become important. In community-based participatory research (CBPR), the ideal is to involve insiders or local communities as equal partners or in leading the initiative. CBPR often deals with issues of concern for marginalized or disenfranchised communities, and research is frequently judged by its position on a continuum

from insider-led to outsider-led or researcher-led (See, e.g., Sagaris, 2020; Hacker, 2013). Governance and urban planning studies often reference Shelly Arnstein's (1969) ladder of participation, which describes eight levels of citizen's decision-making power. Full citizen control is viewed as the ideal, while consultation is considered mere "tokenism," as there are no guarantees that the input will be taken into account. Similarly, Bergmann et al. (2012) divides the intensity of involvement of societal actors into four stages: information transfer by consultation, cooperation, collaboration, and empowerment. Tripp (2005) instead ranks equality and freedom of choice to participate in research on a scale from compulsion to collaboration.

Participatory design emerged from the strong democratic ideals of the early Scandinavian collaborative design movement, where designers worked together with workers' unions, empowering them to influence their workplaces and tools (Bødker et al., 2000). These views seem to have reemerged in the last decade, as participatory design has turned to a broader set of societal challenges (Blomberg & Karasti, 2013). Similar to PAR and CBPR, participatory design and design for social innovation both focus on insider-led approaches, supporting the development of grassroots initiatives and engaging user communities to take action (Björgvinsson, Ehn, & Hillgren, 2010; Manzini, 2015). Even in this context, design has, however, been criticized for experimenting on rather than with the users. Steen (2011) notes that designers often construct projects to test their own ideas and fail to consider the views and interests of their participants. Critical voices have discussed how "truthful" participatory design approaches require open negotiations with the participants (Pihkala & Karasti 2016). As with CBPR, this is said to be particularly important in community-based projects (Kraff, 2019).

The normative democratic ideals of citizen control have, however, been questioned. Drawing on an abundance of examples from urban planning practice, White & Langenheim (2021) effectively turn Arnstein's ladder sideways to show how all steps of the ladder may be needed and useful in different situations and contexts. In particular, the authors note that designed and visualized proposals can make consultation more productive and that mere information and even manipulation may be needed to handle extreme or very limited views. Importantly, they point out that even when experimentation does aim for lasting change, full citizen control is generally not appropriate, practical, or effective. The DLLs have all been clearly research-led, and this dissertation includes a

discussion about when and why this stance may be needed in future-oriented sustainability research.

2.4. Real-life experimentation

Transitions, and why real-life experimentation is needed

Sustainability transitions—meaning broad system-level change—are needed to solve the critical environmental challenges of our time. There is a broad consensus that new technology alone will not be enough and that radical, system-level innovation is needed, such as circular business-models or radically new transport systems (Freeman, 1992; UN, 2015). Likewise, a simultaneous shift is needed towards more sustainable lifestyles to reduce consumption (UN, 2015; Jackson, 2008). This understanding has been central for the development of the DLL approach at KTH.

The Multi-Level Perspective (MLP) and Strategic Niche Management (SNM) literature streams discuss societal transitions considering three analytical levels: niches, where radical innovations occur, system-level regimes, and the exogenous socio-technical landscape–or society. This framework describes transitions in which stable socio-technical regimes are replaced under the dual pressure of slow and gradual changes in the surrounding landscape and the rapid development of new technological niches (Rip & Kemp, 1998; Geels, 2002). This understanding of the mechanics of societal change has been central to the increasing emphasis in research and in society on real-life experimentation, of which the DLLs are an example. Sustainability transitions require the co-evolution of economic, cultural, technological, and institutional systems (Geels, 2004; Rotmans & Loorbach, 2009). Building or changing a socio-technical system has therefore been found to require continuous experimentation and learning (Loorbach, 2007; Welch & Warde, 2015; Caniglia et al., 2020). Solutions to this kind of challenge must be explored in everyday life situations to understand how people appropriate them (Dourish & Button, 1998; Spaargaren, 2011) and to allow us to anticipate the resulting new socio-technical configurations (Loorbach, 2017; Edwards & Bulkeley, 2017). Ceschin (2013) highlights the need for real-life socio-technical experiments when developing sustainable product-service systems intended to promote more-sustainable lifestyles, and suggests that design can support these experiments (ibid.; see

also Ceschin 2014). Furthermore, as Scott et al. (2012) notes, these kinds of experiments require the ability to imagine and construct "entirely new complexes of behaviors, values and things," leading them to suggest "designing change by living change" (p. 285). Since sustainability innovations inherently challenge existing structures and lifestyles (Ceschin & Gaziulusoy, 2016), users and other stakeholders may, however, resist trying them out in the context of their daily lives or current operations.

Rip and Kemp (1998) describe niches as places where dedicated actors nurture and further develop an innovation to better align with the surrounding society and create "configurations that work". SNM specifically investigates how technological niches can be facilitated and supported by incubator programs, or in designated test areas and collaborations. Through the provision of such "safe spaces", people may try out new behaviors, and innovations can be developed to challenge the current regime (Ceschin, 2014; Sengers et al., 2019). As will be discussed later in this dissertation, this view of experimentation thus includes the *normative intent to scale up*. The understanding that socio-technical experimentation is needed to explore potential sustainable lifestyles has been core to the DLLs, which can be considered as safe spaces in this sense.

Modes of experimentation

Reflecting this understanding of transitions, and to address the great environmental and social challenges of our time, many have observed an ongoing shift towards approaches that are based on real-life experimentation, both in research and in society at large (Schäpke et al., 2018; Sengers et al., 2019; Enwall, 2021). The envisioned type of experimentation, however, shares little in common with the traditional concept of highly controlled scientific experiments. The term bounded socio-technical experiment has been introduced for smaller-scale multi-stakeholder projects that introduce innovative technologies or services with the aim of developing new functional socio-technical configurations through learning-by-doing and iterative improvement (Brown et al., 2003). Today, this experimental shift or turn is clearly visible in the growth of approaches such as transformation labs, urban living labs, real-world laboratories, future labs, fablabs, challenge labs, and policy labs (Corsin Jimenez, 2013; McCrory et al., 2020).

Different research fields assign different meanings to the terms experiment

and experimentation. In the words of Donald Schön (1983): "In the most generic sense, to experiment is to act in order to see what action leads to. The most fundamental experimental question is, 'What if?'" (p 145). This implies an experimental intention, although rather open-ended. In contrast, Latour (2004) famously views the whole of society as an unstructured and collective experiment. More in line with Schön, Sengers et al. (2019, p 161) define a sustainability transition experiment as "an inclusive, practice-based and challenge-led initiative, which is designed to promote system innovation through social learning under conditions of uncertainty and ambiguity"—a usage that underscores the fact that such experiments are intentionally designed for this end.

Drawing on pragmatism, Ansell and Bartenberger (2016) describe three main modes of experimentation for sustainability: Controlled, Darwinian, and Generative. Controlled experimentation is mainly deductive, aiming to answer questions, validate hypotheses, or identify causal effects, which requires a highly controlled environment. Darwinian experimentation is mainly inductive and concerned with encouraging many and varied experiments to observe their outcomes (this may include post-hoc observation of societal changes). Generative experiments are mainly abductive and concerned with "learning by doing." Such experiments may involve iteratively refining a prototyped concept to find a working solution. In the transitions literature, bounded socio-technical experiments and design experiments are mainly understood as iterative development processes and thus generative in Ansell and Bartenberger's framing. On the other hand, Ansell and Bartenberger (2016) also describe Strategic Niche Management as mainly Darwinian, since research infrastructures such as living labs generally fund and support multiple experiments or even systems or "ecologies" of experimentation to promote change. Similarly, cities or regions may use "experimental governance" to promote multiple policy interventions and improvement efforts targeting broad societal goals while also including structures for self-monitoring and institutional learning (Castán Broto & Bulkeley, 2013; Sabel & Zeitlin, 2011). This approach responds to environmental volatility and uncertainty with greater flexibility, by decentralizing goal implementation to bottom-up initiatives such as frontline service agencies or citizens' councils (Sabel & Zeitlin, 2011). As these Darwinian approaches broadly aim to develop and to scale up solutions, policy interventions conceptualized as experimental governance are not understood as provisional or bounded experiments by citizens or users, but rather as improvement efforts.

Experimentation to provide actionable knowledge

A pragmatic approach to research values knowledge that is relevant or useful in the real world: i.e., knowledge is valuable if it lends itself to being acted upon in the real world. Argyris (1996) notes that actionable knowledge is not only that which is externally valid or relevant but also provides information on how to create or bring about intended consequences in practice, "in settings beyond those in which it was first created" (p. 392). Mach et al. (2020) argue that in more recent conceptualizations of the term, actionable knowledge does not necessarily include practical guidance but can be in the form of, e.g., data or analyses that are sufficiently accessible to support relevant actors in decision making. Meagher et al. (2008) suggest that societal research impact can be instrumental or conceptual, with instrumental knowledge being directly used for a specific decision or solution while conceptual impacts influence policymakers and practitioners in more complex and indirect ways that are hard to assess. Conceptual impacts may include media coverage, increased awareness, changes in attitudes, or the emergence of new networks.

Against a backdrop of the many calls for new modes of science to support transformative change for sustainability, Caniglia et al. (2021) highlight how different forms of experimentation, as well as transdisciplinarity or co-production, can provide actionable knowledge. Exploratory and open-ended real-world interventions and reflection on them may identify pathways to change and provide actors with possible actions in response to new situations. As will be discussed in this dissertation, this has clearly been the case in the DLLs. Similarly, the living labs literature emphasizes how actors may co-develop actionable knowledge as the labs provide "a means through which to gain experience, demonstrate, and test ideas" (Bulkeley et al., 2016, p. 14). On the other hand, real-life experimentation is always situated, and results from one real-world setting may be highly contextual and not transferable to another.

In a review of 59 sustainability-related research projects, Jahn et al. (2022) found that strictly academic research generated much higher academic impact than did applied, practitioner-oriented, and transdisciplinary research, and that generally there seems to be a trade-off between academic and societal impact.

Experimentation in this dissertation

Overall, the many and partly conflicting understandings of experimentation

may lead to a lack of clarity and confusion. In this dissertation I mainly limit the discussion to spatially or temporally bounded interventions and experimentation setups in real-life contexts. In line with Sengers et al. (2019), I also focus on experimentation that explicitly intends to test or learn about socio-technical configurations, and that ultimately aims to address a sustainability challenge.

The approaches discussed here do not include controlled experiments to estimate causal effects or *continuous* experimental development such as A/B testing or durable installations that seek to optimize a service based on data feedback. I also exclude so-called "natural experiments" in the sense of observational studies, where the experimental conditions have been determined by nature or by factors outside of the researchers' control, such as political decisions.

2.5. Living Labs

The concept of living labs can be situated within a broader field of real-life testing and experimentation approaches. Living labs are the most widespread mode of real-life experimentation in the literature, and an overview of this literature has relevance for many other similar concepts or methodologies, such as urban living labs, real-world laboratories, future labs, and transformation labs. There is a great deal of overlap between these approaches, and in this dissertation the concept of a *lab* is used broadly and discussed in more general terms[3]. In the discussion below this broad relevance for many types of real-life experimentation research is implicit, while I specifically highlight important differences between the specific variations.

Many researchers describe the concept of living labs as having roots both in the Scandinavian tradition of user involvement in design processes and in von Hippel's (1986) work on user-driven innovation (see, e.g., Schuurman, 2015). In Europe, a strong living labs movement began in the early 2000s, driven by European Union (EU) policy measures to accelerate innovation processes built on user involvement, which led to establishing the European Network of Living Labs (ENoLL) as a platform to foster shared methodologies. Ballon et al. (2007)

3 As an example of this, Fritzsche (2018) uses the term open laboratories as a catch-all phrase for all "lab" approaches that allow companies to engage with external actors.

suggest that living labs can be positioned between early-stage prototyping and field trials and the more mature commercial concepts tested in pilots.

Sustainable Labs North-Western Europe (SUSLabsNWE) has proposed Sustainable Living Labs (SLL) as a triple-helix co-creation approach to innovate solutions for reducing the environmental impacts of our society (see, e.g., van Timmeren & Keyson, 2016; Romero Herrera, 2017a). SLLs focus on understanding how users appropriate technologies (especially ICT solutions in the home) to potentially enable more sustainable lifestyles or seek to engage local users and stakeholders in developing sustainable solutions (van Timmeren & Keyson, 2016; Liedtke et al., 2015). This more user-centered approach is more explicitly inspired by design, with an emphasis on users self-reporting their energy consumption practices and co-creating and testing solutions.

Living labs as infrastructures

Living labs are often described as providing an experimentation environment or a milieu for organizing user participation in innovation processes, which may be a physical place, such as a city or a university (Bergvall-Kåreborn et al., 2009; Leminen & Westerlund, 2014). In line with the concept of open innovation, living labs have also been increasingly described as multi-actor networks or innovation arenas that facilitate multi-actor collaboration and open innovation (Schuurman, 2015). In line with, and in some cases inspired by, Strategic Niche Management, these mainstream European living labs are understood as *research infrastructures* that host or support multiple experiments. Ansell and Bartenberger (2016) describe this as a Darwinian approach to experimentation, since these infrastructures aim to promote an "ecology" of experiments in order to collectively further development (see, e.g., Marvin et al., 2018; Sengers et al., 2018). Many living labs thus effective encompass both physical and institutional infrastructure and commonly make a business out of providing test facilities or networking opportunities for commercial stakeholders. ENoLL (2020) has described living labs as "open innovation environments attracting inwards investment". These labs place much weight on operations management, funding structures and business models to sustain their operations (see, e.g., Mastelic, Sahakian & Bonazzi, 2015) and these factors are part of the defining criteria for ENoLL members. Nevertheless, Ståhlbröst (2012) found that living labs may also be established to carry out a single innovation project. In an assessment of ENoLL's members Schuurman (2015) found that almost 40% appeared to be

inactive, and that several of these labs seem to have been single-project labs.

Labs as a mode of urban governance

The current living labs literature, and especially the strand concerning urban living labs (ULL) increasingly highlight their role as a mode of experimental urban governance (Bulkeley & Castan Broto, 2013; Marvin et al., 2018; Caprotti & Cowley, 2017; Evans, Karvonen & Raven, 2016). ULLs are concerned with the management and development of Smart Cities (Laurent & Pontille, 2018), and the focus is therefore often on optimizing the sustainable performance of urban infrastructure through the use of digital technology and data-driven approaches (Hollands, 2015). In addition to these technocentric approaches, other ULLs are concerned with collective learning through means such as community engagement and and stakeholder consultation, and some ULLs are geared towards supporting and engaging grassroot communities (Bulkeley et al., 2016). ULLs are predominantly large-scale, and the "lab" is often a city or even a region.

Diversity and confusion

Living labs have been described various as a methodology or planning approach, an organization, infrastructure, ecosystem or network, as a physical or virtual environment, or as a combination of the above. Overall, living labs struggle with a wide diversity of definitions and understandings (Dutilleul, 2010; Leminen, 2015; Luederitz, 2016), leading Louna Hakkarainen (2017) to liken the concept to "a wet bar of soap." In their literature review, Dimitri Schuurman and colleagues describe living labs as "an empty box" that could contain any methodology or research approach (Schuurman et al., 2015). Today, a growing number of similar research and innovation structures likewise aim to govern or accelerate transitions towards sustainability, denoted with terms such as urban living labs (Voytenko, 2016), real-world laboratories (Schäpke et al., 2018) or challenge labs (Larsson & Holmberg, 2018). These methods have a broad overlap, and the same experimental setup may be described using more than one of these terms in different publications.

Limited user participation in technocentric experimentation

The active involvement of users is commonly described as a core principle of

living labs (Westerlund & Leminen, 2014; ENoLL, 2020). Because the European Commission has highlighted user-driven open innovation, living labs are commonly understood as a means to involve end users to have an impact an innovation process through their interaction with new artefacts in real-life settings (ENoLL, 2015; Priday & Pedell, 2017). How users are involved varies greatly, however (Feurstein and Schumacher, 2008; Hossain et al., 2019; Puerari et al., 2018). With their roots in open innovation (Chesbrough, 2003), living labs are often shaped by technocentric or managerial points of view. Leminen et al. (2014) identify four user roles: informant, tester, contributor, and co-creator. Users have been found to participate in activities such as using technological services or taking part in training sessions and meetings (Guzman et al., 2013). Kommonen and Botero (2013) found that most living lab initiatives claimed to be user-driven, but were instead "geared towards organizing and streamlining user involvement activities for a narrow product development process" (p 5). Thus, commercial product development projects may not emphasize actively engaging local communities in a living lab, or their input may be disregarded (ibid.). Mulder and Stappers (2009) note that living labs often harvest results from "demonstration-and-reaction" approaches, while long-term or deep involvement is rare.

As Marres and Stark (2020) observe, many living labs and other real-life experimentation setups are mainly "modifications in the environments in society so as to render the setting capable of data capture, analysis, and feed-back—that is, to equip it as a test environment [p. 436, italics in original]," which means that testing can be taking place at any time, and without one's knowledge. The European living labs movement is built on an understanding of living labs as "a set of methods and a milieu for leveraging user-technology reactions and interactions" (ENoLL, no date). Steen and van Beuren (2017) found that many ULLs relied upon sourced user data that was actively or passively provided by users.

Limited user participation in living labs

At an ideological level, living labs are also strongly connected to design practice, and especially participatory design, which many point to as their earliest origins (Schuurman et al., 2015). The term *co-design* is often included in living lab methodologies, and users are said to be involved in living labs, but no clear co-creation methodology is put forward (ibid.). Mulder and Stappers (2009)

have proposed more actively engaging living lab participants at the front end of research using generative techniques taken from the field of design, such as making "collages, diagrams, models, and other visualizations" (p. 5) to support self-observation and reflection. The authors note, however, that it is challenging for living labs to involve users in a generative capacity as co-creators. In marketing and management, user co-creation is mostly understood as involving users as informants in later product development stages, which is likely to have limited the use of design in living labs (Kommonen & Botero, 2013; Hakkarainen, 2017). In SUSLabsNWE, even though the focus was mainly on ICT and how people may develop an "ecological rationality," there was also an aim to use design methods to understand users' appropriation of technologies and their impact on daily life practices and sustainability (Romero Herrera, 2017b). A closer reading, however, shows that this was limited in practice. In the book *Living Labs - Design and Assessment of Sustainable Living*, a design researcher describes embedding design practice "within a more established quantitative environmental science and civil engineering approach" (Keyson et al., 2017, p. 300). The project is overall described in a relatively technocentric fashion, and design methods are adapted to suit this context. The methodology, labeled "Empirical Research Through Design" (Keyson et al., 2017b), is described as a highly structured and controlled process of rapidly developing and testing cumulative sets of working prototypes to reach acceptance of new technologies. Even qualitative user data is collected using digital and automated tools for logging and self-reporting.

User participation only in late stages

Building on Sanders & Stappers' (2008) map of the participatory design landscape, Marc Pallot and his colleagues (2010) created a domain landscape to position living labs according to the level of co-creation with users and the level to which social perspectives are included. Using this framework to assess 14 living labs, Salminen et al. (2011) found 12 were merely technological testbeds. Steen and van Beuren (2017) found that few urban living labs included pre-design research activities, and no labs included users in this phase; instead, active user involvement took place mainly in the testing, implementation and commercialization phases of a general innovation process (see also Mulder and Stappers, 2009; and Hakkarainen, 2017). Herth et al. (2022) investigated five living labs and found that they all described their activities as "commercialization."

Limited real-life experimentation in living labs

Another core principle of living labs is that innovation activities should take place in natural, real-life settings (ENoLL, 2015; Westerlund & Leminen, 2014). Living lab environments can, however, be virtual places, buildings, educational institutes, or even whole cities, and it is not clear whether real-life experimentation takes place within the context of people's everyday lives. In a literature review, Hossain et al. (2020) describe several different understandings of the term real-life environment, including everyday life usage contexts, physical experiential or creative facilities for designing and experiencing products and services, and the lab in a more abstract sense as a user-centric space in which to engage real users and other stakeholders in collaborative learning. The latter two understandings do not seem to denote people's everyday lives but merely "real-life" meetings between users and relevant actors. When users have been more actively involved in activities, this generally has taken place in workshops and not in real-life environments (Compagnucci et al., 2021; Steen & van Beuren, 2017; Kommonen & Botero, 2013).

In an anthology about SUSLabsNWE (van Timmeren & Keyson, 2016), there are some prominent exceptions, such as the well-known "splashing" study and the experiments with a concept named the "Power Chord", but in most cases, more open-ended experiments do not seem to have been set in real-life contexts. Many of the examples of projects using design methods seem to have been entirely conceptual student projects, and in some cases design researchers have engaged users through creative workshops and games. Several other studies in SUSLabs seem to have relied mainly on surveys, feedback harvested through mobile phone apps, or on passively sourced sensor data.

In my own reading of website texts and media communications for three urban living labs featured as examples in an urban living lab handbook, one ULL[4] was limited to only hosting student projects, workshops and a few public events. The only physical interventions that seem to have taken place were the placement of a park bench and a small community garden, and sponsoring a local chapter of the global charity Cycling Without Age. In another ULL[5], a French city which has been a well-known IT cluster since the mid-nineties was first labelled a living lab, and a few years later re-labelled a ULL. Lab commu-

4 (http://urbangrolab.nl, 2020)

5 (https://www.issy.com, 2020)

nications, such as booklets and online presentations, list and describe diverse activities that span from the billion Euro project of building a whole new part of the city, to the development of a mobile phone app to pay for parking, and a strategy for more customer friendly city administration. With the exception of a street survey and a community dialog meeting, I cannot find that users have been actively involved.

There are also lab formats that do not explicitly include prototyping or real-life experimentation but instead aim to engage and empower relevant actors in real-life dialog. Urban transformation labs (Nevens et al., 2013) are conceived of as spaces for dialog between a variety of stakeholders and user communities for framing socioecological challenges or city planning efforts. These dialogues build on extensive pre-studies and system-analysis to develop awareness amongst participants that they are part of a system and need to be the change that they want to see. These labs conceptualize real-life experimentation as a step that in practice takes place outside the "lab", which is mostly viewed as an arena for meetings and workshops to energize "frontrunners".

Overall, there is a broad tendency for living lab approaches and similar methodologies to either view the living lab as a *physical meeting place for co-creation activities*, or as *a real-life testbed where usage data is collected*. The real-life exploration and experimentation that takes place in a DLLs does not fit within either of these perspectives.

In the light of the limited user involvement in other approaches, the proposed DLL methodology poses an alternative. The DLLs have focused on the front end of research, as Mulders and Stappers (2009) propose, but have also taken a significant step beyond to actively involve potential users in more open explorations.

3. Methodology

3.1. Research perspectives

Practice-Based Design Research

First put forth by Frayling (1993), research through design (RtD) means conducting research by use of design practice and by carrying out design projects. RtD practices employ strategies of doing or making, often through the use of designed artifacts that serve to both reveal and to embody possible futures (Zimmerman et al., 2007; see also Forlizzi & Zimmerman, 2008).

In many cases the term *research through design* has been used more or less interchangeably with the closely related term *practice-based design research* (PBDR). Practice-based design research, however, emphasizes the use of *design practice as a means for research*, or in some cases that the research is carried out by designer-practitioner-researchers (Vaughan, 2017). As these qualities are central to the DLL methodology, I will mainly use the term practice-based design research in this dissertation. Overall, Mistra SAMS, CoKitchen, and the other studies that I have taken part in shared the explicit aim of integrating design methods to address complex and systemic sustainability issues, to demonstrate possible sustainable lifestyles, and to engage people and actors in learning processes.

Design and action-oriented research

Design and the social sciences are both broad and messy subjects that employ a plethora of methods that are the subject of ongoing debate. Some researchers describe practice-based design research as strongly connected, or even identical, to action research in the social sciences (Swann, 2002; Koskinen et

al., 2011). Action research has been described as action-oriented, experimental, and participative (Reason & Bradbery, 2008) and as following a cycle of planning, acting, observing, and reflecting (Zuber-Skerrit, 2018). As Cal Swann (2002) outlines, action-oriented research aims to change a situation for the better. Experimental, cyclical, and iterative approaches and the integral role of reflection are all central characteristics of design, but not all design subscribes to the often-stated democratic aims of action research. As will be discussed in more detail in 6.4, action research has been described as explicitly concerned with emancipation and social liberation (Bradbury, 2015), with a strong focus on close collaboration among agents within the research environment (Coughlan & Coghlan, 2002). However, this is the case only for some approaches to design research.

Therefore, while I acknowledge their similarities, I view PBDR and action research as distinct fields. Design, with its connection to art, involves nonverbal and visual thinking and places more emphasis on creativity and intuition, the future and speculation into what might be. Still, design is inherently action oriented, and the two fields share roots in pragmatism. Both have been described as new or "third" paradigms for research, and both share the challenges that come with this position. Furthermore, many of the rich methodological contributions of action research can be useful for design research. The research presented in this dissertation has clearly built on design practice, which is inherently action oriented.

Design research as a reflexive research practice

RtD simultaneously aims for and leads to both empirical and methodological results (Zimmerman et al., 2010) and to theory for design (Zimmerman & Stoltenberg, 2014). When performing reflective research through the use of design practice, it is possible to access tacit knowledge about the practice itself, which can, in turn, be used to enrich the practice. Ehn and Ullmark (2017) describe Practice-Based Design Research as reflective and theoretical work alongside practice. This means that the same design project may produce both domain knowledge (knowledge about an investigated field or solution) and ideas and knowledge about how to design (often referred to as prescriptive design) (Binder & Redström, 2006; Stappers, 2007; Zimmerman et al., 2010; Koskinen et al., 2011). In fact, design research in the field of human-computer interaction (HCI) indicates that many of the more important contributions to design

practice methodologies have been the results of RtD projects (Zimmerman et al., 2010) rather than research directly aiming to produce prescriptive design knowledge. My work is characterized by this practice of using design research methods to produce (domain) knowledge, alongside more theoretical reflection and methodological development.

Mistra SAMS and CoKitchen are typical cases of RtD, in that they have produced both domain knowledge on issues such as urban mobility and co-living and knowledge on the designerly methodologies employed. This means that I have built my PhD research as a parallel process alongside my role as an operational resource in Mistra SAMS, CoKitchen, and other studies. Also, because my engagement in Mistra SAMS and CoKitchen spanned several years, my understanding of the design methodologies we have been using has emerged in parallel with the experimentation in the projects. As Donald Schön (1985) outlines, while designing interventions and conducting research, I reflected both in and on these actions. There have been reflections made and noted both "in-action" while carrying out design research and later "on" these actions, especially when planning subsequent project stages, conducting analysis for written publications, or when presenting our work. The most important factors often became clear at times when I needed to explain what we were doing to the team or to people outside it. Such presentations and explanations increasingly called on us to give a name to our methods and to provide theories and references to back them up. Thus, while implementing, staging, and conducting research within the living labs, I gradually came to understand the labs as "designerly".

I view my research as *reflexive practice* (see, e.g., Finlay, 2002). This includes all the difficulties and reflexive requirements that come with this inside perspective (Alvesson & Sköldberg, 2017; Coghlan, 2004). First person inquiry involves dealing with emergent processes (Coghlan, 2007). This includes exploring and making sense of a research process from the inside while being immersed in a research organization where problems are messy (Schön, 1995). The theme of my PhD research gradually emerged as I explored and made sense of the research processes from the inside. During the several years long projects an understanding coalesced regarding what aspects of what we were doing were either designerly, or seemed new or different.

3.2. The cases and the data collection

It is difficult to provide a complete overview of the primary and secondary data that went into the processes of planning, implementing, and analyzing the DLLs, as well as the parallel process of developing our methods. Even though I had more limited involvement in other projects, my most operational engagement in Mistra SAMS and CoKitchen took up around sixty percent of the time I have been able to spend on research over the full course of my PhD research process. The analysis therefore builds on years of operational experience, discussions, and documentation.

For all the labs, the analysis has been based on the original research applications—in some cases including early project plans, project reports, published research papers, web pages, some news releases, and other communications. The analysis of Car-Free and LEV-Pool, in which I was not personally involved, also builds on years of conversations with my colleagues Sara Ilstedt, Teo Enlund and Mia Hesselgren. For the DLLs where I was operationally involved, my analytical process also included a partial rereading of my personal research notes, meeting notes (including almost a hundred photos of whiteboards), internal planning documents, and operational documents such as meeting agendas, participant information leaflets, and partner agreements. I also reread many internal and external project presentations—mine and others—and many of the published research papers. Over the duration of the projects, I have presented ideas and overviews of methodological choices for the purpose of discussing them with colleagues and research teams. These opportunities for reflecting and formulating methodological issues and choices in texts often led to important steps in the analysis. While writing this dissertation, I also revisited some central email conversations in order to understand tensions, responsibilities, and timings during critical events in DLL planning and implementation, such as when planning three workshops to reconcile differences over living lab planning in Mistra SAMS in September and October 2020.

Table 2 outlines my level of involvement in the DLLs. Here, data collection and analysis refer to the domain knowledge produced: for example, regarding mobility or co-living practices. As described above, I collected and analyzed data on research methodology and organization from all the DLLs.

Table 2. My operative research participation in the DLLs.

	Research design and planning	Practical designing and staging	Recruiting users and partners	Admini-stration and support	Data collection	Analysis
Car-Free	–	–	–	–	–	–
LEV-pool	–	–	–	–	–	–
Playing-Rules	H	R	R	R	H	H
Restaurant Lab	A	P	–		–	–
Work-Closer	H	A	A	H	H	A
CoKitchen	P	H	P	P	A	A
Riksten	H	P	A	–	–	–

–	No involvement
P	Partly or indirectly involved
A	Actively involved
H	Highly involved
R	Solely responsible

Active involvement entails taking an operational role in carrying out a project step. *Highly involved* means that I had a central role and carried out a large part of the work, often as part of a small team. *Solely responsible* means that I was the only person in the research group actively engaged in that project stage, which in some cases included collaborating with partners or commissioning work to external resources.

Table 3 provides an overview of which cases are analyzed in the appended publications. Publications B, D, and F include overviews or cross-case analyses of more than one DLL. The selection of cases was different in each publication to support the discussion of each article's topic. The number of cases has also been limited due to article length limits and the complexity of discussing certain cases. Publications A, C, and E each discuss a single DLL, and even if they contain descriptions of the research method, they are mainly used in the

dissertation as examples of the rich and broad results produced in the DLLs. These three publications focused on presenting domain knowledge concerning mobility, workplace structures, and co-living effects and practices, respectively. No scientific publications have emerged from the Riksten lab thus far.

Table 3. Cases used in the analysis for the publications.

Publications	Car-Free	LEV-pool	Playing Rules	Work Closer	CoKitchen	Riksten
A European Transport Research Review			x			
B DRS International Conference 2020	x	x	x	x		
C Travel Behaviour and Society				x		
D Anthology	x		x	x		
E DRS International Conference 2022					x	
F Proceedings of CINet 2022	x			x	x	

All the DLLs used a mixed methods approach, combining qualitative and quantitative data, as well as several data-collection methods. The varying scale of user participation has influenced how this was carried out in practice, and the following table provides an overview of the DLL's user participant data sources and collection methods (see table 4).

Table 4. Participants, recruitment, data collection methods, time and place for the DLLs.

User-Participants	Participant recruitment	Data-collection methods	Place and duration
"Car-Free"			
3 families. In total 5 adults and 6 children.	Self-recruited through Facebook advertising. 3 out of 72 applicants selected due to their motivation, and to achieve a heterogeneous sample.	6 rounds of bi-monthly semi-structured in-depth interviews, travel-diaries and car-use notes, travel-data collection app.	Lab duration 12 months 2014-2015. Stockholm, Sweden.
"LEV-pool"			
2 separate setups at two large Swedish employers, with a total of 16 care-takers and approx. 170 service users.	"Care-takers" were self-r ecruited using internal communication channels. The service was open for all employees and used by approx. 10 and 7 % of employees.	2x2 rounds of semi-structured in-depth Interviews, 2x2 user and stakeholder workshops, facilitated production of user-created short-movies	Lab duration 6 and 9 months, in 2016 and 2017. Stockholm and Älmhult, Sweden
"Playing-Rules"			
9 individuals in 7 suburban households in 3 areas with different accessibility options, all of whom owned a private car	25 individuals randomly approached outside food-stores in selected suburbs. After logging travel for 1 test-month, a mixed group was selected.	2 short structured interviews to get basic data. 3 semi-structured in-depth Interviews using scenarios and other trigger materials. Use of travel-data collection app	Lab duration 1 month baseline measurements + 6 months lab interventions, 2018. Stockholm, Sweden
"Work-Closer"			
73 participants recruited. Approx. 35 active and 8 highly active NTC users., all with full-time employment and relatively long commutes (around 1 hr).	25 users self-recruited through advertising in a local Facebook-group. 42 recruited through three large employers, for employees living near the NTC. 6 self-recruited (snowballing).	2-3 rounds of participant Interviews. Data from travel-diaries and the booking-app, Stakeholder interviews.	Lab duration 18 months, 2019 - 2020. Stockholm, Sweden.
"CoKitchen"			
5 students the first academic year and 4 (+1) student the second year. In total 39 other students took part in explorations of the service concepts.	24 users self-recruited through KTH Relocation homepage. Heterogenous group selected.	2-3 rounds of in-depth participant interviews each year. 2 surveys. 2 rounds of phone interviews with service users. Logged energy and water-use data.	Lab duration 2 academic years, 2020 - 2022. Stockholm, Sweden.
"Riksten"			
At the time of writing, 14 citizens take part as active co-researchers.	Self-recruited through local Facebook advertising and by snowballing from previously engaged individuals in the area.	TBD	Lab duration from June 2022 - TBD, Stockholm, Sweden.

3.3. How the cases were analyzed

Since I have taken part in several DLLs and also in a number of other real-life experimentation projects during my PhD, I analyzed the emerging DLL-methodology in many iterative steps. More structured analyses have been carried when writing of some of the earlier method-related publications (B, D, and F, and as described above). The analysis has also progressed in several more steps through discussions with the research teams and when drafting internal or external presentations. This is in line with Gray and Malins's (2016) description of analysis in the field of art and design as being iterative, reflexive activities. This iterative analysis has mostly been carried out "on the wall" (Sanders & Stappers, 2012) using methods from design practice, such as affinity clustering using sticky notes and images to identify method characteristics and emerging themes. Especially during the Covid-19 pandemic, I worked on my analysis online using the digital whiteboard service Miro. I coded and interpreted documentation, research notes, and interview transcripts either by transferring excerpts to sticky notes or using digital tools. In some steps, I conducted my analysis using visual mapping on whiteboards (see Figure 2) or using digital visualization tools such as Adobe Illustrator.

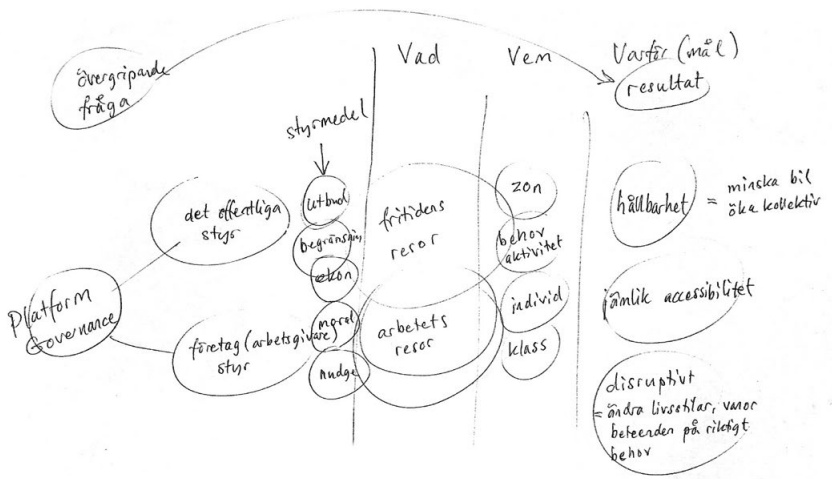


Figure 2. A whiteboard photo illustrating how preliminary and emerging concepts were structured visually while planning labs in Mistra SAMS.

In two cases, a database in MaxQDA was used for coding and creating a

more-structured analysis of selected lab-related documents. Special attention was given to identifying tensions, such as opposing views or differing uses of terminology in documents from different project phases or different project groups, as well as events or interpretations that were ambiguous or unclear.

The more results-oriented publications (A, C, and E) concerning domain-related knowledge produced by the labs were, in some cases, analyzed by interdisciplinary teams using more-conventional methods. For publication C in particular, the qualitative analysis was mainly done using MaxQDA in accordance with the qualitative content analysis method, as described by Schreier (2014). Even here, however, the existence of whiteboard documentation makes it clear that I also used visual methods to structure materials at meetings, and especially to develop insights and clarify relations among emerging themes.

Cross-case analysis

In this dissertation, I have taken a step back to conduct a broader and more structured cross-case analysis to validate and further develop the key characteristics of the DLL approach and and their interrelationships. In line with Stake (1994), as referenced by Piekkari & Welch (2018), I have tried to analyze each case as a whole, as comparisons across cases can easily lead to an analysis of specific dimensions outside of their context, where conditions may vary. The sample of cases consists of all the research projects carried out thus far that have included design researchers from Green Leap and that have been based on real-life experimentation using design interventions. By including a larger number of cases in the cross-case analysis than in the appended research papers, I have also attempted to arrive at more generalizations regarding the processes and outcomes.

In a first step, a more in-depth reading of the rich data from the DLLs let me to find and code an initially larger number of methodological characteristics. The inclusion of more background information about the research contexts and preliminary planning allowed the analysis to cover the important aspects of why and in which situations DLLs have been used. This reading also identified recurring types of research outcomes and impacts, which were included in the analysis. In a second step, I then analytically reduced these characteristics by comparing the DLLs with other real-life experimentation methodologies. I sought to include only those characteristics that seemed to distinguish DLLs

from other real-life experimentation methodologies in the literature. This step was demanding, as it required considering a large number of partly overlapping real-life experimentation approaches, such as pilots, demonstrations, testbeds, urban living labs, transition labs, and real-world laboratories—building on different ontologies, including open innovation, social innovation, and participatory design (Schuurman, 2015; Hossain et al., 2019). These approaches are, furthermore, often umbrella-concepts that include a vast array of specific methods and concepts (Leminen, 2015; Schuurman, 2015) using different terms and definitions (Luederitz, 2016). When reading about a designated type of real-life experimentation methodology, I therefore also read cases studies and searched for other project communications, such as websites or reports. Based on these readings, I suggest that the methodological descriptions in the literature can, in many cases, be considered ideal types, since the way these approaches are implemented in practice varies greatly.

In the end, twenty key characteristics or criteria were selected that include research aims, innovation principles, and typical outcomes. For each of these 20 criteria, I set up a set of indicators to enable validation. A structured cross-case analysis then aimed to identify events that either validated or disaffirmed, fit or did not fit with these criteria, for each of the six DLLs. To strengthen the analysis, I then analyzed a few key characteristics using an added criteria in the form of a negatively formulated counter-criterion. For example, the analysis of "Explicit aim to openly explore an issue or concept" as a key criterion was complemented with an analysis of its negative counter-criterion "No aim to test or evaluate a concept or to answer a specific question." The main cross-case analysis, criteria and findings are summarized in section 5.2.

3.4. Limitations

This dissertation is built on first-person inquiry as a part of a larger research organization. One main limitation has been that in some cases I had little control over and a limited overarching perspective on the studies that I took part in. With LEV-Pool and Car-Free, in which I did not take part, I naturally have little knowledge of the day-to-day developments and understandings at play during the implementation, since external-ly oriented communications and post-hoc narratives tend to present projects as more linear and structured than they may have been in actuality. I also was not part of the pre-planning

stage in CoKitchen. Instead, I worked closely with the project leader and was generally operatively in-volved in decision-making and had a good overview of the research activities. Mistra SAMS had already started when I was hired, and I was also mostly not involved in the pre-planning for the second phase[6]. During implementation I took a very active role in the day-to-day development of the DLLs, but with a few exceptions, I did not enjoy a mandate to devise methodologies. Even though there was a broad understanding that we were prototyping sustainable concepts within the framework of a living lab, researchers held conflicting views, and there was limited discus-sion about research methodology and aims. In Mistra SAMS, this limited my research in two ways: First, the lack of communication between the research groups and my not being a mem-ber of the executive committee meant I had limited insight into research work and discussions outside the living lab design group. Second, except for the practical implementation of Playing-Rules, I was not able to shape the living labs in order to try out hypotheses or answer research questions related to my dissertation. I did have an influence on their planning processes, but es-pecially in the second phase of Mistra SAMS, practical implementation came to be mainly car-ried out by researchers from other fields.

Finally, my understanding of the research perspectives of other program researchers is, of course, limited. My discussions here regarding other researchers' views are based on my inter-pretations of project documents, statements, and activities, and on literature from these respec-tive fields. In some cases, it has been difficult to engage other researchers in discussions regard-ing methodological choices or understandings. The issues at the center of this dissertation re-flect my own research interests, and other project members would most likely choose to discuss and emphasize different aspects or events. Therefore, the accounts and the discussion around the research projects presented in this thesis are mainly from a design perspective, and with a specific focus on the living lab interventions.

6 Even though my insights from SAMS' first phase may have been indirectly influential, as discussed in Section 6.5.

4. Empirical projects

4.1. The proposed DLL methodology

How the designerly living labs started

The Green Leap design group was formed by senior design researchers and practitioners at KTH and specifically aimed to act as a catalyst for change by engaging design as a tool for sustainability transitions. Early funding applications suggested the use of future prototypes to explore the positive sides of more-sustainable lifestyles (Ilstedt & Enlund, 2012).

In 2014, the radically innovative study "A Car-Free Year" was created. An independent research institute approached Green Leap to study the potential of light electric vehicles (LEV) to replace fossil fuel vehicles. This institute also hoped that design would provide more action and visibility than "normal" research approaches. As there was little knowledge about how people might deploy LEVs, a design practitioner suggested to "just take the car away from some people" to see how it might work out. The idea caught on as a way to publicly demonstrate sustainable future lifestyles. Three volunteer families were recruited to live without their cars for a full year. In all, the study was planned and implemented by designer-practitioner-researchers. I argue that Car-Free was the first of what has emerged as a line of Designerly Living Labs.

Explorative studies in early research stages

All the DLLs have aimed to explore more sustainable future lifestyles, addressing this challenge by prototyping possible sustainable concepts in the context of real-life social and societal structures. As Scott et al. (2012, p 285) note, this requires the purposeful construction of "entirely new complexes of behaviors, values and things."

With all the DLLs, there has been an understanding that there was insufficient knowledge to design and develop fully operational solutions. Although some of the prototyped concepts were viewed as promising, the DLLs were mainly intended as provisional first steps in a longer research process. Some of the DLLs identified issues for further research, while others led on to pilots or inspired new policy. The DLLs all aimed to:

- Explore complex sustainability issues in early research stages.
- Frame sustainability challenges, often by revealing related and unknown factors, relations, or matters of concern.
- Identify opportunities and barriers for desired lifestyle changes.
- Engage ordinary people in real-life exploration and demonstration of sustainable lifestyle changes, over time.
- Explore the adaptation and adoption of sustainable concepts, with a focus on social practices and changes on a socio-technical or societal level.
- Identify new types of value, that might be gained by shifting to more sustainable lifestyles.

Learning devices as a doing-first strategy

The DLLs have built on prototyping or *staging*, a part of a more-sustainable possible future. These future prototypes have included the concept of a car-free lifestyle, a vehicle sharing service, shared apartments designed for co-living, and policy interventions to encourage sustainable mobility. This staging has, in some cases, required the orchestration of efforts and resources from several project partners.

As discussed briefly in Publication B, these concepts have been intentionally provisional. When addressing a complex issue or wicked problem, it has been clear that knowledge is lacking, that relevant stakeholders were not ready to take action, or that the issues required framing to allow workable solutions to be developed. Therefore, provisional concepts have been designed as a starting point for learning and as a way to draw attention to the issue. The concepts prototyped in the DLLs in some cases built on lng histories of research but were often ideas that had not been tested or had been tested only in very different contexts. The concrete design interventions were then mostly ideated very quickly and did not generally entail extensive literature reviews or attempts

to align the concept with current user preferences and needs. Instead, the final step of envisioning each lab concept was based on prior knowledge and intuition (Hesselgren, 2019). With the exception of Mistra SAMS, the core ideas were generated before their research applications were funded. All the intervention concepts relied on normative visions of sustainable futures.

Real-life Interventions designed as learning devices

In the DLLs, prototypes of sustainable future concepts have been staged in real-life contexts. The most important difference compared to a more general understanding of living labs is that these concepts were mainly understood as "learning devices" (Sjöman & Hesselgren, 2020) and were not intended to be solutions per se, to be developed and scaled up. Thus, all the DLLs have been understood as bounded experiments and not as attempts to solve issues in the here-and-now. Even though some of the DLLs may look at the first glance like pilots for service concepts, the future prototypes were designed mainly for generative rather than evaluative purposes (see Sanders & Stappers, 2014). In most cases user needs were not researched before the concepts were designed, and the concepts were not evaluated or further refined during the DLLs. Instead, the DLL interventions openly explored how these concepts may fit, conflict with, or be integrated into people's lives and societal structures. The DLLs have thereby successfully identified barriers and pathways for sustainability transitions.

Longitudinal experimentation

All the DLLs were longitudinal studies, a design that allows time for new practices and understandings to evolve and become routine. Because the DLL participants lived with the interventions for several months, using cardboard props or enacting design scripts simply would not work. Therefore, the future prototypes included real vehicles and environments, as well as real money. Longitudinal research design is significantly different from most other examples of practice-oriented design, which are generally limited to short time frames: from one-time attempts to one or a few weeks (see, e.g., Scott et al., 2012; Kuijer, 2014).

Provoke, disturb or enable experimentation with new practices

Inspired by practice-oriented design, the interventions were "deviations" from the norm (Scott et al., 2012) that aimed to prompt adaptation, improvisation, and experimentation (Warde, 2005; Kuijer, 2014), as well as to trigger reflection.

DLL interventions may be strict or soft disruptions by adding or removing elements in people's lives (Hesselgren et al., 2017). The interventions—such as the removal of the participants' cars in Car-Free— enabled, triggered, or forced participants to experiment with changing their practices. Added elements have included both physical artifacts such as cocoon-shaped chairs in the CoKitchen co-living apartment, and service concepts such as that used in LEV-Pool, but also included intangible changes such as a conflict-solving exercise that was offered to the CoKitchen participants or the set of future policy changes that were explored in Playing-Rules. Importantly, even when participants did not make changes, the interventions prompted them to consider possible changes

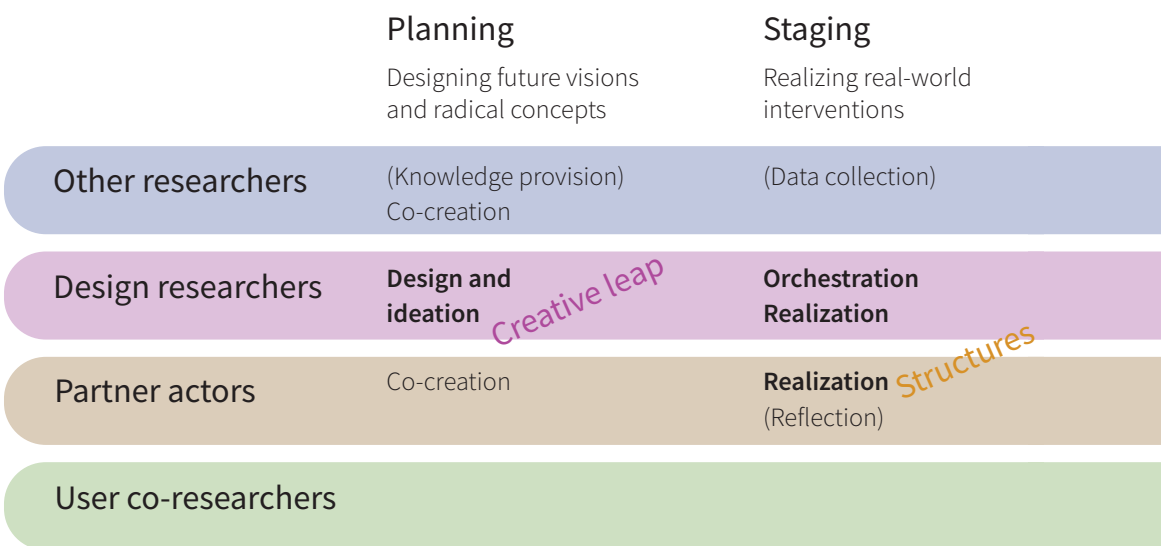
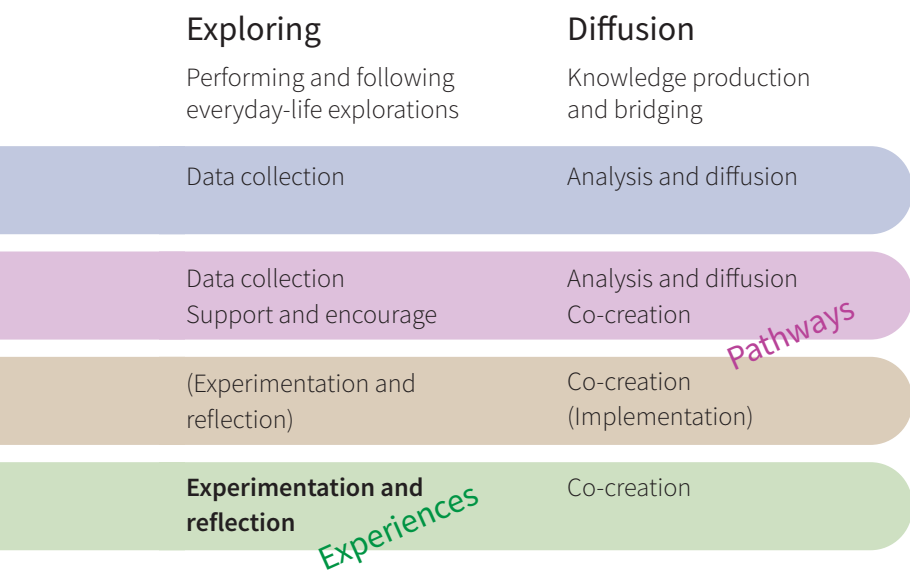


Figure 3. A generalized overview of the main stages, actors' roles and activities in a DLL. Also, the main focus of each of the four stages are highlighted in colored text.

and reflect on their current practices on repeated occasions, an exercise that often revealed unexpected barriers to the envisioned sustainable future.

Engaging users as co-researchers

The DLL user participants were actively engaged "co-researchers" (see, e.g., Publication B; Hesselgren et al., 2017) in exploring the design interventions. Drawing on Brown, Reeves, and Sherwood's (2011) conclusions from field trial methods, the living lab participants were not viewed as passive subjects to be studied but as active investigators of their own practices, who contributed as experts on their lives and experiences (Sleeswijk Visser et al., 2005) in a reflexive capacity (Finlay, 2002), a feature that is further discussed in 6.4.



Summary

The methodology described here is necessarily broad, as different research issues and contexts led to partly differing research designs. The DLL methodology includes a core set of ideas and a general structure, which are described below and illustrated in figure 2.

A DLL:

- Explores a possible future concept that is normatively assumed to support or lead to more sustainable lifestyles.
- The future concept is prototyped or staged in the present day as a bounded real-life intervention, often with support from relevant private or public partners.
- Interventions are designed as provisional tools for learning and not as solutions to be developed, tested, evaluated, and scaled up.
- By realizing or staging the future concept and making it "work", the researchers and partners identify how it conflicts with or does not fit into today's structures.
- A relatively small group of user participants are recruited as co-researchers who are closely involved, who understand the purpose of the lab, and who take part as experts on their lives and experiences.
- When interacting with the intervention, the participants note and reflect on their practices, since they are enabled or even forced to experiment with changing them.
- Learnings often include barriers for or pathways towards sustainable futures.

4.2. Earlier projects - antecedents

This subsection includes case descriptions of A Car-Free Year and KTH LEV-Pool, which were both conducted before my PhD research period.

A Car-Free Year (henceforth *Car-Free*)

Planning and historical background

I argue that Car-Free was the first of the emerging line of designerly living labs at KTH, and in many ways constitutes an ideal type. The small, independent research institute Sustainable Innovation (SUST)[7] approached Green Leap seeking to conduct a research project regarding the potential and limitations of electric vehicles, with the hope that design could contribute to more action and visibility in the sense of easily communicated results. SUST wanted to learn about "how far electrification can take us", including the potential of a new category known as Light Electric Vehicles (LEVs). They also had more-specific questions such as where charging stations would be needed to support the expected growth in electric vehicles. The group concluded, however, that there was insufficient knowledge about how these new electric vehicles would be adopted in order to answer more-detailed questions. One important starting point was a recent SUST report, which concluded that the CO2 footprint from one day of commuting in a diesel car equaled a full year of commuting by electric bike. At this time, a senior design practitioner at Green Leap suggested "just taking the car away from some people" in order to see what would happen. He argued that the idea would make "good TV"—meaning that it would likely result in stories that would be attractive to the media. The project idea was thus to replace participants' cars with rented LEVs. To get beyond the initial novelty and allow time for new practices to form, the study was planned to span a full year. A joint application for research funding was made, stating the explicit aim to "challenge the car norm."

Implementation and method

SUST took an active role in making the study feasible, managing practical issues such as removing and storing the families' cars and making sure that

7 The acronym SUST is commonly used, and originally derived from SUstainable Science and Technology

all associated costs were covered. At this time, LEVs were not yet available for rental, but leases were arranged through a collaboration with a commercial partner. However, efforts were made to design the experience of selecting and leasing the LEVs to make it feel like a regular service offering. The cost was subsidized to a price level that was considered plausible in a future where such services had become common. The families' routines and travel needs were not researched before the study, and in an interview, the senior design practitioner who headed up the project explained, "No, we don't work that way, we are not the common type of researchers."

Facebook advertising led to expressed interest from 72 families seeking to make changes towards a more sustainable lifestyle. Three highly motivated families with varying demographics were selected to live without their car for a full year. Overall, the researchers were looking for both commitment and frictions (Hesselgren, 2019). Families with children were selected, as it was well known that this group finds it harder to manage without a car. Furthermore, two families owned country homes that they habitually visited by car. All family members' travel was tracked using a mobile phone app. The families also filled out travel diaries and were interviewed each month during the course of the year. Inspired by accounts from practice-oriented design, families were given a "punch card" that allowed them to take a small number of car rides when alternative transportation modes were unfeasible. This was included to pick up "breakdowns," when the families did not find a way to solve a transport need, to thus balance out the strict intervention of removing the car. Frameworks from practice theory were also used in the analysis.



Figure 4. One of the families, with two electric cargo bikes (Photo: Viktor Gårdsäter)

In the application for research funding, the project was clearly divided into a first part, "A Car-Free Year," and a second part, "The Car-Free City", which aimed to use the knowledge produced to model the impacts that car-free living would have on a city. A Car-Free Year was explicitly planned as a qualitative study but included the aim of measuring and comparing how the families' travel patterns changed. This, however, turned out to be difficult due to technical limitations. In all, the project took three years to finish, with preparations taking almost a year and the Car-Free City portion, along with other analysis and writing, also taking a year.

Results and outcomes

The study was challenging for the participants and effectively revealed many barriers to car-free lifestyles. Life without a car conflicted with societal norms, such as the expectation of driving children to activities, the lack of societal support for disposing of refuse, and insufficient cycling infrastructure. When winter cycling turned out to be a major challenge, the researchers arranged for a coaching session with a seasoned bicyclist, as well as support for purchasing suitable clothing. Transporting children to their activities turned out to be an unexpectedly complex problem, since letting children travel alone by public transportation was clearly against societal norm. Parents initially did not feel safe doing so, but in the end the children themselves decided to try public transportation, and found that they gained independence as they learned to travel without their parents. Altogether, the families found the shift easier than they expected, and even though they spent an estimated 10-15% more time on travel, they subjectively felt that they gained more time in their lives. The participants also identified other unexpected positive outcomes from replacing their car, such as spending quality family time when traveling by train.

In the second project stage, after the initial analysis, a design agency was engaged to design concept solutions as visualized scenarios of improvements and services needed in a car-free society. These concepts were then presented to relevant actors on several occasions, including to Stockholm City Council traffic planners and policymakers, a large employer, members of parliament, property developers, and an NGO. One important message was the power of societal norms, including suggestions to design cycling lanes, markings, and signage to make cyclists feel as safe and as respected as car drivers, and to make public transportation more child-friendly and disrupt the norm of driving children to activities. The DLL gained a lot of public attention in the local and

national media, aided by the production of two films about the project. Car-Free also inspired a very similar research study in Finland (see Laakso, 2017), and similar experiments with car-free families have been conducted in several Swedish municipalities, with recruitment for two more in progress at the time of writing, eight years after the initial project. (For an overview of Car-Free, see Hesselgren & Hasselqvist, 2016).

KTH LEV-Pool

Planning and historical background

This DLL was centered around a workplace carpool-service using LEVs and was introduced at one large workplace for two six-month periods and a second large workplace for a nine-month period. The project started from the general understanding that both mobility practices and the types of vehicles used must change in order to sufficiently reduce CO2 emissions, with a specific interest in the adoption of Light Electric Vehicles (LEV). The project was led by the Integrated Transport Research Lab (ITRL) at KTH as a way to test out designing and implementing a product-service system (PSS). The senior researcher that initiated the project was a strong proponent of service design, and Car-Free was considered very successful, due to the large amount of attention it garnered from media and policymakers. This led to the integration of design researchers from Green Leap and the recruitment of a vehicle designer as project leader.

The project was based on the understanding that both smaller vehicles and behavior changes would be needed to reach Sweden's CO2 reduction targets. Two large Swedish employers were engaged, both of whom dealt with the problem of too many employees commuting by car, even though both organizations were working to actively to reduce their carbon footprint, as well as the problem of the large amount of space needed for parking. A pre-study interviewed employees from these organizations about their car-driving behaviors and how car-use for commuting might be reduced. The research group had identified the need to make errands during the workday as one reason that employees drove to work, and wondered whether shared vehicles at work might enable more of them to commute by public transportation. The Green Leap design researchers were interested in exploring the use of LEVs, as well as emerging sharing practices.

The two employer organizations were the Swedish municipality of Botkyrka

(an outer suburb in the greater Stockholm region) and IKEA in Älmhult (in southern Sweden). Both areas—especially Älmhult—are relatively dispersed geographically, and most residents rely on cars for local travel. A car-sharing company, Hertz Sunfleet, agreed to provide access to their booking system and mobile phone application to book the service. Hertz Sunfleet wanted to get access to academic research networks and was also hoping to find new business models for electric vehicles. The research funding application highlighted the fact that no studies had looked at the real-world use of smaller electric vehicles. Also, building on the positive experiences from Car-Free, an NGO interested in more-sustainable driving also came on board to communicate the results. In all, these stakeholders were also motivated to participate as a way to show that it was possible to change and to be publicly involved in a future-oriented project (Sopjani, 2021).

Fairly late in the planning stage, two design researchers came up with the idea of engaging some users as "caretakers" to charge the vehicles at home. This idea came to be of central importance; 17 employees were recruited for this role, which included their leasing an LEV (a Renault Twizy quadricycle) and using it both for commuting and for other transportation needs. The leasing cost was subsidized on the premise that their coworkers could book and use the vehicles for errands during the workday.



Figure 5, The Renault Twizy quadricycles, parked in Älmhult.

The design researchers saw the caretaker concept as an interesting twist, since having to bring the car to work every day would make the intervention more "strict" (see Hesselgren et al., 2017). A service design agency was engaged to conduct a pre-study to improve the user experience for the service concept before the intervention was launched. This study suggested several redesigns to the booking system, but Hertz did not agree to invest in making any changes.

In all, because preparations took a full year and the three testing periods were carried out consecutively, the project spanned three years.

Implementation and method

All employees at the two sites had the option to book the service, and 85 people tried out the vehicles during the lab, making a total of 3,700 bookings. Two-thirds of the travel was made by the caretakers, mostly traveling to and from work. Similar to most living labs, but in contrast to the other DLLs, LEV-Pool was initially planned as an iterative cycle of designing, improving, and evaluating a PSS. Service design workshops were held with the users, but ultimately no changes were made to the service during the project, except for rectifying technical malfunctions. Instead, the project research turned out to be more exploratory than planned, with the Green Leap researchers aiming to support the design of future mobility systems while other members of the project group were aiming to test and evaluate the service concept—a discrepancy that caused some tension. Views regarding the role of the users also varied, and a conference paper that emerged from the project discussed how users were both actively engaged in reflecting and providing input and passively exposed to disruptive changes in their everyday life context (Sopjani et al., 2016b). Data were collected from the booking application, travel logs, surveys, workshops, and interviews, in a mixed-method approach. This work was mainly mainly carried out by the design researchers, who shared an exploratory mindset and a focus on social practices.

The caretakers were self-recruited based on certain basic requirements such as having a place to charge the vehicle at home. The users who found the offer attractive turned out to be environmentally conscious individuals, with a strong interest in adopting new technologies (ibid.). Recruiting caretakers was challenging, since not very many people had the right conditions to use the service; many employees lived too far from work for LEVs to be practical (see Hesselgren, 2019).

Results and outcomes

The research findings were broad and over time resulted in several scientific papers, notably two papers about disruptive innovation and a typology of user engagement and contributions. One paper discussed how the designerly methods may have strengthened stakeholder commitment, as they got involved at a concrete level (see Sopjani et al., 2016a). One design paper discussed addressing sustainability issues by bridging actionable insights to relevant stakeholders (see Hesselgren et al., 2017).

There were many unexpected findings regarding how sharing and the small LEVs conflicted with daily practices such as workplace norms and regulations, user skills, and expectations regarding the vehicle. Overall, the number of bookings and trips was low, and much of the analysis concerned explanations for this finding, underscoring how new services must be carefully integrated on several levels if they are to be successful. The large number of factors involved meant the concept could not be conclusively evaluated in terms of economy or CO2 savings. The new service and vehicles were slowly integrated into the users' everyday lives over a long period of time. Before using the vehicles, caretakers were optimistic about replacing the majority of their trips with the Twizys. They had concerns about its size and range, but it turned out that these factors were not generally a hindrance. The caretakers made adaptations to their daily practices to try and overcome these limitations or else became accustomed to them.

Hertz Sunfleet had hopes for a proof of concept that could be scaled up commercially, but the fact that they did not make any changes to the booking service indicates that they were not ready to invest at the outset of the project. However, Hertz became more interested over time and ended up creating a new business model based on making corporate vehicle fleets bookable by citizens in the area outside of business hours. This new service has been successful and is now used in several Swedish municipalities.

4.3. The Restaurant Lab

The Restaurant Lab has not been included as a case in any of the appended publications for this dissertation, but it held great potential and presents an interesting case. Because the Restaurant Lab failed to get funding, I could not

follow the project in detail and no structured data collection was carried out. Therefore, the Restaurant lab is not included in the main cross-case analysis. Still, I spent an equivalent of two months' full-time work over the course of two years on this project, which led to methodological insights.

Planning and historical background

The project was initiated in March 2018 by two energetic and enterprising design practitioners with a strong interest in sustainability, working together with two well-known Swedish chefs. One had previously founded Stockholm's arguably most sustainable chain of lunch restaurants, introducing ideas such as determining the menu based on the availability of food close to its expiration date and letting the customers pay for their meals by weight. The core idea behind the Restaurant Lab was to take these practices further and build the world's most sustainable restaurant. The team aimed to "take a frog-leap into the future" and push the limits for how sustainable a restaurant system could be, especially in terms of food waste and energy and water consumption. The project would make a business out of providing a testbed and demonstration space for companies and researchers to experiment with new sustainable solutions. The team behind the Restaurant Lab was a DLL dream team, but it turned out to be difficult to find project partners or funding. In-person meetings were held with more than 20 different companies and organizations that expressed interest in the concept, but ultimately failed to commit to supporting the project financially.

The team finally got a letter of intent to participate in the initiative from Sweden's largest restaurant wholesale merchant, Martin & Servera. At this point, the Restaurant Lab team agreed to take over a vacant restaurant space at the KTH campus. The funding application described how Martin & Servera could challenge their own organization by interacting with the prototype for a future sustainable restaurant, to identify internal barriers and future business opportunities.

Implementation and method

In the end, Martin & Servera backed out, and two more applications for research funding were turned down. Still, the team did not give up and instead funded the project privately, refurbished the restaurant themselves using

recycled and homemade materials, opening the Restaurant Lab in August 2019.

The team gradually expanded their network and invited chefs, researchers, and business representatives to meetings, seminars, and a podcast, as well as holding talks for other research programs and networks. The team carried out two smaller research projects for Chalmers University on the topic of sustainable food packaging.

Results and outcomes

The Restaurant Lab successfully pushed the limits for sustainability. Total food waste, which was meticulously measured at all points, was reduced to around ten grammes per meal served, and the total CO2 footprint was reduced to an average of 0.43 kg per lunch—well below the UN's "One Planet Plate" (WWF, 2021) target. KTH never provided support for the lab, even though it was present at campus and a number of meetings and seminars were held to discuss such opportunities. When Covid-19 hit, the Restaurant Lab survived for some time on the deserted campus, supporting Stockholm hospitals with crowdfunded lunch bags, but closed permanently in November 2021.



Figure 6. Photo collage, with interiors designed and partly built by the project members themselves from recycled goods, and displaying raw vegetables as decoration.

Overall, taking part in planning the lab provided some important learnings, especially from discussing stakeholder motivations and ways for research partners to get involved in the lab. The early planning stages relied heavily on the designers' capacity for building a team and a shared vision. It was also inspiring to follow the two designers in their hands-on staging of a future prototype

on a shoestring budget. The restaurant was first an immersive and sensory restaurant experience, and later developed into an increasingly functional lab environment.

4.4. CoKitchen

Planning and historical background

CoKitchen differs from the other DLLs in being situated within a more traditional living lab infrastructure, the KTH Live-In Lab[8] (KLL). KLL comprises of two buildings on the KTH campus and includes 300 single studio apartments for students and a 100 m2 testbed area that is exempted from most building code regulations. At the time that CoKitchen was ideated, KLL's main partner organizations were researching technological solutions such as sensor networks to optimize ventilation and energy use.

At a meeting at KLL, the founder of Green Leap met a manager from Academic Housing (AH), a state-owned academic property company. AH wished to investigate co-living as a sustainability solution for student housing but had problematic experiences with earlier types of larger, more dorm-like sharing concepts and thus had concerns about the social aspects and related administrative costs. Co-living was a very new concept that had only been proposed in 2014, but it was starting to garner attention internationally. Recent Swedish surveys had, however, indicated that students had negative associations with dorms, expressing concerns about sharing kitchens and especially bathrooms, which only 6% thought was acceptable to share. With all the potential social complexity involved, the design researcher suggested "just trying it out in real life."

At this time, building code and tenure regulations severely limited co-living in Sweden, and KLL's testbed area was therefore a perfect place for an experiment of this kind. The plan was to remodel four single student apartments into one co-living apartment and let students try it out. The application for research funding stressed the aim of challenging the Swedish norm of single households by "provoking new behaviors, understandings and reflection," as well as producing knowledge to influence building regulations.

8 KTH Live-In Lab, https://www.liveinlab.kth.se

CoKitchen was planned as an interdisciplinary collaboration between AH, a service design agency, an architecture firm, two KTH design researchers, and two sustainability analysts who specialized in lifecycle assessments (LCA). I was included in the research group from the first meeting following the project kickoff and was given an operational role in planning and staging the lab interventions.



Figure 7. The first group of inhabitants in the experiment apartment.

Since social practices and relations take time to evolve, students would live in the concept apartment for a full academic year. To allow the project to get off the ground more quickly and to gain some initial findings, the project was extended to remodel the apartments in two stages. Since there was very little time before the summer when the testbed apartments would be empty, the first remodeling was very basic and entailed little planning and analysis. Still, as there was little existing knowledge, we hoped to get useful input for the second remodeling. The original four single student apartments were joined through a large shared room by simply removing some walls (see Figure 8). In the second remodeling, the architects had more time to design a real co-living apartment.

The actual rebuilding work was commissioned to a construction company by the KLL management, and as other research was ongoing in the testbed area, the apartments were also fitted with sensors and other technologies to optimize heating and ventilation.



Figure 8: The original testbed floorplan, and the two rebuilds (Ilstedt & Sjöman, 2022)

KLL management commissioned the actual remodeling work to a construction company, and because other studies were ongoing in the testbed area, the apartments were also fitted with sensors and other technologies to optimize heating and ventilation.

The design researchers and the service design agency planned a pre-study that included two sets of interviews conducted by the design agency. The pre-study focused on finding opportunities for service interventions to enable more sustainable living and to support the students in reducing social conflicts and tensions. The findings, however, seem to have had little influence on the design of the co-living apartment itself. After discussing the findings in the group, the service design agency created fourteen service concepts. The ones that were implemented included house cleaning schedules, home-delivered dinner kits to help break the ice, better support for recycling, and solutions to help the students handle conflicts.

The two rebuilds were a progression towards more shared and less private space. Late in the process of designing the second apartment layout, a slightly larger double room was added to accommodate a fifth resident. In the words of the architects, this reduced the amount of private space to be “very close to the minimum,” and the change was made in order to provoke more learnings

about potential social conflicts. Several features were added with the explicit aim of exploring specific issues or opportunities, such as “semi-private spaces” or shared storage.

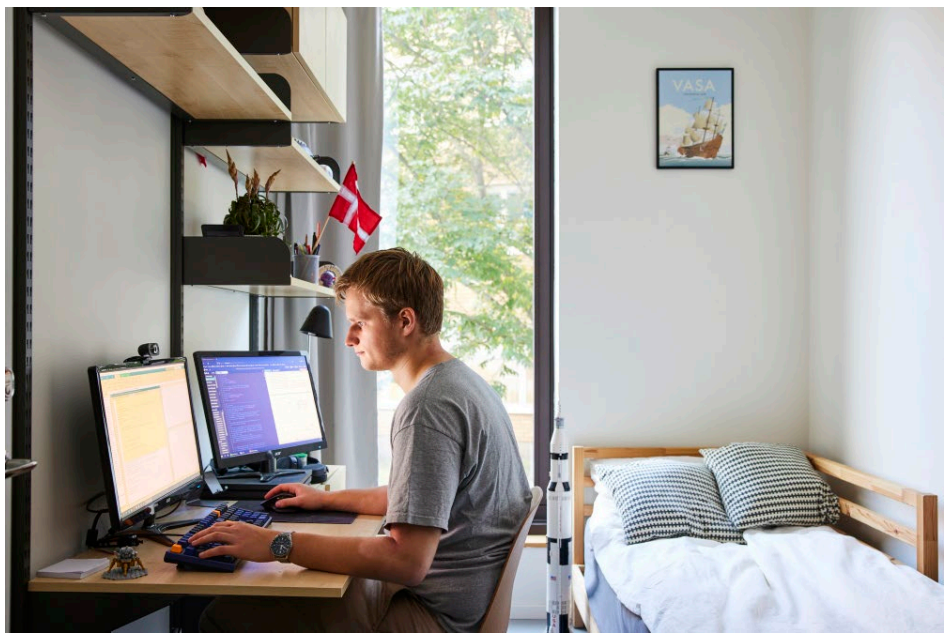


Figure 9. A student studying in one of the minimal bedrooms after the second rebuild. Photo: Fredrik Persson

In all, compared to a typical Swedish 21 m² single student apartment, each student now had access to 69 m² of the total 100, more than tripling their living space.

Throughout the project, different project aspects such as service concepts, construction materials, and the use of sensor data were all discussed jointly, and the research partners openly voiced opinions and ideas. Overall, the architects found this frustrating and requested a more structured process.

Implementation and method

The two sets of five students were recruited by advertising on the KTH homepage. The selection did not follow specific criteria but mainly aimed to get a

mixed group without outliers. For instance, we chose not to include a student who described his long and very active engagement in a sustainability-oriented NGO. In both years, residents were interviewed on move-in, in the middle of the academic year, and at the end of their stay.

Sensors in the apartment continuously tracked water and energy-consumption, as well as many other factors; however, due to incorrect installation during the remodelings and a lack of comparable data sources, this data turned out to be difficult to use.

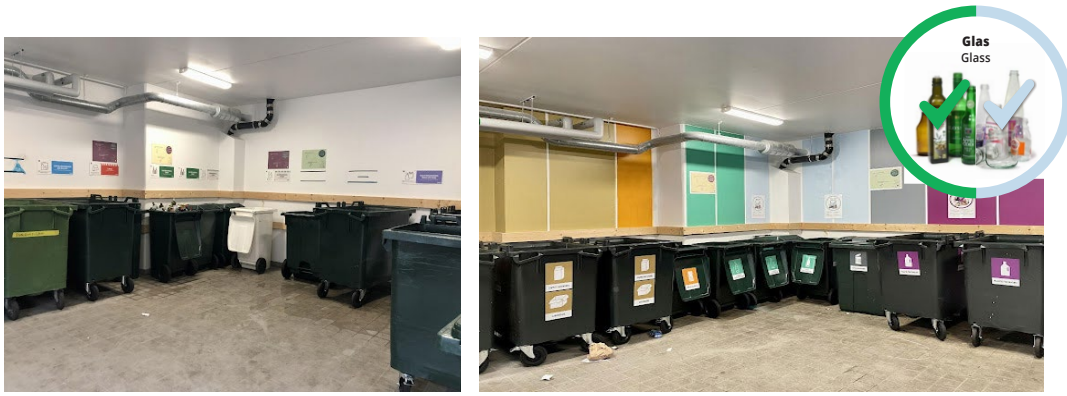


Figure 10. A magnetic cleaning schedule, with personalized 'dolls' that were moved to indicate responsibility for a 'cleaning shift'.

The design researchers carried out most of the operational work of staging the service prototypes, such as designing and producing the cleaning schedules and ordering the dinner kits. Overall, practical work such as recruiting participants and staging the interventions took a lot of time, and the promotion of good recycling practices is a good example of this. First, the architects were requested to include a space for recycling in the kitchen, and the design researchers selected and purchased suitable recycling bags. To further support the students in taking up recycling, I also designed new and more instructive visual labels, using a new color scheme that was being promoted as a new national standard at the time. Still, since the pre-study had indicated that sorting packages in the recycling room was far from intuitive for the foreign exchange students, this led us to discuss how the categories used for labeling the bags could be extended

to cover this space too. To do so, I produced a concept visualization suggesting that the recycling rooms be repainted using the same color scheme. Nevertheless, implementing this turned out to be challenging due to the school's regulations regarding maintenance plans and procurement, and it took several months to get the work done (See figures 11-12).

In the fall, after the first rebuild, the design researchers also held a co-design workshop with fifteen KTH students living on campus to learn more about students' housing preferences. Due to many external factors, a more proper analysis of the workshop outcomes was delayed, and for the most part this input was not used for designing the new apartment. The results turned out to strongly indicate that national surveys about student preferences were wrong. When given understandable and realistic choices that included co-living, all fifteen students preferred to share at least some of the functions of the home.



Figures 11 and 12. The recycling room before and after repainting, and an example of the new signage to the right.

After the second remodeling, the service concepts were tried out again in a second iteration. This time we started out with prototyping a concept for an introductory workshop. The idea was to present social and sustainable routines and to coach the students on handling conflicts, at the time of moving in. We sent out an invitation to the testbed apartment, as well as to ten triplet apartments on campus, but got very few responses.

As these were all the shared apartments on campus, we could not simply send the invitation out to more people. In the end, I personally visited the two buildings where the triplets were located in the evening three days before the workshop, knocked on all doors, asked the students who answered to take part,

and left informational leaflets with hand-written greetings on the doors where no-one was home. The next day, six more apartments volunteered to take part. In all, 25 students attended the workshop, where we also handed out cleaning schedules and recycling bags.

CoKitchen was led by a design researcher, and all partners worked quite independently using their specific competences and methods, although we held regular meetings. Overall, the design researchers and the service design agency collaborated more closely than the other partners. This collaboration was helped by a shared design mindset, although the design agency worked more quickly and with less rigor. For example, the design agency took part in planning and carrying out the introductory workshop and did a set of telephone interviews afterwards. These interviews were not transcribed, but the insights were clearly presented and covered all required areas.

With the exception of the design researchers and the service design agency, the partners did not work closely together, but anyone could request a specific issue to be included in the project partners' work. Although AH was clearly interested in the outcomes, their project manager did not request any specific actions or aims, and the experimentation was very open-ended. The architects complained that AH did not "clearly express what kind of apartments they want" or specify what they wanted to have studied in the project, and requested that AH provide calculations of potential costs and revenues. Instead, the project manager from AH wanted a co-living setup that "pushed things a bit" in order to learn more about well-being and social aspects. This included experimenting with shared bathrooms, even though other parts of her organization saw this as unfeasible given the current regulations and student preferences.

Results and outcomes

The students lived in the apartment for a full academic year and were requested to experiment with service concepts such as the house-cleaning schedules and recycling bags. Overall, co-living was found to have a strong potential to reduce loneliness. The residents stated that they had formed more supportive and "family-like" friendships. Sharing was experienced almost entirely as positive. Earlier surveys had indicated that students did not wish to share bathrooms, but the co-living participants did not mention this as an issue at all. The minimal private rooms led one of the students to purchase a tiny side table for

his laptop, and another participant found a new way to fold her clothes so she could store them more compactly. Overall, the students found that they used the shared spaces more, and said they would not trade the large kitchen or living room for larger private bedrooms. The shared kitchen was described as "the heart of the apartment," and the large table was used extensively for both social get-togethers and studying.

The service concepts turned out to be helpful and were described as important for building social relations. The vegetarian home-delivery dinner kits in particular helped the diverse groups of students build relationships with each other. During recruitment for the first year, the research group first had some concerns about a Chinese couple, as they felt the language and cultural barrier might be steep, but in the end these tensions turned out to be a great asset for the study, as building social relations became a little more challenging. During the weekly dinner that the residents were required to share, these barriers were bridged, and the group soon started to take initiatives on their own. During the first semester they went on joint excursions, visiting a neighboring town and holding a forest barbeque party, as well as celebrating birthdays together.

> *"With this combination of people it does not come all naturally. We got a kickstart. Maybe for other people that don't get along together all that well, this would help them bond and start to make the ambience better."*

All the building materials were included in a lifecycle analysis, which found that co-living excels both environmentally and economically, but AH found that their current way of performing budget calculations was not suited for co-living. Co-living apartments significantly improve the ratio of living area to total building area, but building regulations stipulate that carbon emissions and costs should be calculated based on the building area, disregarding how many people live there.

The project was met with a broad media interest and was featured on both local and national TV and in newspapers. At the time of writing, KTH and AH have received funding for a second project. This time they are working closely with another Swedish research project on shared housing and aiming to host a policy lab to advance changes in building code and tenure regulations (for more detailed accounts of the research results, see Ilstedt & Sjöman, 2022; KTH, 2023).

4.5. Mistra SAMS part 1, 2017-2020

Introduction

The eight-year research program Mistra SAMS has been my largest engagement during my PhD studies. This transdisciplinary research program received funding from twelve partner organizations, including several research institutes, a large technology company, two municipalities, the Swedish Transport Administration (VTI), and a mobility-as-a-service (MaaS) operator startup. The overall research aim of Mistra SAMS was to explore the possible transition to more-sustainable travel through the implementation of Accessibility Service Innovations (ASI) enabled on a digital platform. The Mistra SAMS team included around 20 researchers, the largest groups of whom came from KTH and VTI. The team included researchers from the fields of sustainability, urban planning, governance, sociology, and design[9]. Some researchers from Green Leap and ITRL had previously taken part in Car-Free or LEV-Pool, which played an important role in the decision to include design-methods in the Mistra SAMS program plan, and a senior design practitioner who had played a leading role in both of those previous projects was made part of Mistra SAMS' executive committee. Mistra SAMS' first phase ran from 2017-2020, during which time two DLLs were carried out in which I played a significant operational role—Playing-Rules and Work Closer.

Overall background and organization

The research funding application presented a plan to use a living lab-approach and design methods to prototype and explore combinations of ASIs on a multi-sided digital platform. Instead of "evaluating various technological configurations," there was an explicit aim to "understand the social, behavioral and institutional dimensions" of implementing these sustainability innovations. In the application text, the terms living-lab, pilot and prototyping were, however, used almost interchangeably, indicating that some program researchers may have viewed the living labs mostly as pilots with the added intent to engage users and stakeholders in co-creation. The first four-year part of Mistra SAMS included ambitious plans for an iterative sequence of three living labs, but also

9 For a complete list of partner organizations and researchers, see www.sams.kth.se.

included several other types of studies. According to the project schedule, work package three (WP3) was to first research "promising and disruptive" ASIs and digital platforms globally. Then the largest work package, WP4, would research user needs and preferences and institutional conditions for change.

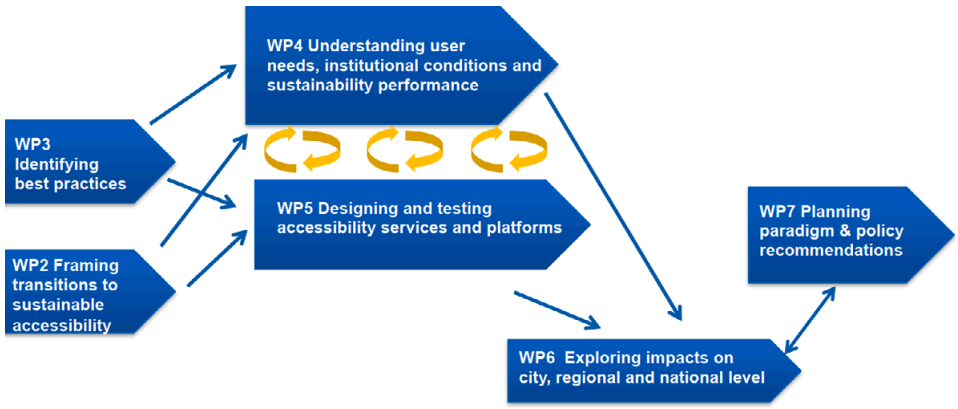


Figure 13. The original plan for the collaboration between the seven work-packages in Mistra SAMS first phase.

Work package five (WP5) would have primary responsibility for setting up the living labs and designing the service concepts to be prototyped. WP5 included me and the senior design practitioner from Green Leap to address the prototyping work. Furthermore, the service design agency Transformator were engaged to support the use of service design methodology.

The living lab planning process

Mistra SAMS met with challenges from the start. The program had planned to follow and collaborate with the MaaS startup Ubigo, but the launch of their planned pilot in Stockholm was delayed and later cancelled. WP3 and WP4 were both delayed for several months, and WP5 had to plan the labs without input on promising service concepts or user needs. It became clear that the question of what to research in the living labs had been left very open in the program plan and that there were many different research aims among the team members. In an attempt to create a shared understanding of the central concept of the digital platform and to stake out a design space for the living labs, the lead design practitioner made a whiteboard illustration of a hamburger that came to be widely used throughout the program.

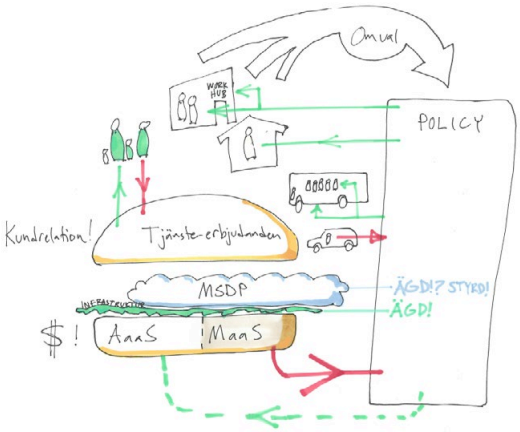


Figure 14. "The hamburger". An example of metaphorical concepts illustrations to promote a shared understanding of a complex system (Teo Enlund, 2017).

Overall, the process of deciding what to explore in the living labs took a full year. WP5 presented several alternative plans for living lab interventions that were discussed at program workshops, but it was hard for the large and diverse research group to move towards a decision. In a later interview, the lead design practitioner described this as a very difficult situation, as it was hard to follow the tangled developments.

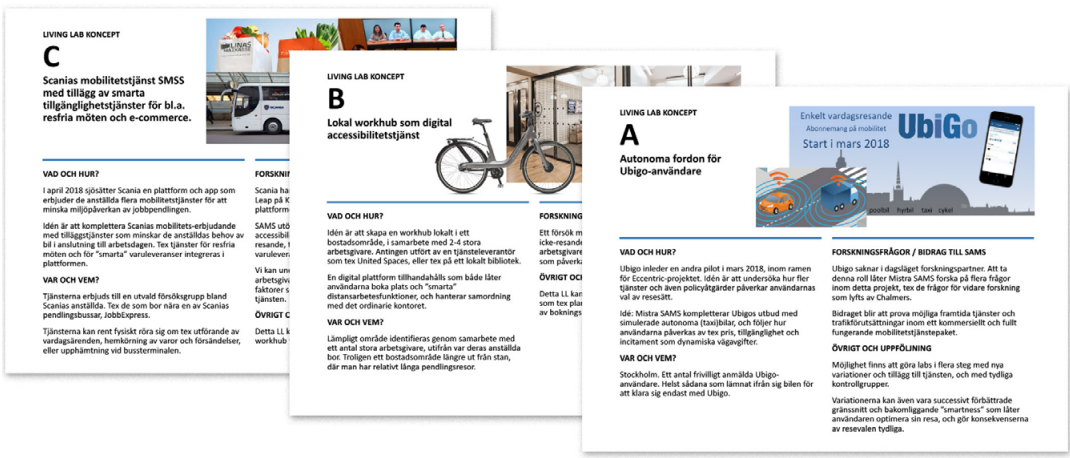


Figure 15. Workshop materials. Three early service concepts for the living labs.

It was also difficult for the service design agency Transformator to engage in what was essentially a process of framing the research issue. In the end, Trans-

formator was commissioned to carry out a pre-study on user needs, but the results were mostly not relevant for the design of the lab interventions. Nevertheless, the agency identified the issue of hidden costs and monetary incentives, which was later incorporated into the living lab planning (see figure 16).

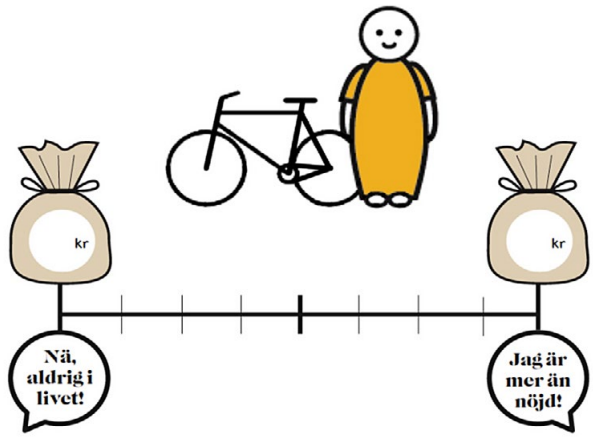


Figure 16. An illustration from Transformator's internal presentation in February 2018: How a cyclist can be motivated by monetary incentives, on a scale from "No, never!" to "Yes, this is more than satisfactory!".

Six months into the project, the number of potentially interesting research areas and questions was still expanding rather than converging. There were still plans to carry out three or even four living labs but no decisions about their content. As each of the earlier DLLs had taken several years to implement, this was looking increasingly unrealistic. At this time, there was a decision to first go ahead with a smaller "living lab #0," with idea that a smaller living lab on a narrow issue might kickstart internal collaborative processes. At a workshop, the researchers identified seven possible broad themes for the living labs, but it was difficult to go from these broad areas of interest to selecting concrete lab interventions. At this point, the issue of economic incentives was found to be suitably simple for the quick and small Living Lab #0. Pushed by a set deadline, WP5 finally got the go-ahead.

However, at this point, when the lab interventions were to be designed and physically staged, no progress was made. As time was clearly running out for a lab to get started before the summer holidays and as the other members of WP5 were occupied elsewhere in the program, I took the initiative and came to design and implement Living Lab #0 almost entirely on my own. Later on,

for participant interviews and for the analysis, I worked closely together with a colleague in WP5.

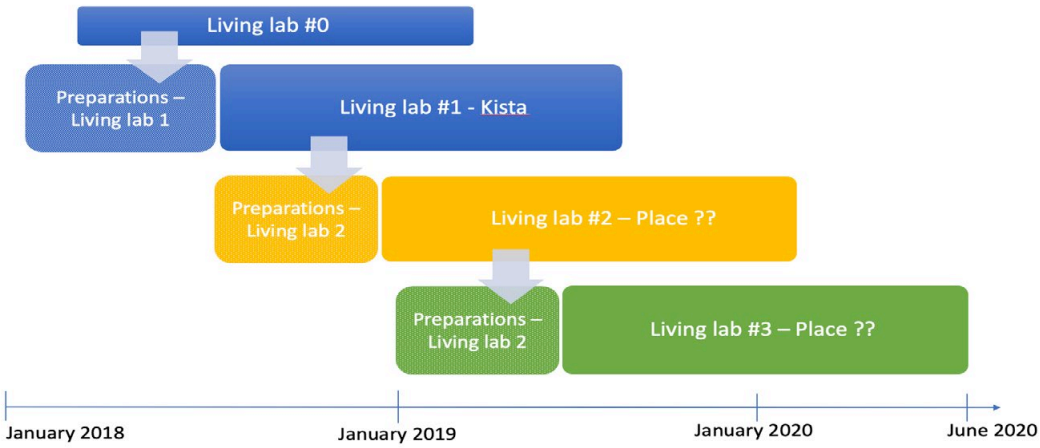


Figure 17. A presentation slide from January 2018, nine months into the program, illustrating the ambitious time-plan to carry out a total of four living labs in less than three years. In the end, only two labs were set up, and the first and intentionally smaller scale "Living lab #0" was later referred to as living lab #1.

Living lab #0, Future playing-rules for mobility.

Implementation and method

Living Lab #0 did not aim to design and test specific economic incentives. Instead, the plan was to use the incentives as trigger materials to elicit reflection on and understandings about travel-related costs. Given these aims, consulting the literature would not provide much support, and in the end, three ideas for incentives were quickly selected to include both cars, cycling, and public transportation and to reward sustainable behavior. The main aim was to design a future scenario that would come across as reasonably probable for the participants. I then aimed to design the interventions as a zero-sum game so that participants who made no changes to their current travel practices would have the same total travel costs as they did before but would save real money if they travelled more sustainably.

Three "future economic incentives"[10]*:*

- Using public transportation at half the cost when travelling off-peak.
- Earning a reward of 1 SEK (approx. €0.10) per kilometer for bicycling to work and for daily errands.
- A mobile phone app displayed the cost of each trip, and participants would "pay" for owning and using their car only at a fixed rate per kilometer. This rate was individually calculated to cover each specific car's fixed and variable costs such that the car would have no cost for the participant if it was not used.

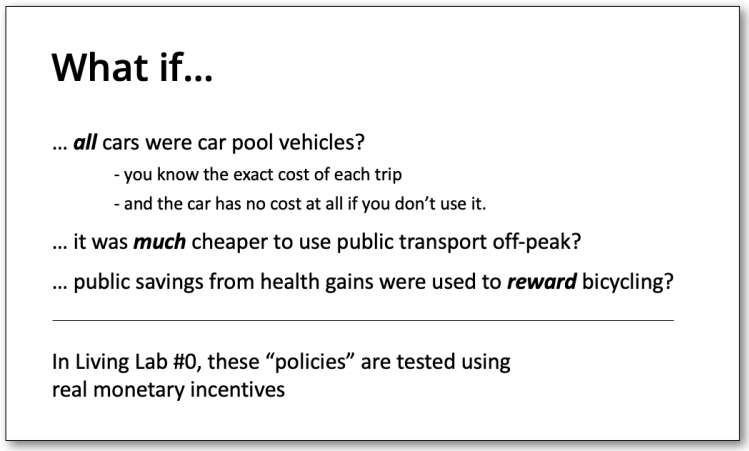


Figure 18. A Powerpoint slide from an internal presentation at a researcher workshop in Mistra SAMS, in August 2018.

This plan, and an associated budget, were presented to the program management group, and after some discussion there was a new go-ahead to recruit participants. At this point the car cost intervention was fleshed out in greater detail, and a consultant was brought in to provide estimates of the participants' cars' depreciation in value. We also needed a solution to track the participants' travel, and a newly developed mobile phone travel survey app was selected.

In Mistra SAMS we had agreed to take baseline measurements of the participants' travel patterns before implementing any interventions. In line with this more structured approach, participants were recruited randomly to give us a broader set of impressions than we would have got using the self-recruitment

10 For a detailed description, see Sjöman et al., 2019.

approach and the intrinsically motivated “forerunner” types in Car-Free and LEV-Pool. To allow a joint analysis at a later point in time, car owners were approached in parking areas outside grocery stores in the same three suburbs where WP4 was planning a survey. A majority of the people approached agreed to take part in the study, and 30 potential participants were recruited.



Hej bilägare!

KTH VETENSKAP OCH KONST

Vill du hjälpa oss forska kring framtidens vardagsresande?

- Prova en smart “självlärande” app
- Få total överblick på ditt resande och dess kostnader
- Möjlighet att spara pengar - verkliga pengar rakt in på kontot!
- Påverka framtidens politiska lösningar och gör en insats för miljöforskningen

I forskningsprogrammet Mistra SAMS undersöker forskare vid KTH resvanor och åsikter kring vardagens resande. Denna undersökning fokuserar på bilägande, men innefattar också kollektivtrafik och cykling.

Undersökningen görs i två delar:

Under en månad använder deltagarna en smart app som loggar och mäter alla resor. Du kommer att behöva berätta för appen vart du åker och ibland med vilket färdmedel, men den lär sig snabbt dina vanor. Två gånger, både efter en vecka och vid mätningens slut, får du en belöning. Du kan välja två biobesök för dig och en vän, och det kommer även att finnas andra alternativ, som tex trisslotter).

Efter denna resvanemätning får deltagarna en analys av sitt resande, och detaljerad information om sina kostnader. Här ingår bl a en summering av dina resor med kollektivtrafiken och en professionell uppskattning av bilens värdeminskning över tid.

Hälften av deltagarna får därefter möjlighet delta i undersökningens andra del som löper över sex månader. Här erbjuds möjligheter att spara stora och verkliga pengar på att göra smarta anpassningar i vardagen.

Är du intresserad av att delta och hjälpa oss i vår forskning? Fyll i vår intresseanmälan!

Skicka ett email till: info@sams.kth.se

MISTRA Stiftelsen för miljöstrategisk forskning vti

Figure 19. ”Hi Car-owner!”. A leaflet I designed to recruit user-participants for living lab #0.

Brief phone interviews were made to collect basic data about participants’ family situation and car-related costs. In several cases, partners and grown children also used the family car, meaning that these family members had to sign up too. In the end, 25 people agreed to measure their travel for one test month and to take part in the subsequent study if selected.

Overall, Mistra SAMS aimed to focus on ordinary, day-to-day travel. Interviewing these 25 people about their car costs yielded the immediate insight that many seemed to use their cars for long summer vacation trips. Thus, after a brief discussion in WP5, we decided to exclude such vacation trips from the data, as they would risk making day-to-day driving look insignificant. This decision led to an administrative burden from having to inquire about any longer trips and requiring participants to manually remove them from their

logged data. Also, these trips turned out to be an important motivation for car ownership, and having data on them would have been useful.

After the initial baseline measurements, we selected nine participants representing the three suburbs and exhibiting a range of travel patterns and household compositions. All their travel over the course of six months was logged, and we carried out three sets of semi-structured in-depth interviews. Most of the interviews were conducted in the participants' homes, using their respective travel data and personalized travel scenarios as trigger materials to prompt discussion.

The mobile phone travel logging app was intended to track the participants' every movement and to automatically detect and log distance and mode of travel. When we finally received the first batch of travel data, it came as a surprise that that it was in the form of raw data and required programming in order to compile simple statistics on each participant's travel during the baseline month. It took many calls and discussions to get a researcher connected to the Mistra SAMS program to program a basic tool that let us extract the travel distances that were required to calculate participants' monthly rewards and savings.



Figure 20. Photos of car odometers, sent in by the participants.

Overviewing the first two month's travel with participants during the interviews, we realized that the app's automatic detection of travel had not worked as well as intended. Car trips were sometimes logged as train rides, long walks at nighttime appeared mysteriously, and lengthy trips were logged as a series of shorter ones. With support from the app developers, we gradually taught the participants how to use more advanced functions in the mobile phone app to correct some of the data themselves, while I corrected other errors after data delivery. In parallel, we also managed other problems, such as participants getting disconnected after changing phones or phone numbers. The nine participants spent much more time than expected correcting their data and

making support calls when the app did not work. Their diligence and patience was likely due both to their frequent personal contact with the researchers and their expressed sense of being involved in the research. They also may have felt more obligated to correct the app's mistakes because the data would be used to pay them real money. To make sure that the figures did not turn out entirely wrong, we also requested the participants to send photos of their odometers three times during the project. Later on, the app developer provided access to a web-based tool to directly review logged data via screen dumps of the mobile app visual interface. This allowed us to scroll through the participants' travels and more easily spot errors (see Figure 21).

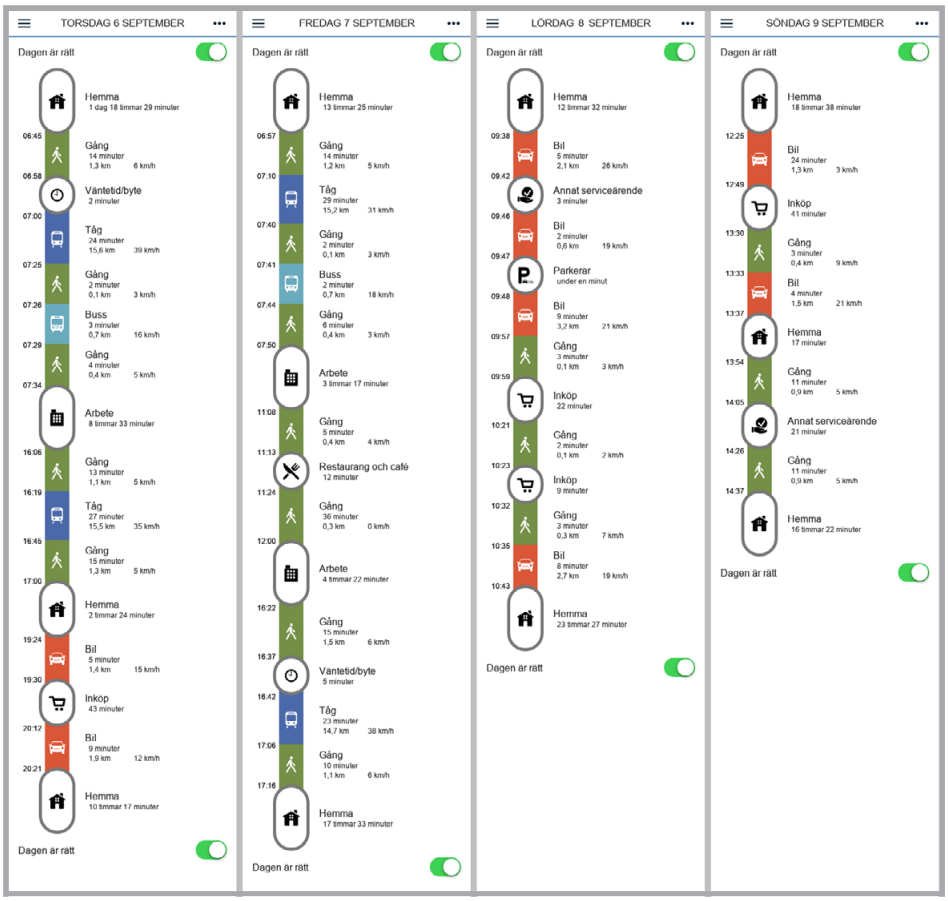


Figure 21. Excerpt from a series of screen dumps of travel data, retrieved from the visual user interface of the mobile phone travel logging app. This image displays four days of travel, Thursday to Sunday.

Results and outcomes

Regarding participants' understanding of their travel costs, their responses were often not rational, and the comparisons they made would always favor car travel by excluding the fixed costs of ownership. Individualized prompts, such as a scenario that involved planning a skiing holiday, confirmed that the participants only considered the cost of buying gas as a parameter when deciding whether to go by car, which made alternative travel modes seem expensive in comparison. It also became clear that many participants presented seemingly rational but contradictory and sometimes changing explanations for why they needed their cars (see Publication A). Six out of the nine participants also exhibited unexpectedly large seasonal variation in their travel patterns. These participants travelled more than ten times more distance by car in July than during November. Because existing national and regional travel surveys do not show differences of this magnitude, this could indicate that those surveys do not sufficiently account for vacation travel. Furthermore, our follow-up interviews indicated that the definition of a "trip" that most Swedish travel surveys use does not align with the participants' understandings and actual travel practices. Another finding was that participants in lower income brackets expressed frustration and unease when seeing the full cost of using their car but argued that they had no other choice, and one said that he would switch to a more fuel-efficient car.

> *"It has forced me to realize that the car probably costs more than my house … and it's really just childish to speed off first from traffic lights. My next car will be a small and very fuel efficient one."*

Other participants simply stated that they could afford the convenience, meaning that no one actually drove less. Through the participants' logged behaviors and their stories, it became clear that only congestion and parking difficulties would reduce their driving, and that most of them were not ready for the re-learning process needed for change (for a more detailed account, see Publication A). Another finding that emerged through the combination of logged travel data and interviews was that there are mid-volume travel flows such as the mass of people going to the same suburban shopping malls at roughly the same time on Saturdays. Unique individual travel needs and destinations were relatively rare, and Swedish public transportation is well organized to support the very large transit flows of workday commutes, but these mid-sized flows

were identified as a potential opportunity for the expansion of mobility services.

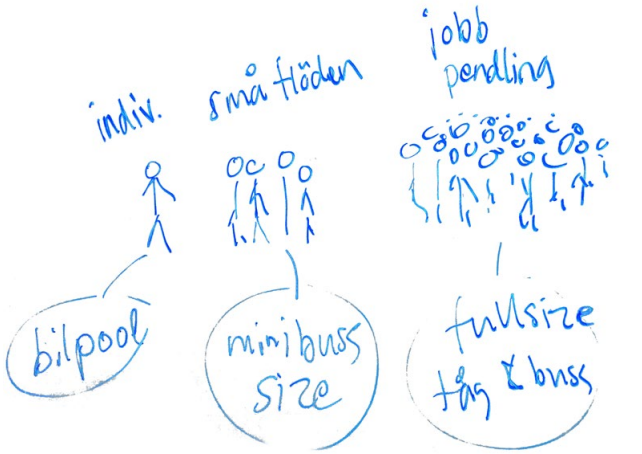


Figure 22. A whiteboard illustration of individual, middle-sized and large traffic flows.

Overall, the findings from Playing-Rules contributed to many different research fields that were not closely related to our previous research. As we had not been able to plan for this, the program researchers with relevant knowledge were tied up elsewhere. Getting these findings published therefore required reading up on several different streams of literature in order to discuss them in context.

Living lab #1, Work closer, travel smarter

The continued living lab planning process

At the same time that the Playing-Rules lab was being planned, the drawn-out process of deciding on the content of the larger living labs continued. WP5 formulated another detailed written proposal with three alternative setups. However, input from research colleagues and partner organizations were mainly suggestions for including additional perspectives and did not support the process of narrowing down its decision.

The decision-making process was continually set back by the implicit understanding that the living labs' research topics should be decided on jointly. A formal decision to set aside a budget would be taken by the program's board, but this seemed to first require the research group to reach a consensus, something that proved difficult given the many diverse research interests. The

process steps were not planned in advance, and new steps continued to be added until a decision could be made (see fig. 23).

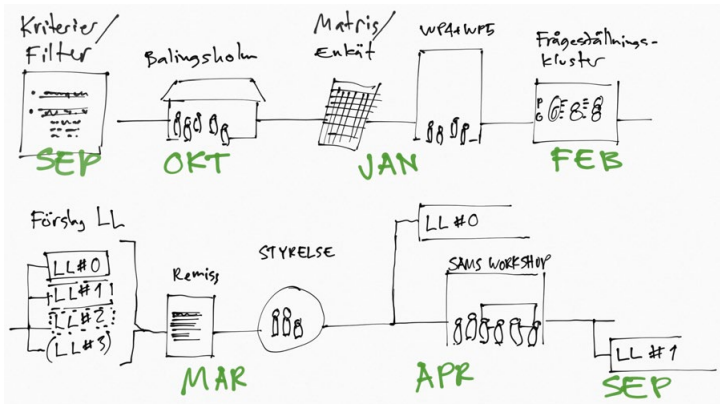


Figure 23. A post-hoc visualization of the main steps in the decision-making process for the research topics and concepts for the living labs in Mistra SAMS. A timeline from September 2017 to September 2018.

The many views and research objectives were eventually reconciled through a compromise. Living Lab #1 would first stage the idea of a non-travel-service to reduce commuting in the form of a "super-local workhub". The lab would provisionally stage a shared office space in an outer suburb where many citizens had long daily commutes, to function as an "accessibility service". Then, complementary mobility and delivery services would be added later on. In line with the program plan, all the services would be made available for booking on a digital platform. In later research publications, the shared office space would be referred to as a Neighborhood Telecommuting Centre (NTC). Telecommuting was not new to the research group, and staging a space for remote work had been one of the earliest concepts in the process. At this time, co-working spaces were becoming popular, but only in very central city locations. The NTC, in contrast, was to be staged in Tullinge, a villa suburb around 25 kilometers from Stockholm city. Living Lab #2 was left open to be decided later, but in the end, there was never time or resources to implement a third DLL in the first program phase.

Implementation and method

When it was time to plan and stage the NTC service concept, I was fully occupied with the unexpectedly challenging data collection and analysis for Living

Lab #0. This time, after another long process of getting a clear mandate and budget, the lead design practitioner staged the NTC almost singlehandedly. An architect was contracted, a building contractor was hired, services were set up, and office furniture, equipment and a digital lock system were acquired. After four months, a complete co-working office environment was in place, while the program managers took care of engaging the partner organizations, including legal aspects. The interior design was both stylish and clearly provisional (see Figures 24, 25 and 26).



Figure 24. A photo of users at the NTC. April 2019.

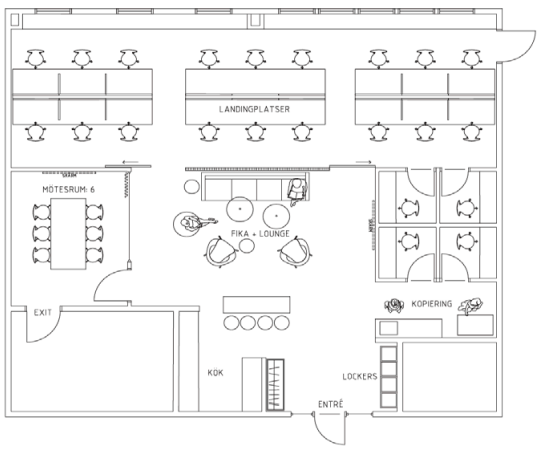


Figure 25. The architect's floor plan for the NTC. White Arkitekter, 2018.

User participants were to be recruited from two large employers who were partners in the research program. This, however, turned out to be more difficult than expected. Even though both employers had senior managers on the Mistra SAMS board, the internal process of allowing a small number of employees to test working remotely from the NTC took almost nine months. For this reason, we recruited an initial group of 20 participants from many different employers by advertising in local Facebook groups.



Figure 26. Photos from the NTC prototype: W-Fi installation, interiors of a "zoom-booth" and a coffee corner. These installations are clearly provisional but still provide the required functionality.

A digital platform was still needed for booking the new services, and a replacement had to be found for the earlier partner Ubigo. At this stage, the program management chose to start a collaboration with another startup company that offered a digital booking platform for corporate travel. As their current web-based booking service lacked many functions that we needed, a project was initiated to devise a software development plan. A lot of resources were put into producing a requirement specification, but in the end, this new functionality was never developed. Instead, the program purchased a commercial license to use the company's standard booking platform only as a basic solution for booking the desks at the NTC.

As the research program aimed to study the combination of accessibility and mobility services on a digital platform, electric bikes were acquired with the intention of designing a bike-sharing service. This, however, turned out to be difficult, as it became clear that this type of service should ideally be located closer to people's homes. With few additional resources for staging or running

the living lab, data collection and day-to-day administration of the NTC was all that the four part-time members in WP5 could manage. The bikes were simply made available for participants to use, and the plans to add other services on the same platform were never implemented.



Figure 27. The electric bikes that were rented by the program and made free to use for the research participants.

Results and outcomes

In an initial set of interviews, applicants expressed expectations of using the NTC a lot, mainly to reduce time spent commuting. Early feedback from participants indicated that working from the NTC reduced stress significantly and felt different from working from home. Working at the NTC even led one participant to completely re-evaluate her situation, and after only about a month she quit her job and took a job closer to home for slightly less pay.

> *"It felt so nice to hang out with local people. Cycling to work makes me feel so much better. Before, I had like two lives: one life in the city, and one here. Now it's more like – it feels safer to be closer to my children. I can go home at lunchtime if I have to pick something up."*

Something that came as a surprise was that participants described the NTC as a place to concentrate and get work done. Here, they were not disturbed by colleagues' requests. The people at the NTC were those that happened to live close to it, and with only one exception, they were not coworkers. This also meant that the well-equipped meeting room was never used. This shouldn't have come as a surprise, but it did, and it later we found that it was interesting news for actors in the coworking business too.



Figure 28. A woman working at the NTC. August 2019.

It also became clear that many of those who applied to participate never showed up at the NTC or only used it rarely. Contrary to their own expectations, only a few participants had working conditions that allowed them to use the NTC more than 2-3 days per month. Many believed that they would soon start to use it more, as they were, for example, in the midst of a reorganization, or temporarily had a higher workload, or were supporting a new coworker in training, and it took months before they realized that even if those specific situations ended, they got replaced by new barriers. For the few participants that were in a position to use the NTC more often, working closer to home led to increased their well-being and also led to other changes, such as shopping more locally and using their cars less overall. In a few cases, this led participants to purchase electric bikes.

Due to the limited use of the NTC, the project focus gradually turned from understanding commuting and travel practices to identifying the large number of barriers to remote work, including workplace norms, team collaboration, insurance issues and managerial routines. When reporting on these results, a reviewer for Travel Behaviour and Society commented that a resulting paper (Publication C) was “more about the psychology of working from home, than about effects on travel shifts.” The paper eventually got published, after adding the notice that “we believe that research into means for avoiding transport will often have to include more interdisciplinary perspectives.”

Overall, the majority of participants found it hard to use the NTC, and it was never very well-utilized. To draw out more findings I came up with the idea of challenging a subset of participants to work at the NTC at least three days per week for a period of two weeks, and to report any difficulties they experienced in a printed diary. Follow-up interviews then helped to identify and validate the main hindrances for remote work: finding that issues such as team collaboration, physical meetings, company policy and management routines made it hard to be away from the company office. Other barriers could have been addressed more easily. For some participants, remote work required many online meetings, and at the NTC this meant that they had to sit in a small "Zoom booth" for several hours per day so as not to disturb others. As the NTC was a temporary prototype, these Zoom booths were not very comfortable and poorly ventilated.

In Work Closer, different views about the lab methodology in Mistra SAMS became visible when the program management decided to have a quantitative evaluation made to find out whether the NTC concept contributed to sustainability. The design researchers disagreed with this, but a PhD student from another department was commissioned to conduct the evaluation using travel diaries. As most employees were not allowed to work remotely more than one day per week, the low booking rate affected the evaluation outcome negatively. The evaluation also found that the users had office spaces at their employers' offices that now went partly unused. Moreover, as most users usually commuted by train, the reduction in CO_2 emissions was small. In summary, the evaluation found that the travel avoided did not make up for heating the many empty office seats, meaning that the NTC was not sustainable.

Over time, the partner organizations gradually became more interested in the NTC. Two Stockholm Municipality employees greatly appreciated the NTC prototype as a place to work for a few hours on the way to or from external meetings, which were often booked at the beginning or end of the workday. These participants later lobbied for a similar solution to be implemented in their own organization by having all the organization's many offices around the city offer open workplaces for internal use. This concept was presented as part of the city's plans for 2021, but was never implemented for reasons that are not know but may include the normalization of working from home during the Covid-19 pandemic.

Work Closer also received a lot of positive attention from the funder Mistra, as well as from both local and higher-level policymakers, including a visit by a secretary of state.

Epilogue

In the middle of February 2020, the Covid-19 pandemic struck, and almost overnight all the participants who had found it so hard to use the NTC were ordered to work only from home, leaving both their employers' main offices and the NTC empty. In an immense effort on the part of Swedish employers, all barriers to remote work were lowered in a few weeks. Inadequate policies were changed rapidly or simply ignored. Our participants, who were mainly well-off white-collar workers, rebuilt guestrooms into home offices, often with equipment supplied by their employers. The living lab was kept running during the whole year in hopes that the pandemic would abate, but when public health restrictions were partly loosened after the summer, only a handful of dedicated users returned.

It is interesting to consider what might have happened if the Covid pandemic had never happened. Before that point, there was a growing commercial interest in co-working, but only in central city locations and mainly for net-working purposes. In Stockholm, there was also a growing interest in ways to reduce commuting, connected to increasing traffic congestion and the partly insufficient public transport. The NTC identified a number of barriers for distance work, as well as several key insights for the design and placement of NTCs. The concept had worked up interest both from employers, policymakers and co-working actors, to whom these findings could have been communicated, possibly leading to action.

Coming from a "piloting" research culture, and inspired by the niche manage ment literature, parts of the Mistra SAMS program management had been working to generate interest for scaling up the NTC concept during the second phase of the program. The municipality of Botkyrka later provided a budget to turn the NTC into a pilot project for one more year, together with a commercial startup co-working company. After Covid, however, employees in the area have largely kept working from home, and the commercial service did not manage to attract paying customers. Instead, a similar concept has been launched success-fully in a lower-income area with smaller apartments and younger residents.

4.6. Mistra SAMS part 2, 2021–2024

Introduction

In 2020, the Mistra SAMS program applied for and received funding for another four years, from 2021 to 2024. Around 20 researchers were engaged—most of them carrying over from the first term. The lead design practitioner who had been on the management team in the first phase had left KTH, and was replaced by another design researcher from Green Leap. In the second phase, I was heavily involved in the planning stages for a third living lab in Riksten, a suburb of Stockholm, but since May 2022 I have no longer worked at the program. The Riksten living lab was launched in June 2022 and is still ongoing at the time of writing. It is not clear at this time whether this lab will fall under the Designerly Living Labs approach proposed in this dissertation.

Overall background and organization

The second part of Mistra SAMS was designed to be more integrated and was planned around a joint, central living lab that would provide empirical data for the whole program. The researchers formulated three work packages that were meant to supply the lab with three complementary research perspectives: Citizen, Market, and Policy Actors (see Figure 29). Additionally, the work package Living Lab Design was planned to facilitate the joint process of planning and implementing the shared living lab.

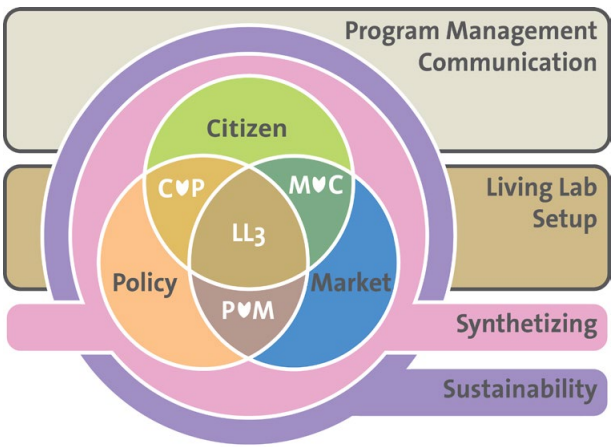


Figure 29. The “Snail”. An illustrated overview of the SAMS 2 organization. In the center is the shared living lab, with the ‘research perspectives’ around it. Teo Enlund.

As the lab was planned to use a design-driven methodology, Living Lab Design's three group members were me, the other design researcher from Green Leap, and a consultant who specialized in leading creative processes. Since the work package was only meant to facilitate the joint process, we had only one half-time equivalent of staffing allocated for work in this group, of which my efforts took up the largest part.

Two of these work packages also planned for other lab formats of their own. Policy Actors planned a policy lab: a series of workshops aiming at clarifying the role of policy actors in the envisioned mobility transition and developing their transformative capacity. Market planned a similar workshop series with market actors and representatives from relevant authorities, municipalities, and regions, which was referred to as a business model lab. Together, the three perspectives also reviewed the existing literature about mobility services and related technologies and interviewed stakeholders from existing implementations, producing a joint report.

LL#3, Riksten

In the Riksten living lab, I only took part in the planning stages. This case description will focus in on events and actions that relate to methodological issues and conflicts concerning the design of the lab.

With the four research groups and perspectives came different research cultures, understandings, and aims. Citizen sought to initiate and build citizen engagement around sustainable mobility issues and to support citizens in taking insider-led and personal action. More in line with the aims of the overall research program, Market sought to invite a large number of market actors to offer services on a digital platform as a way to explore new business models. Public Actors planned to follow the actions of public actors and to support them in building capacity for managing the advent of smart mobility, which was only vaguely connected to the living lab. The fourth research group, Living Lab Design, included the program's two design researchers and aimed to stage a sustainable future scenario, including the design and exploration of more future-oriented mobility services. It is worth noting that none of the groups had any clear responsibilities for the practical staging of the lab, and with the diverse and partly conflicting research interests and ideals, it would take eighteen months until even a very limited living lab could be started.

Living Lab Design did not have a mandate to make plans for the lab or to make any decisions. Instead, our task was to host a series of workshops to facilitate joint planning. This work wound up being quite a challenge. Altogether, Living Lab Design hosted eight such workshops, with workshop materials carefully designed to support a shared understanding of the methods to be used in the lab and decision-making about the type of service concepts to be included. Some of the workshops presented overview information about the Riksten area and sought to shape shared understandings of the types of mobility services that were needed. However, this process made conflicting research interests increasingly visible. An in-depth preliminary study on mobility provision in the area focused on factors that were mostly outside the program's scope, such as new road construction. Several researchers from Citizen and Public Actors had backgrounds in fields such as sociology and discussed grassroot approaches such as a communal bike garage. They also had plans to involve the municipal community dialog team to support citizens in developing a community-based solution for transporting children to and from activities. Market included more researchers with engineering backgrounds and was looking to follow commercial partners in developing new sustainable business models. Living Lab Design was planning to design more future-oriented sustainable service concepts and to invite user participants to explore them. In the program plan, the living labs concept was first described broadly, based on literature about ULLs. Another section then included more detailed plans on how the Mistra SAMS living labs would use a collaborative design approach, based on practice-oriented design interventions (with specific reference to Publication B). Nevertheless, at one joint workshop other researchers presented literature from the social sciences and participatory action research, suggesting that bottom-up approaches are critical for implementing change and to promote social justice. Still, the program's management either seemed not to notice these stark differences or did not view them as cause for concern.

For these reasons, the Living Lab Design group came to focus on bridging these conflicting views in order to allow planning to move forward. In this phase, I was the only design researcher with a significant amount of time allocated to Living Lab Design, and I also had a specific role in the program plan of documenting and seeking to improve the overall research program methodology. During my 18 months in the program, Living Lab Design held a series of workshops for the program, for which I produced visualized intervention concepts and co-creative workshop materials to facilitate the planning process. I also

provided specifications for resources and skills that were needed in order to implement the living lab in practice. During this time, I was also partly engaged in the Market group, where I was responsible for contacting potential service provider partners.

Exposing differing research ontologies and mindsets

Joint workshops and seminars revealed several indications of divergent views regarding the lab, such as Citizen being cautious about partnering with commercial actors while Market discussed business ecosystems. In Living Lab Design, we found that these tacit conflicts stymied practical-level planning for the living lab. As an example, engaging market actors to provide their services in the area would require large numbers of users, but Citizen was planned and budgeted to closely follow only a small number of highly engaged forerunners. The startup company who provided the digital booking platform had developed additional functionalities when taking part in other research projects, and was again engaged to develop a tailored platform solution and app for the living lab. Due to the lack of clarity as to what the living lab was to look like, it took a year until they could receive the specifications for the functions that were needed. In all, the different groups all worked on their own research activities in parallel, but planning for the shared living lab was at a standstill.

Eight months into the program, Living Lab Design planned and hosted a workshop specifically to bridge these differences. For this event, I constructed a workshop material in the form of a four-field matrix, depicting four intentionally strongly contrasting research design positions (see Figure 30) based upon ideas and views that had been expressed in internal documents and in comments at previous workshops and meetings. The “serving” corner was in line with more mainstream notions of commercial pilots and living labs, aiming to develop, improve, and test suitable combinations of existing service concepts to align with local user needs. The “enabling” corner, in contrast, denoted an aim to change conditions for service providers through regulations and incentives on a digital platform, and to track how this ecosystem of actors responded. The “provoking” corner was more in line with the definition of DLLs used here, aiming to stage more-disruptive and challenging service concepts based on a sustainable vision of the future. Finally, the “engaging” corner was rooted in insider-led approaches, aiming to engage and support citizens and local actors in taking their own initiatives and to follow their learning process. In this

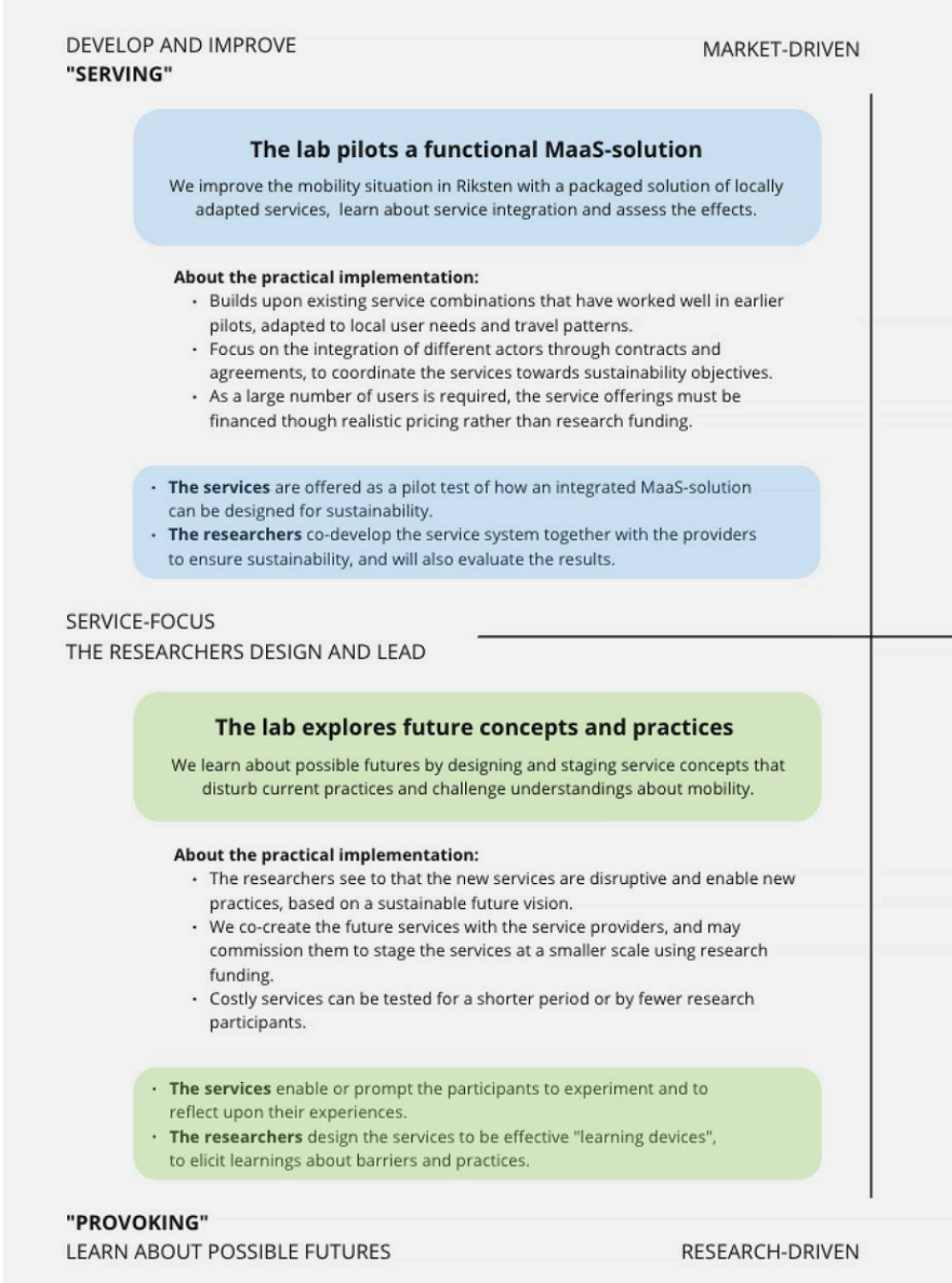
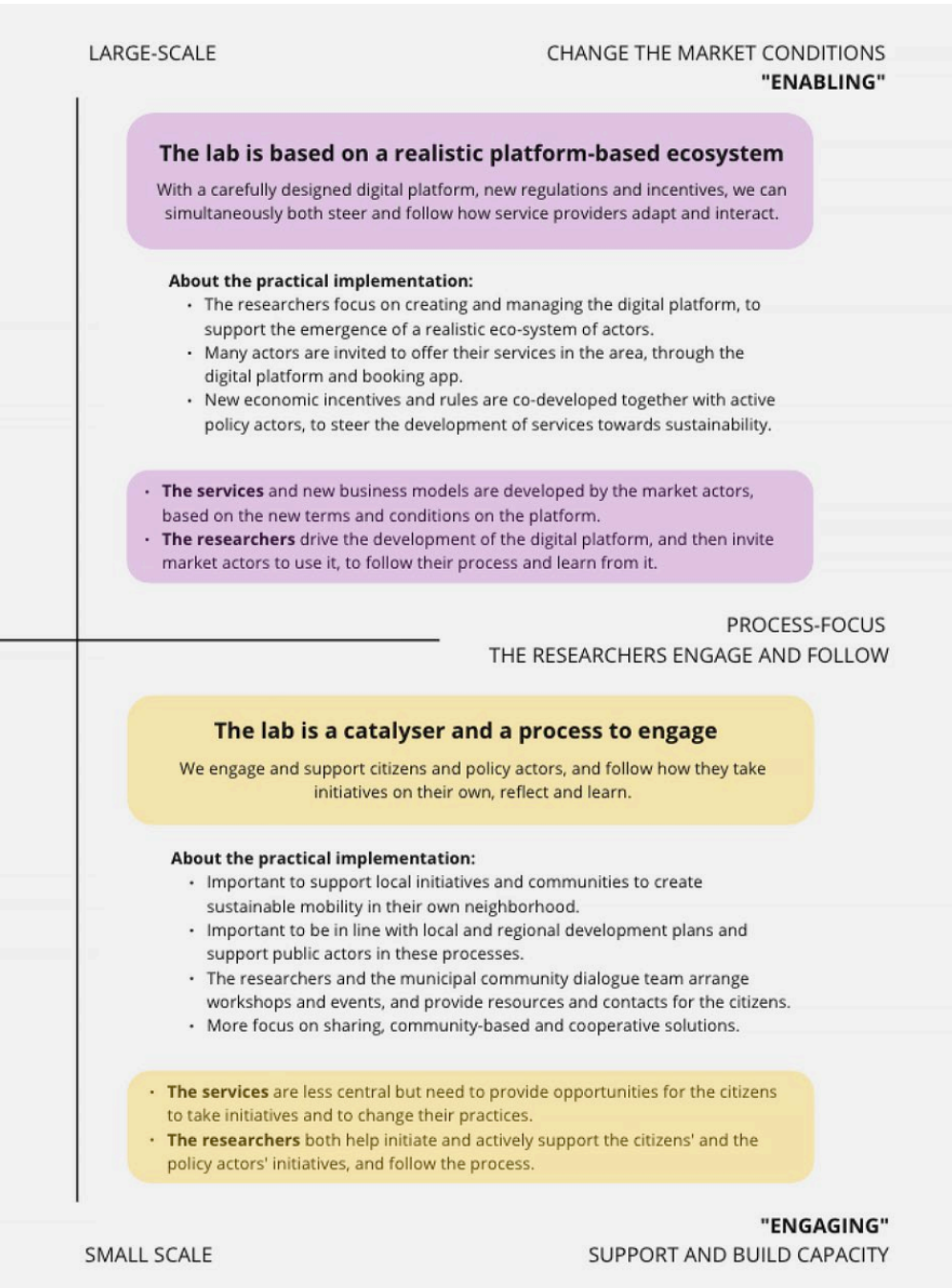


Figure 30. Workshop material. A typology of four types of real-life experimentation in mobility research based on different research aims, cultures, and values.


LARGE-SCALE
CHANGE THE MARKET CONDITIONS
"ENABLING"
The lab is based on a realistic platform-based ecosystem
With a carefully designed digital platform, new regulations and incentives, we can simultaneously both steer and follow how service providers adapt and interact.
About the practical implementation:
• The researchers focus on creating and managing the digital platform, to support the emergence of a realistic eco-system of actors.
• Many actors are invited to offer their services in the area, through the digital platform and booking app.
• New economic incentives and rules are co-developed together with active policy actors, to steer the development of services towards sustainability.
• The services and new business models are developed by the market actors, based on the new terms and conditions on the platform.
• The researchers drive the development of the digital platform, and then invite market actors to use it, to follow their process and learn from it.
PROCESS-FOCUS
THE RESEARCHERS ENGAGE AND FOLLOW
The lab is a catalyser and a process to engage
We engage and support citizens and policy actors, and follow how they take initiatives on their own, reflect and learn.
About the practical implementation:
• Important to support local initiatives and communities to create sustainable mobility in their own neighborhood.
• Important to be in line with local and regional development plans and support public actors in these processes.
• The researchers and the municipal community dialogue team arrange workshops and events, and provide resources and contacts for the citizens.
• More focus on sharing, community-based and cooperative solutions.
• The services are less central but need to provide opportunities for the citizens to take initiatives and to change their practices.
• The researchers both help initiate and actively support the citizens' and the policy actors' initiatives, and follow the process.
"ENGAGING"
SMALL SCALE
SUPPORT AND BUILD CAPACITY

corner, the planned mobility services were less central. During this workshop, the four research groups each had to select one of the four corners of the matrix and answer a set of practical questions about researcher roles, citizen and stakeholder participation, and required resources.

At the workshop, the opposed research views and aims became readily visible, as the four groups selected three different corners of the matrix. Citizen and Public Actors both chose the "Engaging" corner, which was not unexpected, as the groups' researchers were mostly from the same department and fairly aligned. Living Lab Design favored the "provoking" corner. Market selected the "Enabling" corner but argued that in practice, all four living lab types could be combined. The fourth, unused "serving" corner of the matrix was designed to reflect the apparent views of the regional public transport provider (SLL), which was an important partner to the research program at the time. In all, the four research design positions partly reflected the research group's perspectives, but also included research ideals and ideological positions, and led to conflicting practical requirements.

After the workshop, a new plan was made for the lab interventions that seemed to be an acceptable compromise for all. The living lab was provisionally narrowed to only include the Citizen perspective's aims and plans in the first step, since these aims were small-scale and required little collaboration with external actors. Also, Citizen was the largest and most well-organized research group and had a clear responsibility and designated resources for recruiting participants. The other groups lacked explicitly agreed-upon responsibilities for practical action. Market had expressedvery different research interests that would require engaging many service providers, but those aims had been postponed and the group had no plans at present to make this happen. The three members of Living Lab Design had limited resources and mandates and were only responsible for facilitating the planning process. In all, it became clear over time that the new plan did not actually bridge the different views of the lab. Even though an overall plan was now in place, the decisions and actions required to implement had not yet been made, and discussions came to a halt.

Design-driven stakeholder engagement

Early in the process, I had contacted several potential mobility service provider partners to understand their business interests, hoping to engage them in the

living lab experimentation. Some providers' interests were in line with our ideas, but because the research program could not settle on the format, scale, or content of the living lab, we could not go forward with discussing collaborations. In November 2021, when planning still appeared to be stuck, I offered to host two co-design workshops to engage these potential partners hands-on by encouraging them to actively consider potential roles for their organizations in a more sustainable near future. The workshop discussions centered around the prompt of a future vision document that described how a different economic reality had emerged for the provision of mobility services in the year 2026 (see figure 31).



Figure 31. "What if?" The 2026 future vision I produced for the co-design workshops. New mobility services aimed at individual travel are subsidized so they are viable in the outer suburbs and not only in the city center, in line with today's subsidies for public transportation.

The first workshop, concerning bicycling services, worked out unexpectedly well. Representatives from three different service providers immediately began discussing how their services could potentially be combined to fulfill different

types of travel needs in an outer suburb like Riksten. Using maps and cards that depicted local user need groups (see Figure 32), they discussed geographical zones and "stations" for their vehicles. They also spontaneously discussed how they could save on costs by using a shared service organization, in light of the lower population density in the outer suburbs.



Figure 32. Two photos from the first co-design workshop, concerning different bicycling services in the context of Riksten, an outer suburb to Stockholm.

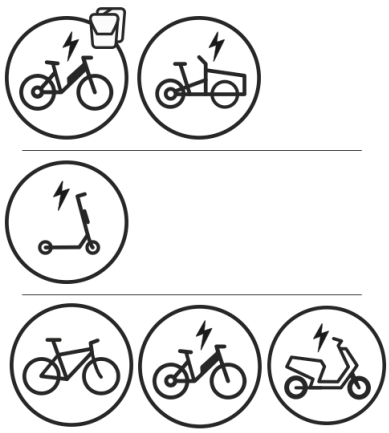


Figure 33. Graphic icons for the three bicycle service concepts that emerged from the workshops: from top to bottom, shared neighborhood cargo bikes, free-floating e-scooters for spontaneous travel, and three types of vehicles included in a leasing service for commuters.

We prepared a detailed presentation of the resulting service concepts (see Figure 33) to the executive committee and were later informed that the discus-

sion had been positive but also rather cautious. Some members had concerns about collaborating with an e-scooter provider, and there were conflicting views around the scale of the living lab. To further support decision-making, a subsequent more-detailed concept for the living lab included a concrete solution for the latter issue, suggesting four possible and compatible levels of user participation (see Figure 34). I also made additional contact with the three bicycling services providers to clarify their interests and requirements for participating in the lab.

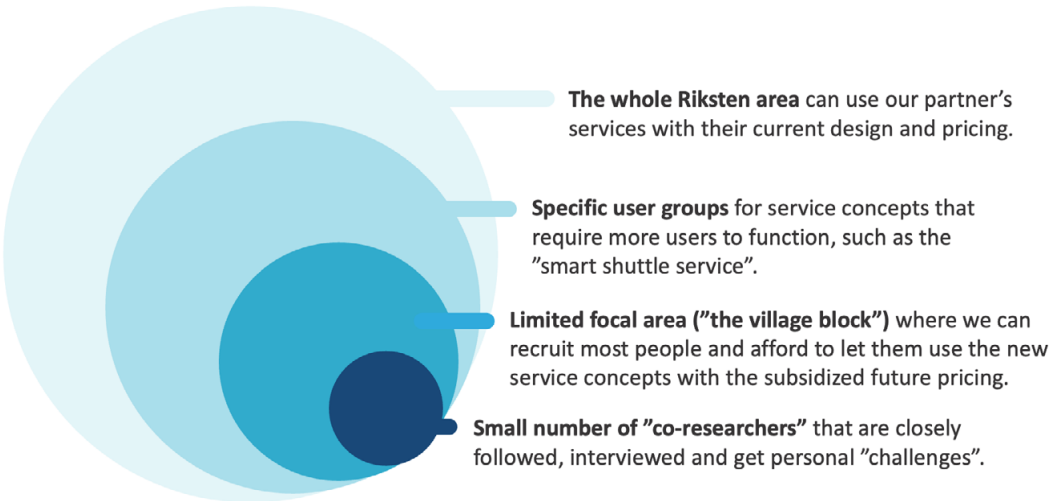


Figure 34. A visualization presenting an overview of the four suggested levels of user participation for the Riksten living lab.

At this time, however, no further action was taken by the program to involve service providers in the lab. The program management was waiting for action from the regional public transport provider (SLL), which was planning a commercial pilot in the same area.

There were hopes that SLL would come to an agreement with several service providers to launch their services in Riksten at no cost to Mistra SAMS, and this likely further limited decision-making. The core idea behind the research program was to study how new combinations of mobility and accessibility services could be enabled through a shared digital platform. Most service concepts would, however, require external collaboration and a larger number of users. Therefore, including these types of services was postponed until later. The researchers in the Market group were very interested in potential future

business models for services but focused their efforts on their separate Business Model Lab. Here, service providers were instead invited to discuss the issue in a series of online seminars.

While still planning for living lab participants to experiment with new mobility services, I suggested an intervention design that was in line with the 2026 future scenario. During the living lab, using the private car would be made more expensive, but in return, a 'mobility pot' would subsidize the participants' use of the new mobility services. The term mobility pot caught on in the program, leading to a decision to spend a few weeks reading up on the subject of mobility pots, economic incentives, and theories of motivation. In the end, the program decided to instead ask participants to make a personal "mobility budget" and to limit their travel to be within the planetary boundaries.

Epilogue

A few months later, all the mobility service providers that SLL had worked to engage, including the three I had invited to the workshops, one by one decided not to take part in the Riksten experiment. The Citizen group was well organized and had a clear responsibility for recruiting lab participants. Consequently, with time running out in the spring 2022, the smaller number of participants that was planned and budgeted for was recruited. No new mobility service concepts were developed, but in a collaboration with the previously contacted bike-sharing company, the research program purchased three electric bikes that came with digitally controlled bicycle locks to be used for sharing solutions. The bikes were made available for participants to book and rent. The digital platform and mobile phone app were used for booking the electric bikes but were also redesigned to further the "active citizenship" of the sustainability-oriented participants. The added app functionality included a tool that allowed participants to set up personal goals to reduce their private car use and a social forum for discussion and mutual support. As the bikes were expensive, a decision was made to subsidize the service in line with the 2026 future vision.

In 2022, just before the summer, a very small-scale and provisional launch was made in Riksten. At around this time, I left the program to focus on the final phase of CoKitchen and on my dissertation. At the time of writing, in April 2023, the bike-sharing company has become more involved, and a new type

of shuttle bus service has been launched, partly based on a concept from the second service provider workshops. The public transportation provider, SLL, has also advanced in their negotiations with a number of commercial service providers and will most probably soon launch a commercial pilot in the same area.

5. Analysis

This chapter reports on the main steps in the analysis during this PhD. Section 5.1 outlines the earlier stages of analysis, with reference to the resulting publications. Section 5.2 presents an account of the cross-case analysis for the six empirical cases that was carried out for the writing of this dissertation, as well as the most important results.

5.1. Earlier steps in the analysis

First steps in identifying key DLL characteristics (Publication B)

During spring 2018, I carried out a small literature review about living labs and formulated my initial thoughts on how the living lab methodology we were planning to use in Mistra SAMS differed from the commonly used definitions in the literature. After this, I presented a first draft report on this emerging design research methodology in a joint program planning document that was produced at this time (see figure 35).

Here, the living labs method was described as "interventions in a real-life setting, designed to simulate a possible future scenario." At a program conference in April, the main differences between the design-driven methods we planned to use and the mainstream European understanding of living labs were discussed.

A more-structured analysis was then conducted in the autumn of 2019, while working on Work Closer and before the Covid-19 outbreak. The resulting conference paper was the first published attempt to describe the emerging methodology and shared characteristics of the various research projects. This analysis identified eight key characteristics of DLLs (see Sjöman & Hesselgren, 2020).

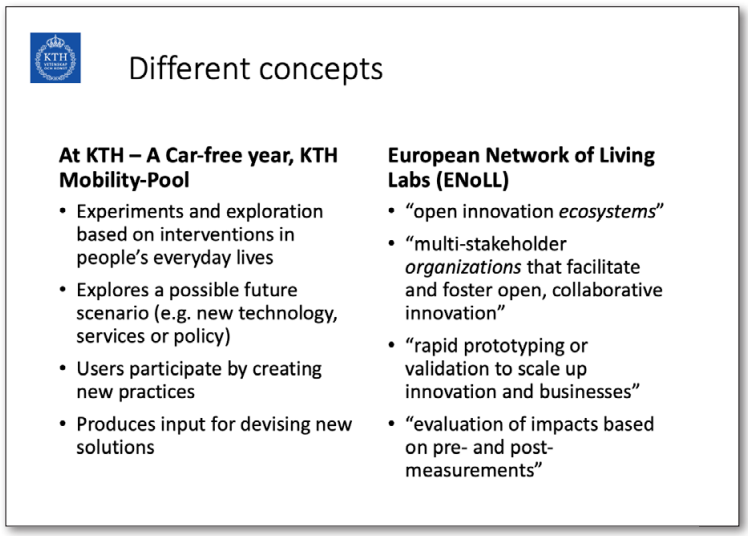


Figure 35. A Powerpoint slide from an internal presentation that I held to the researchers in Mistra SAMS in April 2018 (left column translated from Swedish).

Learning vs. solving research aims, and modes of participation (Publication D)

In the second stage of analysis, some of these characteristics were found to be related, since they shared a connection to early research stages (see Figure 36). These results were presented in Publication D. The DLLs open-ended and mainly exploratory research design appeared to have resulted from addressing similar research situations by use of design practice.

Explore & frame	**Innovate**	**Design/develop**	**Test & evaluate**
Main aim is *learning*			Main aim is *solving*
Early research stages			Later research stages
Open-ended exploration			Answer well-defined questions
Mostly design or research-led			Mostly market or policy-led
Mostly small scale			Mostly large scale

Figure 36. Typical characteristics of research in early or later research stages. (Sjöman & Hesselgren, 2022)

Later research stages of detailed development and implementation generally favor the use of evaluative methods, but the DLLs have not emphasized evaluation of the implemented concepts. It also follows quite naturally that in early stages, studies are mostly researcher-led, since market and policy actors are typically not ready to invest, while in later research stages projects are often led and funded by market or policy actors. Publication D describes these general methodological differences as strongly linked to the overall purpose of the research, which can lean more towards either "learning" or "solving."

Although the main aim of the DLLs has been learning, several of them also aimed to make a societal impact, and have influenced or resulted in activities with a solving aim in subsequent stages. The DLL experiments were understood as bounded, however, and not as developments that aimed to solve specific issues in the here and now.

Publication D also compared how users participated in the DLLs versus other real-life experimentation methodologies from the literature, which is discussed in greater depth in Section 6.4.

A reversed process, that uses prototypes in early-stage exploration (Publication F)

In a peer-reviewed conference article for the Continuous Innovation Network (CINet) conference, I presented a further step in the analysis, focusing on how the DLLs' process compares with that used in mainstream European living labs. This paper discusses how living labs generally focus on later developmental stages of implementation and commercialization, and that activities in real-world settings take place primarily in these stages. The DLLs, in contrast, had been carried out in early research stages and explicitly focused on exploratory activities set in people's everyday lives. Publication F included a visual overview of this "reversed" process. Importantly, the DLLs have used prototyping for exploratory purposes, while other design methodologies commonly use prototyping for evaluative purposes (Sjöman, 2022). This is further discussed in Sections 6.1 and 6.2.

5.2. The cross-case analysis

The cross-case analysis is structured around 20 criteria in the shape of key charactertistics that were identified and developed based on the earlier stages of analysis outlined above, and on revisiting the original data. The data was then scrutinized across the cases, aiming to disaffirm or validate these characteristics and also to enrich my understanding of their application and entanglements in practice

Method Characteristics / Research Principles

1. Addressing sustainability issues in early research stages

The analysis of research funding applications and initial project discussions regarding aims and stakeholder interests confirms that all the DLLs addressed sustainability challenges in an early research stage. Car-Free, Playing-Rules, and Work Closer in particular reflected an explicit understanding that there was insufficient knowledge to design and develop a functional solution, and that such a solution would take the form of socio-technical system change, partly on a societal level. Thus, the intervention concepts were designed with little or no prior research or analysis, as a means to start a broad exploration and help frame the issues.

LEV-Pool was an exception to these principles. Even though its early planning documents state that "there are no experiences yet from combining electrification with a reduction in vehicle size," LEV-Pool explicitly aimed to iteratively develop and evaluate a functional service concept, an aim more in line with market driven living labs or pilots. On the other hand, the commercial partner did not carry out any of the developments suggested during the project and clearly was not ready to invest in a service of this type before the study. Workshops were held with participants and stakeholders both during and after the experiment ended, but their aim was "not to solve problems, or to make the service more efficient" (Hesselgren, 2019).

CoKitchen also differed in some ways, given that the concept was clearly met with interest from industry; nevertheless, there was also a shared understanding that co-living might be socially challenging and conflict with current societal norms, and that these dynamics needed to be explored. Co-living also conflicted

with many urban structures and regulations at the time, but these were not the main concerns for CoKitchen. As with the other DLLs, the first apartment remodeling was done very quickly and without prior research, and in some ways went against the general understanding of user needs at the time. However, the second remodeling was designed by architects with experience from shared housing, to provide good conditions for co-living, even though some parts were planned specifically as tools for learning and not to fit perceived user needs.

2. Aim to openly explore an issue or concept
3. No aim to test and evaluate a concept or to 'answer' a specific question (negative version of criterion '2', included to further validate '2')

The analysis indicates that a mainly exploratory research design is a key characteristic of designerly living labs. The final analysis carried out for this dissertation identified the importance of how this type of designerly research practices first evolved at KTH. From the start, there has been a strong focus on the idea of prototyping a part of a possible sustainable future. In discussions with researchers from other fields, both internally in the project groups and in other contexts, one often heard statements such as: "But this is just like any living lab or pilot—you are making people test and evaluate a service concept." Here, the central difference lies in the intention to use the concept as a means to explore a future rather than to develop, test, and evaluate a specific innovation. In this vein, Car-Free's service concept for LEV rental was not designed to meet identified user needs in the present, but to explore future user needs. The design of the rental service concept front end emphasized generating the realistic illusion that participants were using a common, established service, and thus making the experience more immersive was prioritized over optimizing its functionality. There was also no iterative development of the service. The DLL explored the concept of car-free living with the help of light electric vehicles, but few of its findings concerned the vehicles or the rental-service concept as such. Instead, the data analysis identified unfit societal structures, such as conflicting social norms and a lack of bicycle infrastructure and delivery services. The second part of the project, "A Car-Free Society," then aimed to model the effects of scaling up car-free life in a future where those barriers had been removed, and to generate concept solutions to inspire policymakers. Overall, the DLL interventions have been designed and staged to offer participants a realistic experience, including a logical background story—such as the future vision

stories that were used to frame the Playing-Rules and Riksten interventions.

In line with this reasoning, it could be argued that LEV-Pool was not clearly a DLL. Part of the research group engaged in an open-ended exploration of user practices and understandings, but the study as a whole was planned as a relatively typical living lab that sought to develop, test, and evaluate a service concept. However, the user needs that were identified were not then used to finetune the design or improve the service concept during the DLL (even though a similar concept was later scaled up to a successful commercial service). Moreover, the central "caretaker" concept was ideated during the late planning stages—not to fit with identified user needs but to offer an interesting "twist" that might provoke more interactions and "elicit more findings" (see Hesselgren, 2019).

While planning Playing-Rules, the group was quite clear that its aim was not, per se, to evaluate the three economic interventions. These three interventions were selected quickly and did not build on existing literature. No analysis was made to see which of the economic incentives might be more efficient. Instead, they were selected to provoke participants to reflect more broadly on the cost of daily travel. Also, the simultaneous exploration of three different interventions, the small and diverse group of participants, the close researcher-participant relationship, along with other factors, made a structured evaluation impossible.

Differences concerning this learning-focused mindset in the DLLs led to a lack of clarity and internal tensions in some projects. In Work Closer, the designers that staged the NTC service concept shared the understanding that it was meant as a tool for learning, but some members of the Mistra SAMS executive committee had a different understanding, which led to the decision to carry out a quantitative evaluation of the concept.

4. Aim is 'learning' more than 'solving' (another perspective of criteria '1' and '2').

As discussed in Publication D, the question of whether a study is conceptualized as aimed at learning or solving is important. Even though there has been some confusion, all the DLLs (with the partial exception of LEV-Pool) were understood as open-ended experiments that were bounded in time and not intended to develop a solution in the present.

This contrast between learning and solving aims was not always black or white, however, as the DLLs were always conceptualized as seeking pathways for sustainable development. At one end of the spectrum, the CoKitchen co-living apartment was seen as promising and was clearly designed to be functional. It was also improved over two iterations, and the study aimed to produce knowledge so that a specific organization could develop the concept in the near future. Nevertheless, the apartment was not only designed to meet stated or experienced user needs. For example, the apartment was rebuilt to accommodate five students rather than four with the deliberate intent of provoking the inhabitants and thus providing additional points of learning. The bathrooms were shared, even though previous research had clearly indicated that students were keen to have private bathrooms. Overall, the DLLs have all included features or twists intended to challenge, disrupt, or provoke the user participants. In Playing-Rules, at the other end of the spectrum, showing car owners the real cost of their car ownership was clearly not intended as a solution, and in Car-Free, no attention was given to designing the participants' LEV rental as a functional service concept. The main focus was on taking the car away, which again, is clearly not a "solution" from the user perspective. Little or no research was made before designing the DLL interventions, even though several of the concepts built on knowledge from previous projects. With the exception of CoKitchen, the DLLs did not include iterative cycles to finetune the concepts. The co-living concept also violated current building and tenure regulations, which meant it was not designed for adoption as a viable solution in the present. Across the spectrum, I argue that all the interventions were designed mainly as learning devices.

By the early-stage framing of an issue, each of the DLLs has been conceptualized as a first step in a longer process, and in some cases the DLL findings have been used in subsequent studies or pilots. The service concept from LEV Pool inspired the launch of a similar commercial service after the DLL had ended, and in the second stage of Car-Free, after the intervention had ended, visualized concept solutions were developed, intended to serve as actionable input for policy makers. When the NTC service facilities in Work Closer were taken over by a new actor, this was in connection to the research program, but after the DLL had ended. In this case, part of the research team referred to this as the project entering an upscaling stage, while it can also be viewed as a subsequent step taken outside of the living lab.

5. Longitudinal experimentation, set in real-life

All the DLLs, without exception, were planned as longitudinal studies. Planning documents and presentations explicitly described this as a requirement to let the sense of novelty wear off, as well as to give new practices time to evolve and become routine. In all cases, it also turned out to be very important to explore lifestyles and interventions across several seasons. In Car-Free and LEV-Pool, winter travel proved to be challenging, and in Car-Free this required planning a new intervention to support participants by developing their skills and providing them with suitable clothing. In Playing-Rules, the unexpectedly large amount of driving that took place during summer vacation both required some replanning of the study and challenged existing understandings based on national travel survey data.

One integral aspect of the DLLs is making interventions in people's everyday lives as a way to understand socio-technical complexities, garner attention and increased impact. In contrast, as outlined in Section 2.5, mainstream European living labs, along with other similar research formats, may engage citizens and stakeholders in co-creation but do so primarily through workshops, interviews, or community dialogues. In these research contexts, when experiments are conducted in real-life everyday contexts, the aim is mostly to harvest user feedback during testing, evaluation, and commercialization of relatively mature product and service concepts.

6. Design-researchers design and maintain control over the interventions.

All the DLLs have involved prototyping future sustainable concepts deployed in participants' everyday lives. Researchers have then observed how the participants reacted and adapted to that future, adopted new resources, or created new practices. The design researchers therefore aimed to stage an immersive and realistic experience for the participants. The interventions were designed with a logical reasoning or "back end" to support them, which was combined with a carefully designed, realistic "front end" user experience so that they would be perceived as plausible future concepts.

It is clear that the DLL interventions were generally designed quickly and without user input, based on normative understandings of what constitutes sustainable lifestyle practices. Here, LEV-Pool again stands out, as this lab included a service design pre-study whose clear goal was to identify user needs

and support the design of a smooth-functioning and attractive service. However, the preliminary study's recommendations were not implemented. CoKitchen also differs in certain ways, as this DLL included interventions on two levels. In this case, a pre-study was carried out by the service designers and supplied concrete ideas for the design of the service interventions, but it mainly did not influence the design of the co-living apartment per se. The choice to design the interventions mainly without input from the participants will be further discussed in 6.4.

While the DLLs did not generally involve the user participants in the design and planning, most of them did include other public and private stakeholders, who in many cases had several opportunities to directly or indirectly influence the design of the interventions, by taking part in study planning or in staging the interventions. Re-reading the project documentations however, it is clear that this involvement largely failed to materialize. In LEV-Pool, Hertz Sunfleet supplied a booking service but did not invest in the new service concept until after the DLL was concluded. In the first phase of Mistra SAMS, the living lab planning group sent out a carefully prepared document to program partners outlining a number of alternative lab formats and possible interventions. Still, the partners did not indicate any clear preferences and offered little useful input. Overall, stakeholders seem to have been difficult to engage in the early stages of design and planning. Interest instead often emerged only when the DLLs have been staged in the real world, and this interest has tended to strengthen as partner organizations interacted with the interventions.

In the second phase of Mistra SAMS, I tried to engage commercial stakeholders to set up new mobility services in the Riksten lab. Still, this did not mean that these companies were asked to design the lab interventions in the form of new commercially viable services. Instead, they were asked to design new service concepts within the framework of a sustainable future scenario. This will be further discussed in section 7.2 with respect to the concept of "orchestration."

User Activity / Engagement

7. Users are considered co-researchers
8. Users are not engaged in formal co-creation activities
9. Close researcher-participant relation

(Criteria '8' and '9' were included to validate and to provide complementary perspectives of criterion '7'.)

DLLs participants have mostly been self-recruited through advertisements or offers directed at selected groups. In Playing-Rules, car owners were more randomly approached in the street. In all cases the participants freely agreed to take part in the DLLs, without monetary compensation for their engagement and time spent. Reading project documentation, I find that all the DLL participants have been well-informed about the research purpose and intentions. Information presented to participants clearly emphasized the value of participant reflection and experimentation as part of their role as co-researchers. Participants were not involved in study planning or as co-creators in designing or improving the interventions, and did not have decision-making power. Thus, the DLL research has not been insider-led.

The DLL interventions were not co-designed, and in no case were they changed or improved over the course of a DLL based on participant input, except to solve purely technical shortcomings. There were a series of formal co-design workshops in LEV-Pool intended to find ways to reduce car commuting and improve the service concept, but the service concept was not changed to reflect the results of the workshops.

In all cases, the design researchers met most of the participants several times in person—in some cases in their homes—and came to know each other by name. Arguably, there have been close researcher-participant relations. The participants' level of engagement was generally high, as exemplified by the challenging situations that Car-Free participants confronted, the time-consuming data correction work that participants had to do in Playing-Rules, or by the Challenge-takers efforts in Work Closer.

10. Users engage in reflection.
11. Users experiment with changes to real-life practices.
12. Users face challenges or experience difficulties (interventions were disruptive or designed to provoke).

    (Criteria '10', '11' and '12' were found to be strongly connected and were analyzed as a group.)

All the have DLLs emphasized in-depth interviews structured to be open to

emerging topics, in many cases using trigger materials to encourage reflection and storytelling. Participants often discussed or questioned their current lifestyles and barriers and motivations for change, as well as their own understandings or perceptions. Even when the interventions did not lead to much practical experimentation (for various reasons), they still provoked participants to reflect on their current practices and understandings. In many cases, merely considering or unsuccessfully attempting to make changes led to realizations about norms or other barriers that would otherwise have been difficult to anticipate. With the exception of a few participants in Playing-Rules, all DLL participants have experimented with creating new practices, and often in the face of challenges or difficulties.

One thing that became clear in the analysis was the extent to which participants' differing practical circumstances and level of motivation enabled or hindered the emergence of new practices. One of the research publications from LEV-Pool discusses participants' varied roles with respect to the intervention, from "non-users" to "ambassadors." In Work Closer, two participants described having very independent roles at work, and for long periods they used the NTC almost daily. However, most participants described it as challenging to work remotely even one day per week, and some were not allowed to use the NTC at all.

Stakeholder Activity / Engagement

13. Stakeholders are not in control of research methods or aims.
14. Stakeholders learn by being subjected to, or challenged by the lab interventions.

    (Criterion '14' was included to further explore '13'.)

Even when asked for input, private and public stakeholders in many cases found it difficult to engage in the DLLs. In some cases, there were indications that the sustainability issues in question were "on the radar" of private stakeholders, but they had not yet identified new potential business models. Public stakeholders often seem to have been limited by the fact that the issue at hand did not explicitly fall under their scope of responsibility. The role of Academic Housing (AH) in CoKitchen was different. The project leader from AH was deeply involved and indicated that she viewed the lab as a proof-of-concept to

gain traction for co-living internally and to influence regulatory barriers.

When nonacademic stakeholder organizations were involved in the DLLs, the strategy for their involvement was sometimes for them to more or less actively take part in staging the interventions by providing resources. Even this passive role as a resource provider has in many cases provided stakeholders with useful knowledge about how their products or systems worked for users: partly because the use in the DLLs has been different from that of ordinary users, and partly because the studies were well documented. The wintertime use of the Renault quadricycle in LEV-Pool and the use of the beehive chairs in CoKitchen are examples of this. Hertz Sunfleet did not invest in making any changes to their current services or booking system during LEV-Pool, but they were able to use the knowledge produced and to go straight to a pilot after the living lab ended.

In Work Closer, Ericsson and other partner companies participated by providing users for the prototype NTC, although this involvement turned out to be a challenge for Ericsson. Even with support from top management, it took nine months to secure the necessary permissions on multiple levels and resolve issues with resistance from local managers, security concerns, and unclear insurance policies, before its employees were allowed to use the NTC. This intense learning experience uncovered many barriers to implementing this type of remote working solution, but also illustrated the concept's potential value for employees in terms of well-being, productivity, and reduced stress. The municipality of Stockholm made plans to implement a flexible workplace solution in their own organization to better utilize employees' time before and after offsite meetings, but this plan was never implemented following the interruption in business-as-usual caused by the Covid-19 lockdown measures.

In the Riksten DLL in Mistra SAMS' second phase, I worked to involve commercial mobility service providers more actively. The co-design workshops emphasized leading these stakeholders to consider how their respective organizations could do business in a future where private cars had become much more expensive and "smart mobility services" were subsidized in the same way as public transportation is today. Interestingly, the three representatives from different bicycling services immediately started to discuss how their services would complement each other and how they could potentially build a shared service organization to reduce the costs of operating in an outer suburb. In this

case, I never managed to get a mandate or budget to fund their participation. These three, and other potential partners ultimately declined to invest in the project, and Mistra SAMS instead ended up purchasing a small number of electric bikes.

CoKitchen differed in two ways. The stakeholder partner AH had stronger business reasons and clearer aims for supporting the project financially, but it did not provide any resources for staging the future prototype. Instead, the CoKitchen DLL was one of several experiments hosted at the KTH Live-In Lab, and the remodeling was handled by the living lab administration and its construction partners. Still, the CoKitchen prototype have engaged parts of AH's operations in other types of interactions, as it became clear that their current ways of making business calculations were not appropriate for a co-living setup.

Innovation outcomes and impact

15. Enabled innovation activities in subsequent steps (outside the DLL)
16. Aimed to produce 'actionable' knowledge for societal impact.
17. Identified structural barriers for sustainable practices.
18. Produced knowledge of user practices, norms and perceptions.

    (Criteria '15' - '18' were found to be associated with each other and were analyzed as a group).

The DLLs often aimed to provide relevant stakeholders with actionable knowledge, in particular in the form of identifying barriers to sustainable change. The barriers uncovered include inhibitive regulations; inappropriate technologies, services, and infrastructure; and current user understandings, social norms, and practices. Car-Free and Work Closer are the best examples of this, as they received a lot of attention from and were presented to policy-makers at several levels. As with Criteria 13 and 14 above, some DLLs—notably LEV-Pool, CoKitchen, and Work Closer—engaged stakeholders in interacting with the DLL interventions, which provided intense learning experiences. In at least five cases, this led to or clearly influenced subsequent steps of research and development.

19. New types of value gained from sustainable practices were identified and demonstrated

Publication B describes how DLL experimentation led participants to identify new, and often unexpected, forms of value that could be gained by shifting to more sustainable practices. Revisiting the interviews, it was easy to find several examples of this. One participant in Work Closer, after trying the NTC a few times, decided to quit her job on the other side of the city because it felt so good to be connected locally and to be closer to her children. Many participants found that the reduction in commute times from using the NTC allowed them to spend more time with their children, to exercise and practice their hobbies. In Car-Free, one family found traveling by train to their country house provided quality time with the family and realized that the same trip by car was not enjoyable. The students in the CoKitchen co-living apartment found that the friendships they formed there were more familial and supportive, and that they enjoyed sharing food and household equipment. These types of positive values, often social in nature, would be hard to anticipate without living differently, and mostly became visible only after some time.

20. Large outreach relative to the size of the study.

All the living labs except Playing-Rules shared an aim for high visibility. Ever since the earliest developments, the DLLs have aimed not only to explore but to publicly demonstrate possible future sustainable lifestyles. As the lead design practitioner in Mistra SAMS put it, the DLLs provide "good TV" as they are easy for people to relate to. Car-Free was followed by many similar projects in other Swedish municipalities, and a quick web search identified two more such projects current recruiting citizens at the time of writing. Work Closer was highlighted by the research funding agency Mistra and presented as a promising example at an EU-summit in Brussels, while the NTC was visited by a secretary of state. Playing-Rules, on the other hand, had a very small budget and was not followed up on, and therefore likely had very little impact outside the research program.

The DLLs' relative reach or impact compared to other types of research is, of course, difficult to estimate. Nevertheless, Car-Free, LEV-Pool, Work Closer, and CoKitchen all received broad media coverage at both local and national levels.

Results and discussion

The discussion in this dissertation is divided into three chapters. Chapter 6 and 7 address the main research questions of understanding and implementing DLLs.

More specifically, Chapter 6 answers the questions of why, when, and how DLLs have been used, the defining characteristics of the approach, as well as common results and impacts. This chapter also positions DLLs in the literature and in relation to other approaches for real-life experimentation.

Chapter 7 discusses the second broad research theme about how DLLs have been, and should be, implemented in practice. The first part presents challenges and strategies related to the collaborative implementation of DLLs, and the second part discusses designerly capacities for visioning and mediating, and for the practical staging of DLLs.

The third broad research question or theme concerns what is specifically designerly about the DLLs and their implementation. Insights and reflections regarding this overarching research theme concern most aspects of the DLLs and are therefore interwoven throughout all parts of the discussion.

Finally, in Chapter 8 I briefly present some additional reflections from completing a design methodology PhD.

6. Understanding DLLs

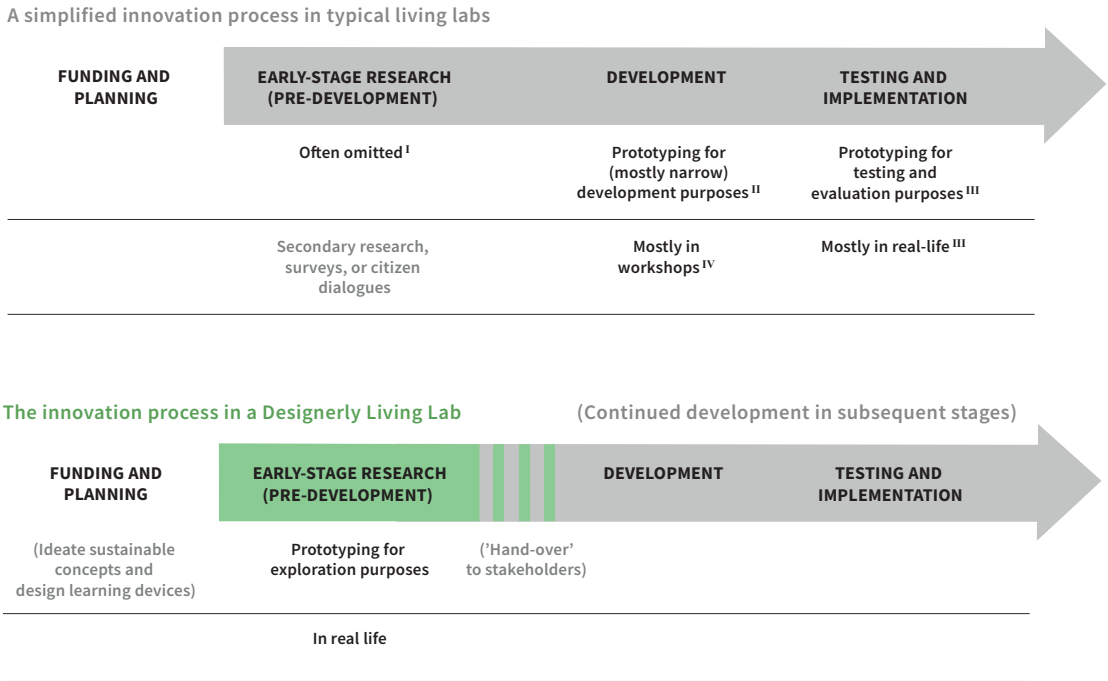
6.1. Prototyping futures in early research stages

Design is concerned with how things ought to be (Simon, 1969), which includes envisioning possible futures. Design is also abductive, as designers create provisional concepts as a form of physical hypothesis. In this way, in situations involving overwhelming complexity and ambiguity, designers can turn to "doing" or "making" something as a way to start learning (Coyne, 1992). Prototypes can be used to probe uncertain or complex contexts where an analytical approach cannot reveal the solution (Chow, 2013). I have found this designerly predisposition for learning through doing-first strategies (Mintzberg & Westley, 2001) to be at the core of the DLL-methodology.

The DLLs were set up to approach complex and clearly socio-technical sustainability issues in an early research stage when:

- a future concept was deemed not possible to develop into a fully working solution at that time due to costs, contextual limitations, complexity, or lack of knowledge.
- the envisioned change was of a more disruptive and systemic nature and outcomes were uncertain.
- relevant private and public stakeholders were not ready to take initiatives.
- potential users were not ready to change their lifestyles.
- there was an aim to provide actionable knowledge for relevant actors.
- social or behavioral factors were understood as important.
- partners hoped that design would provide more "action" and garner public attention
- there was an aim to demonstrate sustainable alternative lifestyles.

As discussed in publication F, by 'doing first' the DLLs have focused on a different part of the overall innovation process than most living labs, and have also engaged user-participants in experimentation for other purposes, see figure 37.



I : Steen & van Beuren, 2017
II : Kommonen & Botero 2013
III : Salminen, 2011; Hossain et al., 2020
IV : Compagnucci, 2021

Figure 37: Top: An innovation process in a typical living lab with prototyping and experimentation taking place in late stages. Bottom: DLL activities are limited to the early stages of the process. Real-life prototyping for exploration purposes, may be followed by later public or private sector developments (adapted from Sjöman, 2022).

The DLL interventions have been specifically planned and designed to gain knowledge of how new socio-material assemblies might fit or conflict with, or be integrated into societal structures. The DLLs have therefore been planned to provide input in anticipation of later design phases and subsequent development elsewhere. Thus, DLLs can also be regarded as an initial pre-development iteration within a longer development process, whereas in mainstream living

labs the pre-development phase is often omitted and users are rarely actively involved (Steen & Van Beuren, 2017).

In mainstream living labs, pre-development research has been found to include exploratory activities using ethnographic methods (Schuurman, 2015). However, since these approaches typically seek to create innovations, these activities explore the current state of affairs (idem.; ENoLL, 2017) to find opportunities. The DLLs have instead researched possible future scenarios to identify necessary pre conditions for the evolution of alternative sustainable practices. Since the required changes were often at a systemic level falling well outside the scope of the lab, the resulting knowledge has been shared with other actors. In several DLLs, the intervention concept has also been viewed as a promising concept, which has been *demonstrated* to further support the required transitions. I thus suggest that the DLLs have been concerned with transformation in a broad sense, rather than with innovation in the narrower sense of creating and exploiting an invention, which is widely held to be the central aim of living labs (see eg. ENoLL, no date).

This way, DLLs can also be viewed as following a reversed process. Chow & Jonas (2010) argue that the dominant HCD process of first studying users should be reversed when a project is open-ended, since narrowing the design space is only relevant in problem solving. In open-ended situations, they note, it is not useful to conduct a user study first, because some user needs and wants can only be identified after a design is introduced. Similarly, practice theory holds that user needs result from practices and not vice-versa (Shove, 2007). Open-ended projects therefore often include designerly "making" strategies, such as visualizations or prototypes that encourage speculation (Buchanan, 1992), as well as "trigger materials" or "probes" aiming to elicit reflection (Kankainen et al., 2012; Mattelmäki, 2008). I argue that the DLLs have clearly played this role and that this type of open-ended exploration is rare in real-life experimentation.

Early-stage explorative prototyping

In living labs, prototypes can be used in earlier stages of development, but then most commonly in workshops or in testing facilities (Compagnucci, 2021). Real-life prototyping has been found to be used mostly in the late stages of testing and implementation, and for evaluative purposes (Salminen, 2011; Hossain et

al., 2020). Viewing the DLLs as a methodology used in the pre-development or pre-design phase of longer innovation or development processes, their use of prototypes in early stages differs from common understandings of prototyping in the design field. From the perspective of Sanders and Stappers' (2014) well-known illustration of design projects (see Figure 38) position prototypes in the evaluative stage, in line with their more customary usage.

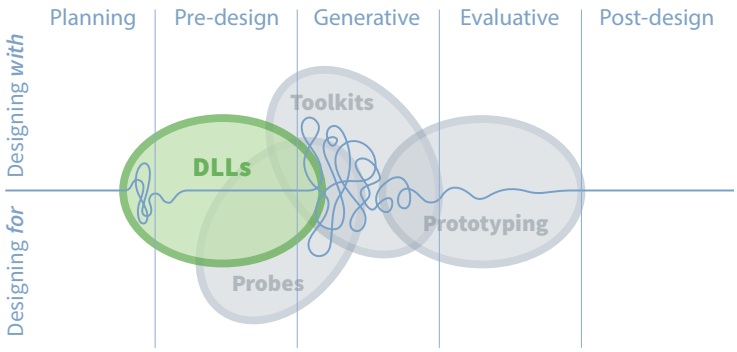


Figure 38. The use of DLLs and other participatory design methods during the design process. (Adapted from Sanders & Stappers, 2014).

A DLL, being mainly concerned with framing the design space can be conceptualized as explorative prototyping in the pre-design phase.

> *To conclude this section, the DLLs have been designed to approach complex sustainability issues in early research stages, at a time when relevant stakeholders were not ready to take initiatives, and the concepts were not yet requested by users.*
>
> *The key characteristics of the DLLs result from this research situation and purpose. reflect this research situation and purpose. Thus, DLL activities have been limited to the pre-development stages of the design process and have sought to provide inputs for subsequent developments, outside the lab. Early-stage explorative prototyping has been used to identify necessary pre conditions for the evolution of alternative sustainable practices.*

6.2. Provisional tools for learning

The research purposes of learning vs. solving

In early research stages, the DLLs have been planned with the explicit aim of openly exploring and learning about an issue. I argue that whether the research aim is to *solve* an issue in the here and now, or to explore and *learn* is an important distinction. Puerari et al. (2018) make a similar distinction between co-creation for the purpose of learning vs. making, where making aims at a specific innovation goal or output, while learning aims at creating knowledge and enabling changes at a socio-technical or societal system level. This distinction is important because when the aim is learning, experimentation is not limited to product or service development per se and is instead open to more general findings. These findings may challenge current concepts, fall outside of the producer's control or reveal more-disruptive paths forward. In contrast, living labs and similar research approaches often describe their research as exploratory, but these explorations are often restricted by narrow aims and limited to the current state of affairs. Kommonen and Botero (2013) find that living labs are often producer-driven and geared towards "a narrow product development process," involving users mainly for testing specific features. In an in-depth review of three large living lab structures that were broadly regarded as examples of best practice, Schuurman (2015) distinguishes between lab activities in exploratory vs. exploitative stages of an open innovation process. Here, all activities are directly related to the process of developing a specific product or service concept. I suggest that this mainstream type of living lab have a solving aim, reflecting a market logic or *technical rationality* (See Bratteteig & Stolterman, 1997). When these approaches include exploration or discovery in describing their methods, it refers to research into the current usage situation, to identify user needs and provide opportunities for private or public stakeholders to innovate (see, e.g., Følstad, 2008). This may take the form of focus groups or citizen dialogues. The term *experimentation* then mostly refers to real-life testing of innovations, followed by evaluation of their impact or their product-market fit. However, it can also refer to co-creation activities in workshops, such as the use of brainstorming or mockups to co-design innovative solutions (ENoLL, 2017). In publicly funded living labs, and notably in ULLs, a similar solving aim may reflect concrete policy objectives, such as making specific improvements to the urban environment.

Bulkeley and Castán Broto (2013) hold that transition experiments always encompass dual aims of learning and making changes. I find that this has been the case in the DLLs, but the experiments have been clearly understood as bounded in space and time. Any lasting real-world change has been envisioned as taking place in later stages and outside the lab, often a few years into the future and on a societal system level.

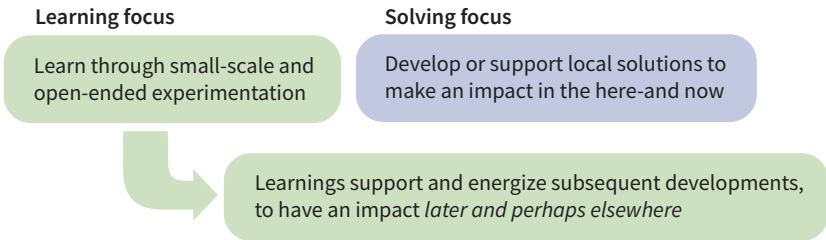


Figure 39. The temporal difference between a learning or solving aim (adapted from Sjöman & Hesselgren, 2022). The DLLs have been bounded experiments that aimed at open-ended learning, with any real-world change envisioned as taking place later and perhaps elsewhere.

In the public realm, living lab development processes are often narrowed by technocratic or neoliberal agendas (Palmås & Busch, 2023). In action research, participatory design (PD), and design for social innovation (DSI), the mainly solving aim is connected to ideals of activating and empowering people to make real changes to their own present-day situation. Building on Puerari, I suggest that open-endedness and the inclusion of higher system levels are the most distinctive features of research with a learning aim, in contrast to a narrow focus on producing a solution. However, because DLLs have been concerned with learning through acts of *making*, I suggest that the term solving is better suited for this distinction.

As Hesselgren et al. (2017) note, several DLLs have shared the explicit aim of making an indirect impact in society, either through providing learning experiences or by demonstrating a concept. In contrast, other real-life experimentation approaches are often concerned with the iterative improvement of an innovation to allow for diffusion (see, e.g., Vergragt & Brown, 2007). Diffusion may include embedding or transformative place-making for uptake of the innovation in a local community, or upscaling in the sense of market introduction and entrepreneurial growth (von Wirth et al., 2019). The DLLs have generally lacked such objectives, and I argue that the interventions instead consisted of provisional concepts that functioned as tools for learning. The intervention

concepts were largely not intended for further development and implementation, even though in many cases they were viewed as promising. I suggest that viewing DLL concepts as provisional tools for learning has been one important reason why we were able to take the step of building them in real life. Almiral and Wareham (2011) find that the mainstream European living labs play a similar role by bringing actors together to reduce risk in pre-commercial development, but lean towards exploitation and diffusion, while the innovations were mainly incremental. I suggest that pilots and the more mainstream notion of living labs typically aim for, and require stronger indications that a concept or a technology is ready for commercial exploitation.

As outlined in the cross-case analysis, the interventions' main function as learning devices seems to be validated, as the DLLs with a few exceptions fit with the following criteria:

- With some exceptions in Mistra SAMS, the DLL concepts have mostly been ideated very quickly, with little or no preparatory research into existing solutions or current user needs, or else they did not aim to fulfill such needs. In some cases, pre-studies have been carried out, but these have had very limited influence on the design of the interventions.
- The DLL research has been open to findings related to higher system levels or that require changes to the research scope.
- The DLLs have been small-scale and have included methodological choices that clearly do not support unbiased evaluation or seeking answers to predefined questions.
- At the time, the DLL interventions were described as designed to provoke or disrupt in order to elicit experimentation and reflection. LEV-Pool aimed to develop and evaluate a service concept but also included a concept "twist" that was designed to provoke more learning experiences.
- With the exception of the two remodelings in CoKitchen, there were no iterative cycles of development or improvement of the DLL concepts.

I find that the first point is in line with Preben Mogensen's (1992) concept of provotyping to understand future needs. Mogensen suggests actively using an initial lack of knowledge to allow challenging an existing practice. If a designer understands the users' situation too well before designing, they risk developing a "mutual taken-for-grantedness".

The DLLs function as backcasting set in real life

The interventions are at the core of the DLLs. Staging future concepts in today's society and then letting potential users and stakeholders interact with them has revealed barriers to those futures, as well as positive gains and motivating factors. These elicited barriers may be things such as norms, regulations, or infrastructure that is lacking, is unsuitable for or conflicts with the hoped-for sustainable future.

The interventions enable or prompt our co-researchers to experiment with changing their practices or to more actively reflect on them. In Publication B, I referred to this identification of barriers and pathways forward as backcasting set in real life (See figure 40).



Figure 40. A visualization of the backcasting process in a DLL. A component of a sustainable future vision is 'staged' in today's world. Potential conflicts with current structures are identified, and can thus be addressed.

In her dissertation, Mia Hesselgren described this as mapping or "pinpointing" points of inertia, tensions and threats that stand in the way of the dissemination and mainstreaming of the DLL participants' new sustainability practices.

Identifying these barriers or tensions has often revealed potential pathways for action. CoKitchen identified barriers to the broad uptake of co-living in the form of unsuitable building and rental regulations; a follow-up research project aims to facilitate the implementation of the required changes.

Practices are not re-created or changed through new products or services alone. Practice change is intertwined with many other developments that may seem like exogenous variables, outside the designer's influence. DLL researchers seeking to enable participants to experiment with changing their practices have, in several cases, identified and needed to address these seemingly external socio-technical system factors. The DLL interventions have included attempts to challenge conflicting social arrangements, understandings, and everyday routines, to bridge competence gaps, and to bypass unsuitable regulations. The DLLs have also made it clear that practice-oriented design often requires making changes to bundles of practices (Hesselgren, 2019).

The Valley of Inconvenience

In presentations, I have often referred to what I have termed "The Valley of Inconvenience" (see Figure 41).

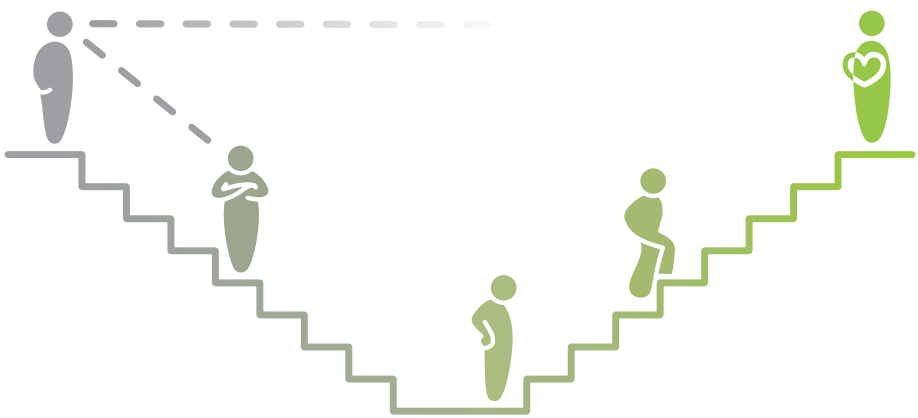


Figure 41. "The valley of inconvenience", a concept that was inspired by an insight from design agency Transformator's prestudy to the Playing-Rules DLL.

Standing on the top of the stairs to the left, I enjoy a high level of convenience, and it is easy to imagine the inconvenience I would suffer if my car was suddenly gone tomorrow morning and I needed to find a bus stop and figure out where to buy a ticket, find a way to get my children to school, and could not fill

my trunk with groceries on my way home. In this case, I can clearly imagine the inconvenience of taking a few steps in a sustainable direction. I perceive this as going down the stairs. It is much harder to anticipate the alternative value I might gain later on after having relearned and found new ways. This value might include time to relax on the train, eliminating the stress of finding parking, or talking to my children on the bus. We struggle to see the new kinds of value (often social value) that comes with living a more sustainable lifestyle, on the other side of the valley. In the DLLs this effect has been very evident. Participants have often found the interventions challenging, as they involve relearning, and in many cases only later on discovered the positive, social value that these changes entailed: such as the children in Car-Free who found that they gained independence when they had to learn to use public transportation (Hesselgren and Hasselqvist, 2016). In CoKitchen, the students described living in the test apartment as more familial and homelike. In Work Closer, many of the NTC users reported they felt less stress and had more time with their children when they worked closer to home. These positive gains are in line with what Rosemary Randall (2009) has called "the happiness tale," referring to the Ecological Footprint movement's narrative of a new low-carbon world potentially bringing us a closer local community, more meaningful work and more quality time with our families (Chambers et al., 2000). In contrast, people more often suspect that sustainability transitions will entail less material wealth, breaking with our way of life, losing in status and identity (ibid.; see also Spaargaren, 2011). Allowing time for participants to emerge on the other side of the valley was an important function of the DLLs—one that is rarely afforded in other modes of research.

> *To summarize section 6.3, one key characteristic of the DLLs is that they have been designed as exploratory studies. The future concepts staged in the DLLs were not designed as solutions to be tested and evaluated but instead as provisional tools for learning and as prompts for experimentation and reflection. I suggest that the choice of learning vs. solving is a key strategic choice when planning a research study.*
>
> *Like backcasting set in real life, the practical staging of the DLL interventions in present-day settings has identified barriers and linkages on multiple levels, such as social norms or regulations, as well as limitations to current infrastructure, products, and technologies. Identifying these*

> *barriers or tensions has revealed pathways for action. The longitudinal exploration allowed new practices to evolve and revealed new types of value to be gained from more-sustainable lifestyles.*

6.3. Experimentation in real-life contexts

As a vehicle to explore sustainability issues, a main difference between DLLs and most other methodologies is that the DLLs have engaged people in experimentation *within the context of the their own everyday lives*.

"Experimentation in real-life contexts" is often described as a core principle of living labs (ENoLL). Still, the main European stream of living labs have been found to use real-life contexts mostly for late-stage testing to validate concepts and to improve their reliability (Følstad, 2008; Salminen, 2011; Steen & van Beuren, 2017). Instead, when aiming for generative input, these research infrastructures have been found to mainly engage user-participants and other stakeholders in workshops, seminars and citizen dialogues (Compagnucci, 2021). In a literature review, Hossain et al. (2020) describe three different understandings of the term "real-life environments" where two imply the provision of environments where 'real-life meetings' between relevant actors and potential users can take place. Designers have a strong preference for 'making' and 'learning by doing', but in co-design these generative activities have been described as mostly taking place in workshops, using visualizations or *generative toolkits* (Sanders & Stappers, 2012, 2014). Similarly, Ilstedt and Wangel (2013) describe how future prototyping using visualizations, digital planning tools or games, may support future-oriented research.

I suggest that it is important to more clearly distinguish between real-life meetings and real-life usage contexts, as they serve very different purposes. A co-design workshop may use a visualized future scenario to enable speculation and consideration, and to provoke reflection. DLLs, in contrast, stage and explore future concepts in the context of the participants' everyday life, which goes a significant step further towards this end. I argue that the broad tendency in living labs and similar methodologies *to either view the living lab as a physical meeting place for co-creation activities, or as a real-life testbed for collecting usage data* precludes real-life exploration. Engaging people to take part in experimentation within the context of their everyday lives can be

difficult, but is necessary for exploring social factors and societal entanglements. Moreover, sustainable concepts and practices are often counterintuitive (Spaargaren, 2011) or go against generalized norms and understandings, which means that initial user reactions e.g. in workshops, can be misleading. Extended lived experiences from real-life exploration clearly produces more valid user reactions and reflections than workshops or surveys.

DLL design researchers have, in some cases, hosted workshops for co-creation, such as in LEV-Pool, but the heart of participant and stakeholder engagement has been their interacting with interventions in everyday life contexts. Designers, and service designers in particular, may use their sensibilities regarding user experiences to conduct pre-studies that serve as input for development. Some of the DLLs have commissioned design agencies to carry out pre-studies for this purpose, but *real-life prototyping remains the core activity*. Also, as seen with the case studies DLLs, the results of these pre-studies were for the most part not used to improve the intervention concepts.

Although there are some prominent examples of generative co-design activities being carried out in everyday life contexts (see, e.g., Hakkarainen, 2017; Botero & Hyysalo, 2013) these authors describe this as rare. In community-based action research and other insider-led approaches, user communities may engage very actively in making changes in their own lives and neighborhoods, but in expert-led research approaches this is uncommon.

Conceptualizing a living lab as an experiment or an infrastructure

Many have criticized living labs for their diversity of conceptual understandings and methods (see, e.g., Hossein et al., 2020). Schuurman (2015) goes so far as to call living labs "methodologically empty boxes." Living labs are described both as a method or approach and as infrastructures or ecosystems (Leminen, 2015). Addressing this confusion, Schuurman (2015) proposed a framework that considers living labs as encompassing three levels of analysis. On the macro level, a constellation of organized stakeholders seek knowledge transfer and retention, on the meso level an innovation project is carried out, and on the micro-level user involvement contributes to innovation in several methodological steps. A more recent publication from ENoLL (2017), however, more clearly places living lab methodologies at the meso level.

Using Ansell and Bartenberger's (2016) terminology for experimentation, I suggest that understanding living labs as infrastructures or constellations is a *Darwinian approach*, as several experiments are used to jointly make progress toward desired developments in a selected field, in line with Niche Management. The understanding of living labs as an approach or methodology on an operational project level is, instead, a *generative approach* that aligns with the notion of bounded socio-technical experiments in the transitions literature (ibid.). If we consider living labs as a Darwinian approach, experimentation consists of the instigation of a portfolio of experiments. Instigation takes place at higher coordinating levels where results are aggregated and evaluated. If we consider living labs as a generative approach, they consist of a set of methods, and experimentation is understood as the operative research activities entailed by the project at hand, to learn-by doing or to iteratively refine a concept towards a working solution.

I suggest that the broad lack of attention to methods for user involvement in the living labs literature (see Schuurman, 2015) implies that decisions regarding methodology are, in practice, generally *not regarded on the infrastructural level*. The actors that manage a living lab infrastructure may not concern themselves with controlling the methods used, when in practice they are not the ones carrying out the research activities. Almiral and Wareham (2011) describe six living labs cases that are all large network structures. These structures focus on enlisting and orchestrating several actors and their technologies, and mediating with user groups at an organizational level. On a societal level, such as national development programs, focus may lie on instigating, evaluating and upscaling experiments, as well as on knowledge transfer to higher levels (Sabel & Zeitlin, 2011). In experimental governance, learning on the infrastructural level may thus concern monitoring and identifying successful strategies for implementation with the goal of developing effective policies. I argue that this distancing between an administrative infrastructure and the projects and experiments that infrastructure hosts or supports is one main cause for the confusion concerning the concept of living labs.

The differing objectives and research aims at these two levels can also lead to conflict, as we see in the many accounts that imply that more learning-oriented and open-ended approaches have been limited by objectives set at the infrastructural level (see, e.g., Botero & Hyysalo, 2013; Hakkarainen, 2017; Tironi, 2020). The actors on the constellation or infrastructural level often act as inter-

mediaries who manage collaboration over networks and may control funding. They can influence what types of research may be conducted and emphasize validation, evaluation, or outcomes that match funders' agendas (Mulder & Stappers, 2009; Steen & van Beuren, 2017; Tironi, 2020). Kommonen and Botero (2013) have described how the differing aims and views of stakeholders have challenged and limited user participation and open-endedness in living labs. Similarly, Hakkarainen and Hyysalo (2016) provide a detailed case study of the mediation efforts needed to bridge these differences.

In contrast, the DLLs have been conceptualized as single experiments, which means that decisions have mostly not been taken at one level and implemented on another. In most cases, the same design researchers who set up the research objectives also designed the future prototypes, made the broader methodological choices, and decided on day-to-day research methods. In Car-Free, LEV-Pool, and CoKitchen, design researchers set up the interventions and had the closest relationships with the participants, which in practice allowed them to control the associated research methodology. Interestingly, CoKitchen was *hosted* by the KTH Live-In Lab, which was a more traditional living lab structure. However, the administrators of this infrastructure did not control research funding and made no attempts to influence the research. Similarly, the large Mistra SAMS research program has also partly functioned as a research infrastructure. Especially in the first phase, the program distributed funding to several different types of studies carried out by different groups of program researchers. At the program or infrastructure level, there was an ambition of jointly deciding which issues should be researched, which made living lab planning very time consuming. However, after decisions were made and budgets had been negotiated, individuals or groups carried out the work of implementing a lab on the ground effectively also controlled the projects' methodological choices. In the second phase of Mistra SAMS, decisions made at the infrastructure level led to confusion, and the same living lab now incorporates three quite different types of research carried out in the same geographical area.

Schuurman (2015) has suggested that living labs that take the form of single experiments are problematic due to the high costs of establishing a research structure, and since they lack a sustained infrastructure or constellation to support knowledge retention and accumulation. Building on experiences with the DLLs, I suggest that these arguments are only partly valid for this particular type of living labs, since these early-stage efforts have been research- or

design-driven and have not been instigated by a stakeholder aiming to sustain the living lab operations or to directly utilize the research findings. The Mistra SAMS research constellation provided funding and an organizational structure for several experiments, but this came at the cost of generating friction due to the conflicting research aims. However, when an experiment requires a physical infrastructure, conducting it under the auspices of a shared research constellation can be a cost-effective solution. This was the case with the KTH Live-In Lab testbed building that hosted CoKitchen. The Restaurant Lab, finally, was *both* an experimental future prototype in line with the definition of DLLs in this dissertation, and also a physical infrastructure that hosted a number of experiments.

Interventions aiming to enable or provoke experimentation

All the DLLs introduced some kind of *intervention* into people's real lives, by prototyping or staging part of a more-sustainable potential future. These interventions sought to disrupt people's routines in order to facilitate experimentation and reflection—a recurring theme in many fields of research, from psychology to the social sciences (Kurz et al., 2015). Deliberate "de-routinization" (Spaargaren, 1997) has been suggested as one way to make people consciously reflect on their practices, which in turn enables change. The design field has brought forth many interventionist approaches. Junginger and Sangiorgis (2009) describe service designers as "enablers" or "facilitators" for new behaviors to emerge. Reflective design (Sengers et al., 2005) has been described as intervening to create or stimulate reflection. Inspired by practice-oriented design (Shove & Watson, 2006; Kuijer, 2014), the DLL interventions have been intentionally designed to enable or provoke participants to reflect and experiment with changing their practices.

Longitudinal real-life experimentation

The DLL experiments were longitudinal in design. Instead of just projecting or imagining designs in use, the DLL concepts have mostly been staged for a full year or more to allow the user-participants to explore them over time. With the exception of a few approaches, such as extended co-design and infrastructuring (Botero & Hyysalo, 2013; Karasti, 2014), real-life experimentation is rarely longitudinal. Even though living lab structures are often in place over a long period of time, the single experiments or research activities that they host are

generally short.

I argue that the DLLs' longitudinal design was of great importance for three reasons: First, many of the practices we explored were highly seasonal, such as the lengthy summer vacation trips that confounded the analysis in Playing-Rules, or the challenge of winter cycling in Car-Free. Second, many findings took time to unfold, for routines to settle or until patterns began to emerge. One example was how some Work Closer participants initially offered many reasons why they weren't using the NTC currently, even though they planned to use it more in the future. However, as the months passed the participants realized that there would always be new, temporary hindrances. Another example is the several months it took for the program partner companies to allow their employees to use the NTC facility, since this future service concept did not align with many of these large organization's structures. These types of findings support Schulman's (2010, in Hillgren et al., 2011) observation that with social innovations, rapid prototyping is insufficient, and long-term exploration is needed to understand systemic aspects. Third, the longitudinal setup also allowed trust to build between researchers and participants. This has been found to be necessary in order for the participants to comfortably discuss deeper meanings or personal issues (Hesselgren, 2019, p. 226). Furthermore, after interviewing the same participants on multiple occasions, I find that their reflexivity and engagement has in many cases developed and deepened over time, as evidenced in their broader and more critical views on their own practices and in their taking the initiative to experiment to a greater extent—developing what I suggest calling a "co-researcher mentality."

Balancing the strictness of the interventions through recruitment strategies and Challenges

Staging interventions in everyday life contexts over long periods of time can be challenging for the participants, and in the analysis of the two first DLLs—Car-Free and LEV-Pool—Hesselgren et al. (2017) have suggested balancing "strict" and "soft" interventions when exploring alternative ways of living. Strict strategies were described as removals (such as removing the families' cars in Car-Free), and soft strategies as add-ons (such as adding shared LEVs to the vehicles that the participants already owned, in LEV-Pool). In her dissertation, Hesselgren (2019) adds that softer interventions may need to be balanced with stricter rules to force experimentation.

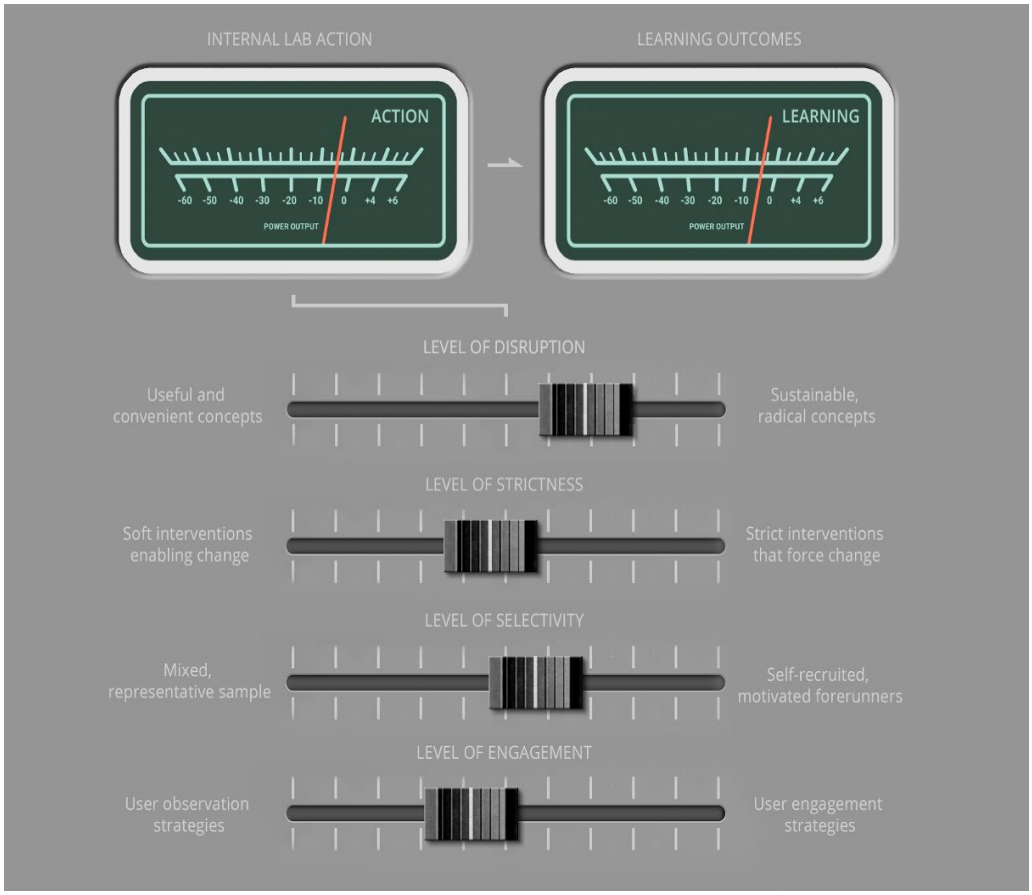


Figure 42. An image of an old-fashioned mixerboard to illustrate the concept of 'balancing' living lab levels of disruptiveness and strictness with strategies for selective recruitment and active user engagement, in order to elicit more learning.

I argue that recruitment strategies also play an important part in this balancing, as stricter interventions require the recruitment of more-motivated user-participants, while softer interventions allow for a broader or more randomly recruited mix of users. I suggest that the more-motivated participants in the DLLs already held understandings and values that could become more widespread in a more sustainable future, and can be viewed as "forerunners." In contrast, the more random mix of participants in Playing-Rules helped identify practices and understandings that serve as barriers to sustainable change. Also, as became clear in Work Closer, even highly motivated participants may find that they lack

the conditions needed for change. I therefore propose the promising concept of "challenges" in the sense of asking a subset of participants to engage in more intensive experimentation for shorter periods within the frame of a longitudinal study. In Work Closer, some of the challenge-takers reported that they more intentionally experimented with different solutions to allow remote work, which also suggests that they had become more engaged in the role of co-researchers. Using challenges for a smaller subset of the DLL participants can thus function as a user engagement strategy and encourage a co-researcher mentality. On the other hand, active user engagement and support must be balanced against more neutrally observing participant reactions and changes that may or may not unfold, and may in some cases require balancing when the researchers and the participants get very familiar over time (Hesselgren, 2019).

Overall, the concept of *balancing* is very central to the DLLs but has been difficult to communicate to non-designer researchers. In an internal presentation to discuss planning for the Riksten lab, I presented an image of a mixer board to illustrate the idea that sensitively and flexibly balancing these factors can strongly influence learning outcomes.

> *To conclude this section, the DLLs build on designed interventions, aiming to enable or provoke user experimentation. Living labs are commonly viewed either as a physical place for co-creation activities, or as a real-life testbed where user data is sourced. In this light, I propose that the longitudinal and very active engagement of user-participants in exploration within the context of their everyday lives are the most distinguishing features of the DLL approach. This researcher-led engagement requires careful balancing of intervention strictness and recruitment strategies. Furthermore, I propose actively engaging user-participants by use of 'Challenges'.*

6.4. Positioning the DLLs with respect to modes of user participation

The DLL-participants' role as co-researchers

The design researchers who have been involved in the DLLs all state that the labs were participatory. Often drawing on Brown, Reeves, and Sherwood's (2011) conclusions on field trial methods, we have argued that the DLL participants were not passive subjects to be studied but active investigators of their own practices, or openly invited as "collaborators in discovering new meanings and values" (Pierce et al., 2015, p. 2088). With the exception of specific co-creation workshops, however, I have chosen not to refer to the participants as co-creators but as co-researchers. As will be further discussed below, I find that they have been engaged in exploration rather than in designing.

The term co-researcher has been used to mean that a user is engaged in the capacity of "experts of their lives and experiences" (Sleeswijk Visser et al., 2005; Sanders & Stappers, 2012), which has clearly been the case in the DLLs. Finlay (2002) suggests that when research participants are viewed as having the *capacity to be reflexive*, they can be co-opted into the research as co-researchers, and that this may also include involving the participants "in a reflexive dialogue during data analysis" (p. 218). This has been the case in most of the DLLs, as the researchers have taken care to inform the co-researchers about the purpose of the studies and encouraged them to reflect on their own actions, lifestyles, and understandings. During interviews and in workshops, participants often indirectly took part in interpretation and analysis by reflecting on their experiences. The term co-researcher is, of course, used by many authors with varying meanings. Latour (2004) suggests that with the increasing rate of change, our entire society is an experiment and we are all co-researchers. In Publication D, I suggested distinguishing between six different roles for user participants in real-life experimentation research, based upon their level of involvement and whether the study aims at learning or solving (Sjöman & Hesselgren, 2022). Here, I add Finlay's (2002) notion of reflexivity to the model, and thus suggest using the term co-researcher with respect to participatory research approaches whose purpose is to explore and learn, using methods that actively involve users or citizens in a reflexive capacity.

Ethically, I argue that we may ask people to participate in experiments, given

that we inform them about what they will experience and why the research is being carried out, to the best of our ability. In our experience, most people find the experience of reflection, experimentation, and learning in a DLL rewarding, which is in line with the findings of Scott et al. (2011) and Brown, Reeves, and Sherwood (2011). We have found that participants generally enjoyed sharing their experiences with researchers. Some participants took initiatives all on their own, and many showed great perseverance when taking part in the longitudinal DLLs, as well as diligence in reporting (Sjöman and Hesselgren, 2020). In the DLLs with fewer participants (Car-Free, Playing-Rules and CoKitchen), researcher-participant relations were very close (see Publication B, also Hesselgren et al., 2017).

High user involvement as power or as level of activity?

The modes of participation to use can be a difficult issue to handle in collaborative research efforts. These methodological choices are related not only to the type of research and its purpose, but also to ideological issues, which may be viewed differently in different research cultures. Distinctions and terminology are not clear, and even when users are referred to using the same term, this may carry very different meanings. The terms participatory design and co-design are sometimes used interchangeably, but while the former carries a strong emphasis on power relations and bottom-up perspectives, the latter generally refers to designer-led co-creation activities.

I find that user participation in real-life experimentation entails important questions of *when* and *how* users are involved. To distinguish between different modes or understandings of participation, it is important to consider what stages of the research process users are involved in. Are they involved in choosing research issues and objectives? Do they take an active part in selecting or co-creating concepts to be explored or tested? Do they co-design and develop these concepts? Or do they reflexively test or explore them? *How* users are involved relates to ethical issues of democracy and power distribution, as well as the level of participant activity and engagement. I find that these two aspects are easily confused and they will thus be discussed in greater detail in the following subsection.

Research ownership and participation

Participatory design (PD) has a long history of longer-term prototyping work in actual, real-world use contexts (Hartswood et al., 2002, Björgvinsson & Hillgren, 2004). To address more complex societal issues, participatory design has been described as creating prototypical "representations" of socio-material assemblies in order to reveal matters of concern (Björgvinsson et al., 2010). The DLL interventions can be seen as prototyping such socio-material assemblies, but there are some important differences. In approaches such as participatory design and design for social innovation (DSI), design is viewed as a means to facilitate grassroots initiatives from citizens or user communities to "solve" issues that affect them (see e.g., Björgvinsson et al., 2010; Hillgren et al., 2010). Here, I suggest that in aiming to make real change in the here-and-now, PD and DSI approaches cannot be understood as *bounded* experiments in the sense suggested by Brown et al. (2008). In contrast, the DLLs have been clearly understood as bounded experiments intended to enable learning and having their main impact later and elsewhere. This leads to some important differences. First, the bounded and provisional nature of the DLLs has allowed exploring and demonstrating future-oriented sustainable concepts that were at least initially, not requested by the user participants. Second, the design initiatives in PD and DSI have aimed to "activate" citizens or user communities to take ownership and to shape, build, and sustain the envisioned change (Manzini, 2014). In the DLLs, in contrast, the design researchers have designed the interventions and then invited citizens to partake temporarily as co-researchers.

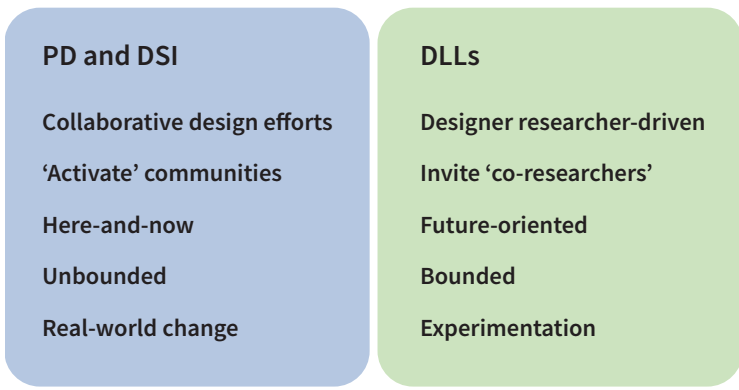


Figure 43. PD and DSI vs. DLLs: Five core differences.

PD and DSI have aimed to foster the active engagement of user communities

and to transfer project ownership to them. In the DLLs, research participants have been invited to partake in the design researchers' projects. Building on the PERL study (European Lifelong Learning Programme) of projects using PD and DSI, Manzini (2014) argues that both approaches can be broadly defined as design initiatives for making social innovation happen, by *activating* citizen participation. As these infrastructures are mainly initiatives taken by designers and design researchers but also open to supporting possibilities that emerge without predefined objectives, Manzini notes that PD and DSI can be considered both bottom-up and top-down approaches. Since all three approaches, including the DLLs, ultimately aim for social or environmental sustainability, they can also be viewed as different modes of design activism.

Methodological choices regarding user involvement may reflect strong opinions regarding what constitutes participation and agency. Equal power distribution or insider-led approaches are often considered an ideal, but in practice, full citizen control is generally not appropriate or practical (White & Langenheim, 2021). In line with Brown et al. (2008), users recognize when they are taking part in a bounded experiment, and I suggest that DLL users may also be empowered by reflection and learning (for a related discussion, see Scott et al., 2011). Conversely, direct attempts at empowering user communities or fostering active citizenship can be viewed as patronizing or manipulative. Also, as Manzini (2014) suggests, since PD and DSI mainly build on a design initiative to activate user communities, these approaches may, in practice, be considered top-down. Furthermore, I argue that sustainable future concepts are not easily co-created by the user communities who are required to change their lifestyles, which I will elaborate below.

Causes for confusion regarding modes of user participation

My review of the literature leads me to suggest that most typologies for user involvement in co-design or user innovation focus on *designing something* or facilitating collective design processes towards this end (see, e.g., Sanders, 2006, 2008; Hermans, 2015; Kohtala et al., 2020). User involvement is generally regarded as partaking in co-designing or co-creating and may therefore be ranked on a scale that runs from passive consumer to professional designer or from passive consumption to active production, or according to the users' influence on decision-making in the design process. Such typologies fail to capture the active exploration and experimentation of co-researchers in the DLLs. On

the other hand, research fields and approaches such as sociology, participatory action research, PD, and DSI all emphasize an ideal of insider-led or community-based research to embed and implement real-world change. Typologies of involvement from these fields focus on power-relations (see, e.g., Arnstein, 1969; Hacker, 2013; Sagaris, 2020) and do not capture the high level of co-researcher activity and engagement in the DLLs. I argue that scales and continuums of user involvement are often misleading when attempting a broader comparison of different approaches to user involvement, because the related research fields and applications have different aims and outcomes, as well as qualitatively different methods. These differences are outlined in figure 44.

Different research traditions and cultures can lead to misunderstandings and tensions regarding modes of user participation. This has been very clear in some of the DLLs, as researchers from fields such as sociology and urban planning have in some cases emphasized power-relations and shared a preference for grassroot approaches, while researchers with backgrounds in engineering expected a structured development process or a pilot test. The concept of co-creation is also used differently in different research domains, and it can be argued that the DLL co-researchers were co-creators of knowledge. In LEV-Pool, Sopjani et al. (2017) suggest that users co-created the system through their usage, and it can be argued that a service only exists in the interaction. In the meaning of co-designing the service concept, however, setting aside efforts at extended communication, user input was not used to improve the concept until after the project had ended, in Hertz Sunfleet's subsequent service launch. Still, the consumption and adoption of new artifacts can be described as creative and innovative processes, and the DLL participants can be viewed as innovators of new everyday practices (Shove and Walker, 2010). Although they have had little influence on the design of the intervention concepts, their exploration and experimentation may indirectly influence larger and long-term developments. Overall, in the DLLs, I argue that we have not invited participants to act as co-creators but as co-researchers.

Why DLL participants cannot lead or co-own the lab, or design the interventions

Innovation and development to progress towards sustainability transitions has been found to differ from other types of societal shifts because sustainability innovations are often perceived as providing limited user benefits compared

to existing solutions, and because there are few incentives for private actors (Geels, 2011). Since sustainable concepts inherently challenge existing structures and lifestyles (Ceschin & Gaziulusoy, 2016), users and other stakeholders may resist trying them out in their daily lives or current operations. Sustainability is also commonly associated with a loss of convenience or with reducing or refraining from common behaviors (see, e.g., Spaargaren, 2011), and requires relearning, and thus people are unlikely to request such changes. This situation is now slowly improving as some consumers are beginning to request more sustainable products and services, but this mostly concerns solutions that reduce the negative impacts of users' consumption. Affluent middle-class consumers may request electric or biofuel cars, but are generally unwilling to make more substantial lifestyle changes. It is difficult for people to envisage alternatives to their own routines or lifestyles, and in the DLLs it has been clear that most participants were not ready to accept the re-learning that was required. Consequently, as Spaargaren (2011) has noted, user-centric or user-driven design approaches such as HCD or co-design are likely inadequate. Instead, the DLLs design researchers have designed and staged sustainable future concepts without user input, aiming to identify pathways towards these futures. Using this approach, the labs were able to identify barriers in the shape of conflicting norms and user perceptions. Building on Sanders and Stappers' framework (2008), the DLLs can be viewed as a practice of designing *for* and

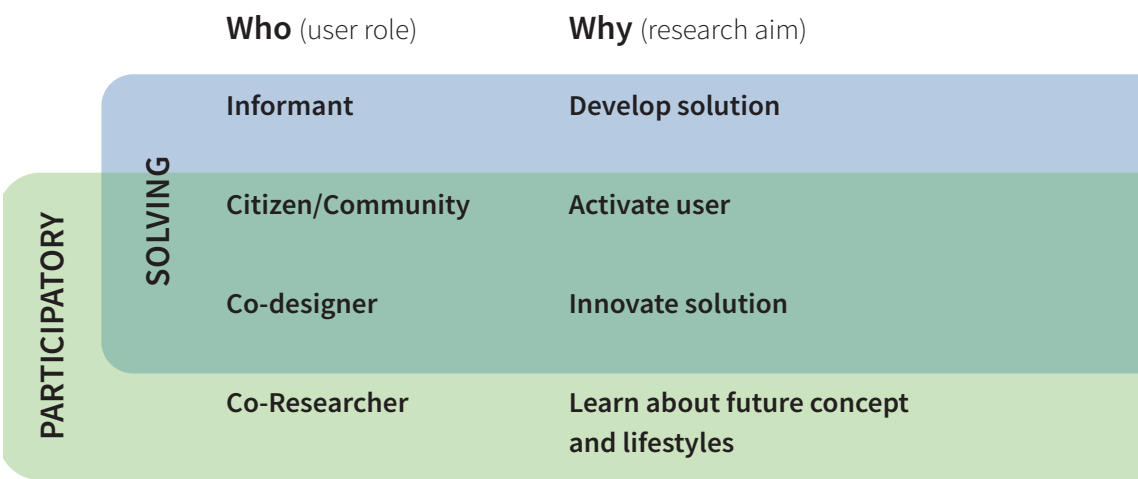


Figure 44. General descriptions of four archetypical modes of user involvement in sustainability research that can be characterized as real-life experimentation. In participatory research modes, users are actively involved. Solving modes of research aim to create solutions and to make change in the here-and-now (see 6.2). User participants in the mainly explorative DLLs are considered co-researchers.

exploring *with*.

The DLLs have been clearly learning-oriented and bounded experiments and have not been intended to resolve real-world problems in the here-and-now. Therefore, there was no ethical need to engage participants as empowered insiders to co-design the interventions. To the contrary, the sustainable futures we aimed to identify pathways towards have often been in conflict with current norms and structures. The interventions have aimed to provoke or enable (Sanders & Stappers, 2014) user participants to experiment, and it is unlikely that users would have called for these future concepts themselves. For this reason, we have in many cases recruited participants that could be called 'forerunners', who were strongly motivated to live more sustainably. Social practice theory posits that user needs are malleable and emerge out of current practices (Shove & Watson, 2006), meaning that future concepts cannot be designed based on the needs that people experience today. Instead, faced with the climate crisis, the DLLs started from the normative view that many of today's practices and lifestyles are unsustainable and must change. In the DLLs we have thus sought to identify ways to live differently, rather than to meet current needs and expectations. This means that HCD approaches would not work for the sustainability-oriented DLLs, since we cannot ask the participants what they want. We also cannot observe their current practices to find out what

How (researcher activities)	**What** (outcomes)
Evaluative (testing, asking, sourcing data)	**Product or service usability** (consumer adoption)
Facilitating (supporting, providing resources)	**Social Innovation** (local or regional structure)
Generative (creative workshops, toolkits)	**Product or service innovation** (adoption and behavior change)
Explorative (real-life experimentation)	**Actionable knowledge** (pathways for transformation)

they may need in a sustainable future. Moreover, in the innovation literature, user participation in concept development has been found to mostly lead to incremental innovation (see, e.g., von Hippel et al., 1999; Heinbokel et al., 1996), and Christopher Lettl (2007) has provided an overview of why users may not be willing or able to contribute to radical innovation.

Trying out future ways of living in the context of today's society may mean facing hurdles or discomfort. Living without a car in Car-Free forced users to experiment with creating new practices, which was initially very challenging. The three families identified several conflicting norms and examples of how society must change to better support car-free living. Taking their cars away was clearly not a "solution" that car owners would have suggested. Instead, the DLL findings identified pathways and actions for society to support a move towards this more-sustainable future. Over time, the families found new ways of living and identified positive gains, but two years after the DLL ended, they had all returned to the comfort and convenience of using their cars (Hesselgren, 2019). I suggest that the families did not choose to remain car-free because the conflicting structures remained in place. Similarly, even though most of the participants wanted to use the Work Closer NTC, current structures made this difficult.

Finally, since it is difficult to engage user communities to explore more radical sustainable futures, this kind of research mostly cannot be insider-led. As an example, at a Mistra SAMS meeting with three partner researchers with experience from several Swedish smart mobility pilots, these researchers described serious difficulties engaging citizens in trying out new ridesharing services and shared vehicles, even in small and closely knit rural areas where trust is supposedly high.

Activating, empowering, enabling, serving or provoking?

Similar to action research in general, many longitudinal real-life experimentation approaches in the field of design work with marginalized groups and support user communities. Initially envisaged as "collaborative design" with worker's unions in the 1970s (Bødker et al., 2000), participatory design often engages with power relations, marginalized voices, and social exclusion (Hillgren et al., 2011). The DLLs instead mostly engaged participants from the middle class, that generally did not provide input to the design of the lab

interventions. Instead, the design researchers have created more or less "strict" interventions to provoke or enable these middle-class participants to question or change their unsustainable practices.

In the last subsection I have presented many reasons why the DLLs were researcher and designer-led, but these research principles may still be perceived as controversial, given the common ideals of equal or insider-led research partnerships. In Mistra SAMS, inspired by Sanders and Stappers' (2014) notion of "serving" and "provoking" modes of design, the research group discussed using different research strategies for different groups of participants. It was thought that more provoking approaches could be used for engaging middle-class participants in experimenting with reducing their car use, while using a "serving" approach would better enable "transportation justice" for lower-income groups such as cleaners and shift workers who often do not own cars. As noted in the epilogue on p. 118, the Riksten living lab later included a concept of this type.

At first glance, there may seem to be little difference between Car-Free and the Citizen research group's experimentation in Riksten. In Car-Free, researchers asked participants to try out a future car-free lifestyle, and in Riksten they asked participants to set up personal goals to reduce their own car use. The core difference is that in Riksten, the researchers planned to engage participants in developing a more-active form of citizenship and in taking personal responsibility for the planet, while in Car-Free the researchers asked participants to explore a design intervention in the form of a future scenario where they could no longer have a private car. Sociology has seen a long debate as to whether behaviors arise from agency or structures. I suggest that the aim of the Citizen researchers approach in Riksten—to *activate* the citizens in taking responsibility and making change—emphasizes individual agency and conscious decision making. The DLL-approach has instead employed a practice-oriented view aiming to change societal structures to *enable* more sustainable lifestyles, which will be discussed in the next section.

Using Practice-Oriented Design, DLLs enable new practices but do not design them

Several design researchers working with practice-oriented design describe methods that include *designing* proto-practices (Scott et al., 2012; Julier, 2007; Kuijer, 2014). Scott et al.'s practice-oriented study included guiding research

participants to critically view and "deconstruct" their current bathing practices. The participants then each designed and implemented new and more-sustainable bathing practices that deviated from the norm, and finally integrated these proto-practices into their everyday lives. In her dissertation on practice-oriented design, Lenneke Kuijer (2014) builds on principles from Shove's (2006) concept of practice-oriented design to further distinguish between using practices as *units of analysis* and *as units of design*. Analyzing five studies that created and used practice-oriented design methods, she describes practice-oriented design as a deliberate process of designing practices. Kuijer (2014) also describes a more integrated design process of suggesting or triggering changes that can then be performed, varied, and further refined. In this view, the designer may suggest that a participant redesign a practice or perform a practice in a new way, according to a script with varying levels of detail. In the latter case, the participant may then improvise and experiment when performing a new practice. Participants thus create new seeds for proto-practices that can then be evaluated and refined in an iterative process.

The DLLs instead build on the view that practices cannot be designed per se (Julier, 2007; Shove, 2012; Mallers & Strengers, 2015). Instead, design may *enable* new practices to emerge through interventions that foster practice innovations. I find that this difference is an important distinction. In designing and exploring sustainability concepts, the DLLs have used a practice-oriented design approach built on designing interventions to make participants experiment with new practices and to reflect on their current ones. By introducing new elements, we have not only offered new opportunities that enabled new practices but also sought to cause a "crisis of routines" (Reckwitz, 2002) by disrupting current practices. Thus, the interventions have broken or replaced links, causing participants to reconfigure existing practices and to create new proto-practices (Shove & Pantzar, 2005; Kuijer, 2014; Hesselgren, 2019). The DLL interventions thus enabled, triggered, or sometimes forced participants to reflect on and experiment with changing their practices. I argue that the DLL designers have not tried to deconstruct current practices before the experiments and made no attempts to *design* proto-practices per se. Instead, practice theory has often been used after the fact to deconstruct and to better understand the proto-practices that emerged during the experiments.

However, Kuijer (2014) also describes examples in which a specific situation has triggered experimentation or improvisation, such as when one participant's

regular showerhead was removed during a study concerning bathing practices. This example is very similar to the DLL methodology, but in Kuijer's description, these instances were still within the framework of an overall process of guiding the participants to *consciously design* new practices. Building on the work of Kuijer, Eijk, and DeJong, Mia Hesselgren (2019) suggests the useful distinction that taking practice as unit of design means taking a *proactive stance* towards generating new practices, meaning that the research is not merely depictive or using practice as unit of analysis. Finally, based on the experiences from Work Closer and Playing-Rules, I argue that the reflection triggered by the realistic and real-life DLL interventions can allow research participants to identify potential alternative practices, even when the interventions did not trigger actual improvisation or experimentation.

> *To conclude this section, user participation is a thorny issue. The type and level of user participation is often assessed based either on power relations or on users' roles in the process of 'designing something'. These typologies neglect user participation in exploratory activities with a learning aim. Building on Hesselgren (2019) and Finlay (2002), I argue that the DLLs are participatory, as the user participants were actively engaged in exploration as reflexive co-researchers. Still, the DLLs have been clearly researcher-led and "owned" by the design researchers, and the participants have not been engaged in selecting the research issues or co-designing the interventions. Building on Sanders and Stappers (2008), the DLLs can thus be viewed as a practice of designing "for" and exploring "with." In the words of Van der Bijl-Brouwer and Dorst (2017, p 6), the users were "invited into the world of the designer" but at the same time–that world was staged inside of theirs.*
>
> *The DLLs are an example of practice-based design, but one key difference in comparison to other approaches under that umbrella is that the DLLs have not intended to deconstruct or design practices as an intellectual exercise. Instead, new practices and understandings have emerged from longitudinal real-life experimentation, prompted by the interventions.*

6.5. Types of research outputs and outcomes from DLLs

Open inquiry to frame or reframe an issue

Like inductive and exploratory research in general, DLLs are concerned with finding emerging patterns from observations and understanding dynamics and behavior, which means that researchers may need to alter the direction of a study (Bernard, 2006). Specifically when tackling wicked problems, Russel (2010) has described the need for flexibility in open and critical inquiry, and for letting the research design remain open to revisions in response to the dynamic complexity of research into human-environmental problems. Mia Hesselgren (2019) notes that when unexpected problems were encountered in LEV-Pool, the team needed to embrace necessary adjustments, further observing that "breakdowns" or misuse often provided important learnings. I find that the DLLs have all been designed as mainly exploratory studies and helped frame issues in early research stages. They did not aim to answer specific research questions but instead stayed intentionally open to whatever was found. As a result, the DLL results have been rich and broad. On the other hand, the rich and often situated knowledge produced may defy generalization or transfer to other contexts.

The DLLs have tended to uncover many unanticipated factors and matters of concern, such as social, technological, or regulatory issues. This differs from more common modes of real-life experimentation that typically have a narrower market- or policy-driven agenda to validate or improve a specific solution (see, e.g., Kommonen & Botero, 2013). I suggest that the more open-ended and vision-driven type of research in the DLLs is important in early research stages, since the initial design will set limits for the design in later developmental phases (Bratteteig & Stolterman, 1997). As the case studies show, unexpected findings or practical problems have in several cases led to the inclusion of new aspects, the reframing of issues, or a change in the research focus. I suggest that this is an example of design's well-known capacity for "problem-making," opening up, or "exploding" known parameters or premises (Vaughan, 2017). As one example of reframing, Work Closer was intended to explore travel-related aspects of working closer to home but ended up focusing on work-related barriers to remote work. Building on Buchanan (1992), I suggest that this kind

of reframing was possible because designers are not specialized in a narrow subject field, and "the problems addressed by designers seldom fall solely within the boundaries of any one of these subject matters" (p. 14). A transportation researcher would likely not have accepted this change in project direction away from a focus on reducing travel or shifting modes of commuting.

I argue that the unexpected findings that led us to reframe research issues were often in the shape of *elicit* barriers or preferences, in the sense that these factors were *uncovered* by the interventions. The many conflicting structures that made it hard for Work Closer participants to use the NTC would likely not have been uncovered through interviews or surveys. In fact, in the first round of interviews, participants expected to use the service much more than they actually did, and it took them several months to realize that there would always be some factor or other stopping them. Similarly, those who tried out co-living in CoKitchen found they loved sharing the kitchen and that sharing the bathroom was no problem at all, even though in national surveys a majority of students state that they do not want to share bathrooms, and kitchens only to a limited extent.

How holistic design research may both connect and challenge other research fields

Scholars have argued that design is inherently holistic (see, e.g., Johansson et al., 2003; Owen, 2007; Stoltenberg, 2008), a trait clearly displayed in the DLLs. As I describe above, staging the future concepts and making them work in today's society has led to the identification of many types of linked factors and issues, both inside and outside of the projects' original scope. I suggest that holistic and open-ended exploration has been an important function of DLLs. Design problems seldom fall within only one specialized subject matter (Buchanan, 1992) and design has been found to be "interdisciplinary per se" (Meier, 2003, as cited in Caplan 2018) and described as a "meta-discipline" that connects others (Caplan, 2018). As exemplified by the barriers for work at the NTC, research that aims to further sustainability may "require crossovers between many research fields" (Geels, 2010). In this vein, I find that interdisciplinary collaboration among different public and private actors and various research disciplines has been both enabled by, and required for, the practical staging of the DLL interventions. Furthermore, as Coyne (2005) notes, making experimental and conceptual artifacts may serve as a critique of knowledge

paradigms. The DLLs' findings in several cases has not only contributed to but also challenged established knowledge in other fields of research. The possible limitations to Swedish national travel surveys uncovered in Playing-Rules is one example of this.

As Ehn and Ullmark (2017) have pointed out, Practice-Based Design Research requires careful balance not to move into other disciplines without having the qualifications. Contributing to a specialist field requires time-consuming reading or interdisciplinary research collaboration, and in the DLLs, this entailed its own set of challenges. The problems with article writeup in Playing-Rules were a clear example of this. The DLL's findings were broad and partly unexpected, and the program could not set aside time for researchers in the appropriate fields to analyze the rich dataset from different perspectives, and to effectively reach relevant research audiences. I have provisionally termed this problem the "design holism-issue." There are indications that this issue was one of the reasons for the more integrated organization used in the second phase of the Mistra SAMS research program. The first outlines of the new program organization were centered around a shared living lab, and the research group was later organized into the three research perspectives that were intended to jointly analyze the lab's results. As described in Section 4.6, this close collaboration proved to be difficult in practice, as will be further discussed in Chapter 7. In contrast, CoKitchen also produced knowledge that contributed to several research fields, including architecture, energy and sustainability, loneliness, building and tenure regulations, as well as business models. However, in this case the interdisciplinary project team was able to assess much of this knowledge and disseminate it to researchers and practitioners in their respective fields.

Actionable knowledge of pathways and barriers

The adoption and consequences of future concepts are hard to foresee. As Hartswood et al. (2002) comment, when new mobility practices emerge as a result of new technologies or solutions, they in turn open new possibilities for further technological development. As described in Section 6.2, staging future concepts in today's society can be likened to backcasting in real-life. In line with Caniglia et al. (2020), I find that the exploratory and open-ended real-life interventions uncovered concrete barriers to and pathways for sustainability transformations. All the DLLs except Playing-Rules have aimed to provide

practical guidance and concrete policy recommendations to relevant actors, in line with the concept of actionable knowledge (Argyris, 1996; Mach et al., 2020). Building on experiences from Car-Free, Mia Hesselgren (2017) uses the term bridging for activities to this end, and describes a strategy of prototyping change at an individual level, to then communicate the experiences of such change to people with power to trigger and direct change (Hesselgren, 2019). Car-Free likely influenced policymakers, as the researchers were invited on several occasions to present specific policy recommendations at several different levels of government (see Hesselgren, 2017). Similarly, a seminar presenting the results from CoKitchen attracted both policymakers, architects, and property developers, and the main partner AH has already implemented changes in new development projects, and a follow-up research project aims to drive regulatory change. Work Closer inspired change at a personal level for many participants, the NTC was visited by policymakers, and also provided employer partners with learnings drawn from interacting with the prototype. For these employers, this led to stronger engagement over time—in one case inspiring an organizational change and in another case leading to the funding of a commercial pilot. In LEV-Pool, the car-sharing partner went on to commercially launch a similar service concept. All the DLLs, except the small and poorly funded Playing-Rules, were also planned as both demonstrations as well as labs, meaning that they were planned for and also received a lot of media coverage. The DLLs in general likely led to societal impacts, but a quick overview of the associated academic papers produced does not indicate a significant academic impact. This is in line with Jahn et al. (2022), who found that there seems to be a trade-off between academic and societal impacts and that the societal impacts of applied, practitioner-oriented, or transdisciplinary research are often difficult to assess.

DLLs are suited for modelling, not evaluation

Living labs have been frequently described as including one or more stages of a development process that ends with the commercialization of a technology, product, or service (Hossain et al., 2020). Real-life experimentation is often used to refine or evaluate a new product or service in the late stages of implementation and commercialization (Steen & van Beuren, 2017). Transition experiments more broadly have been conceptualized as aiming towards upscaling a solution in society (Sengers et al., 2019; Bulkeley & Catan Broto, 2013).

These approaches thus refer to what was described as a "solving" mindset in Publication D. The distinction between aiming for learning or solving is important. When a research project aims to develop and further refine a concept, the process requires evaluation. I argue that this means crossing over a fine line between an open-ended "what if" mindset to a "does it work" evaluative mindset. In some of the DLLs, mixing these two aims or mindsets has led to confusion and conflict.

Why evaluating a 'learning device' does not make sense

As described in Section 4.6, the NTC-prototype in Work Closer was quantitatively evaluated, which turned out to be problematic in many ways. Most users were simply not in a position to work remotely, and this barrier would need to be solved if NTCs were to become more widespread. Also, the evaluation did not take into account the fact that the prototype was provisional and had already yielded a lot of information about how to improve the concept in future implementations. In the end, these aspects were jointly discussed and resolved for the most part in a resulting paper (Publication C), but the process was far from straightforward, and the evaluation results were, in some cases, misinterpreted. This example provides support for the argument that it does not make sense to evaluate the often limited outcomes of a bounded and norm-breaking experiment. Evaluating the prototyped future concepts in a DLL is not logical, as they are not designed primarily to be functional but rather to enable experimentation and to elicit reflection. Furthermore, evaluation demands rigor and favors more passive observation of events in order to reduce bias, and also calls for larger-scale studies. Our small-scale and open-ended experimentation, in contrast, required actively engaging and supporting participants while dealing with unexpected events. All the DLLs logged quantitative data, but with some exceptions, this data has not been suitable for evaluation purposes.

Modelling potential future impacts

The knowledge produced from the DLLs has been conceptualized as input to support actions elsewhere or in subsequent research stages. Such uses could include scaling up, but I believe it is important to emphasize that this does not mean that experimental DLL interventions are made permanent. When Mia Hesselgren (2019) mentions upscaling with respect to the Car-Free DLL, she does not refer to commercialize the vehicle rental service but rather to imagin-

ing how participants' more sustainable "proto-practices" might become mainstream in the future. This imagining allowed the formulation of suggestions and refinements that could be presented to relevant stakeholders. To drive subsequent developments that take place after and outside the DLLs, DLLs have often included a modelling approach. By using the lessons learned on how sustainable concepts could be integrated into peoples' lifestyles, possible future scenarios can be modelled more accurately to investigate the potential large-scale impact of a concept. In Car-Free, this was planned as a specific second stage of the project. In contrast to evaluation, such modelling can be used while still retaining the "what if" mindset associated with learning aims. In Work Closer, other program researchers modelled the potential for carbon emissions reductions if similar services to reduce the need for commuting were scaled up in society. This model concerned a possible future when management, teamwork, and security issues were solved in a way that allowed a major part of the population to work remotely 2-3 days per week and if employers' main offices were downsized accordingly.

> *To conclude this section, I find that the DLLs have produced rich, broad knowledge that has in each case supported the framing of a research issue by identifying unexpected factors, relationships, and matters of concern. These findings have both contributed to and challenged research in other fields. The DLLs have provided demonstrations that have garnered attention both from media and policymakers. There are indications that the DLLs produced actionable knowledge, since the outputs have received interest from relevant stakeholders and in several cases have led to follow-up studies, development projects, or pilots.*
>
> *Regarding the assessment of DLL concepts, I argue that when research has a "solving" aim, this implies that concepts will need to be evaluated to see if they provide the desired effect. In contrast, evaluating the exploratory learning tools in the DLLs does not make sense, since they were not designed to align with current society or to fulfill current user needs. Also, evaluation demands rigor, while the DLLs' open-ended experimentation requires actively engaging and supporting participants. Instead, modelling may be used to assess potential future impacts and to drive subsequent developments.*

7. Implementing DLLs

In this chapter, I discuss learnings from planning, organizing, and implementing the DLLs, in response to the second main research theme. The first subsection has a specific focus on challenges regarding research collaboration, and also include recommendations for the planning of future DLLs or similar approaches. The second subsection presents the design capacities and designerly ways that became visible in the planning and the practical staging of the DLLs. This discussion connects closely to the overarching third research theme regarding relationships with design practice.

7.1. Challenges and strategies for research collaboration

Confusion and conflicts in the DLLs

When setting up a real-life experimentation project or study, researchers must consider a wide range of research designs that may be more- or less-well suited to the research aims and to the situation at hand. In the DLLs, it was clear that research involving several research disciplines can make decisions on research designs and aims difficult. Broad concepts such as *real life* and *experimentation* can be interpreted very differently, and ideals and meanings regarding concepts such as *user involvement*, *user-centricity* or *participation* vary greatly across different fields of research and practice. Another reason for confusion is the multiple meanings and definitions of named methodologies, such as pilots, demonstrators, and living labs, an issue further amplified by the gap between rhetoric and actual practice in research (see Section 2.3 of the literature review). These and other differences were discussed in Chapter 6, including a discussion on open-endedness and framing in Section 6.1, the distinction between learning and solving in Section 6.2, and different modes of and views

regarding user participation emphasizing agency or structure in Section 6.4.

In some cases, multi-actor collaboration with other project partners has also been challenging. Because LEV-Pool involved a variety of actors, such as industrial partners and municipalities, pooling resources and competencies was challenging from an administrative perspective (Sopjani et al., 2016a; Sopjani, 2021). Initially, aims and interests such as sustainability goals were only aligned at an abstract level, and the various actors held diverse views regarding the type and purpose of the exploration and the expected outcomes. Furthermore, roles and responsibilities were not clear (p. 52).

In Mistra SAMS, planning and decision-making became very difficult given that the aim was to plan the research objectives and methodology and to design and explore the lab interventions *jointly*, as a group. As discussed in Section 6.2, the understanding of the interventions mainly as 'research-devices' differs from other, more common research approaches, and these differences were not obvious. Although most of the Mistra SAMS researchers had been involved in the lengthy process of writing the funding-application, the four-field matrix workshop material (Figure 30, p. 98) made it clear that four very different research aims and views regarding methodology were represented among the program researchers and partners. I suggest that the decision-making difficulties the program faced were largely due to these underlying differences in values and research cultures, which were poorly articulated and for the most part unspoken. The tacit nature of these differences can make it difficult to resolve them, since this requires time and targeted efforts. As described in the epilogue to the discussion of Mistra SAMS on p. 104, subprojects in line with at least three of the four different types of research outlined in the four-field matrix were eventually implemented in parallel in Riksten. These different research types also required different sets of user participants, which correspond to the levels I suggested at that time (see Figure 34 on p. 103). This supports the argument presented in Section 6.3, that in Mistra SAMS, the program was more akin to a funding infrastructure, and that at this level, methodology is not a primary concern.

The need for teamwork and physical proximity

Research suggests that the physical proximity of team members is a key factor in transdisciplinary research (Pak & Choi, 2007). In a broad review of research

on teamwork, Kiesler and Cummings (2002) emphasize that there are decades of conclusive evidence for the importance of physical proximity, especially in order to "resolve differences of opinion, perspectives and expectations" (p 72) in teams that include a diversity of expertise. In Mistra SAMS' second phase, the executive committee had their offices in different departments and institutes and met only briefly, twice a month; furthermore, in the aftermath of the Covid pandemic, most of these meetings were remote. Teamwork among the four research groups was often closer, but only a small number of workshops and events were held each year that included all the groups, and these were also often held remotely over Zoom or Microsoft Teams. I argue that this physical division limited *informal dialog* and made it difficult to resolve differences of opinion regarding the living labs. Playing-Rules was largely implemented by only two researchers and thus there was little need to resolve conflicts. However, having access to other, more experienced researchers in the program could have provided valuable support. Lack of physical proximity seems to have been less of an obvious problem in the earlier DLLs, but based on experiences with LEV-Pool, Sopjani (2021) recommends in-person communication between collaborating actors, such as co-creative sessions.

Interdisciplinary and transdisciplinary research collaborations

It has been suggested that transdisciplinary research is needed in order to further sustainability transitions (Caniglia et al., 2021; Fazey et al., 2020). In Mistra SAMS, and particularly in its second phase, the original aim was a transdisciplinary collaboration, meaning that different disciplines would work in an integrated and holistic manner to move *beyond* their disciplinary boundaries (Choi & Pak, 2006). This requires a united intellectual framework (Jose and Ramakrishna, 2021). In Mistra SAMS, however, I argue that such a united framework for the lab methodology was never developed. In LEV-Pool, researchers' views were also in conflict at times, but the funders had knowledge of and experience with working with design, a designer was recruited as the project manager, and much of the groundwork was conducted by design researchers. The project thus became design-led in practice, which may have led to the apparent shift from the original intention to iteratively develop a service system towards a more-exploratory research design.

Car-Free and CoKitchen were clearly design-led, and differences in researchers'

cultures and methodological understandings do not seem to have hampered the planning and staging of the lab interventions. Only design researchers were involved in the day-to-day operations of Car-Free, and in CoKitchen the collaboration was multidisciplinary or interdisciplinary rather than transdisciplinary, in the sense that the different disciplines mostly stayed within their boundaries, even though their efforts were coordinated. The overall research design was in this case planned by the design researchers, and the other partners worked independently by adding their specific competences and methods. The clear roles and responsibilities in CoKitchen thereby allowed for efficient implementation. There was also little overlap between the research disciplines involved, which likely limited unproductive and tacit conflicts. To further clarify, in CoKitchen, interdisciplinary collaboration was fairly straightforward, since we clearly had different and complementary competences and areas of responsibility within the common aim. Sopjani's (2021) work supports this conclusion, finding that in three similar experimentally oriented living labs, including LEV-Pool, *complementary competencies and resources* were important enablers for collaboration. In contrast, in the planning phases for Mistra SAMS there was a great deal of overlap between the competences of the design researchers, social scientists, engineers, and city planners, but underlying differences regarding research aims and values. Additionally, private- and public-sector actors were not involved in these stages to any practical degree.

In several of the DLLs, research collaboration has been needed to analyze and disseminate the rich and diverse results. In Car-Free, LEV-Pool, and CoKitchen interdisciplinary collaborations were planned for the analysis, for modelling potential societal impacts, and for developing actionable concepts for diffusing or effectively presenting results to relevant stakeholders (see Hesselgren et al., 2017). In the first phase of the Mistra SAMS program, in contrast, very little joint analysis took place, and at the time of writing, the second phase is still ongoing.

A guiding vision may provide conditions for design

Design has been found to depend on shared visions, but also to lead by creating visions of the future (Kotter, 1996; Verganti, 2009). Most of the DLLs had a core idea in place before the projects were launched. I suggest that these concepts provided a shared vision for the DLLs that helped integrate the efforts. Wrigley et al. (2020) emphasize that having a shared vision is the most im-

portant condition for design work, and describe a case where a designer sensed conflicting visions at a company. The designer visualized and presented two clearly different alternative visions, leading the directors to separate out these businesses. Another study of two highly successful Swedish design startups (Radón et al., 2013) found that the funders' visons and sensitivity to an aspired user experience were critical in directing the creative work. However, Peng (1994) found that creating a shared vision is challenging in design projects.

In clinical research, having a shared vision has been found to be important for integrating research disciplines (Choi & Pak, 2007). In LEV-Pool, Sopjani (2021) found that abstract visions did not support collaboration. In Mistra SAMS, the overall vision stated in the program plan was "reaching the set CO2 reduction targets." I argue that this was too broad of a vision to guide the selection of a specific core idea for the living lab interventions. Later on in the process, when ideas have been selected for the future concepts to be staged in each lab and had thus become tangible and concrete, I find that they have served as guiding visions to integrate the teams. Similarly, Sopjani (2021) notes that in LEV-Pool, even though alignment was lacking at the outset, the staging of the service concept functioned as a boundary object, and at this practical level, the actors had to come into alignment. At this point, the vision turned to something more akin to Stolterman's (1991) concept of an *operative image*, allowing for concrete discussion and the fixing of certain aspects.

Bergmann et al. (2012) found that shared objectives and issues was the most important factor for successful transdisciplinary collaboration, and Defila et al. (2008, quoted in Bergmann et al., 2012) further argues that subprojects or groups have to show how their research questions derive from the projects' overall objectives and issues. In Mistra SAMS' second phase, this was clearly not the case. For example, in the program plan, the tasks and objectives of the Citizen group did not have a clear relation to the overall research issue of investigating mobility and accessibility services enabled through a digital platform. This group's research mainly concerned the development of "active citizenship" with regards to sustainability.

Through interviews with 110 designers and design managers, Björklund et al. (2020) find that designers are often concerned with silo-breaking and struggle to get mandates, to get decisions made, and achieve get buy-in for design methods. The experiences from Mistra SAMS confirm that finding. The un-

derstanding that the living labs were to be planned jointly by the whole group meant that no-one had a mandate or responsibility to ensure the lab functioned as a whole. There are many indications that the design practitioners struggled to resolve this situation, to mediate between conflicting views, and to engage researchers from other groups.

Across many fields of design, the term "design by committee" is used in the sense that when multiple stakeholders with divergent interests are involved, the resulting designs reflect a compromise with no clear goals satisfied (Purao et al., 2008). Henning (2006) suggests that "design by committee" is often the result of merging multiple specifications into one. The larger LEV-Pool and Riksten DLLs needed to satisfy a wide variety of interests and specifications, and in Riksten, I argue that this clearly led to detrimental compromises in all three labs. With a shared and tangible core idea and vision, and a clearly stated decision to use a design-driven research approach, a larger group of researchers and stakeholders could likely be engaged. The final report from LEV-Pool also emphasizes the need to set aside resources for project management. One promising way forward would be for most actors in larger research projects to take a more clearly supportive role in planning and staging interventions through their knowledge and resources, orchestrated by clear leadership and a shared vision. As Purao et al. (2008) note, in order to avoid a "design by committee" the World-Wide Web Consortium (W3C) recommend a two-tier approach in which a minimal workgroup would design a solution and a larger group of experts or stakeholders would more passively support and comment. This sort of setup was recommended by the project manager in LEV-Pool. It might also have worked in Mistra SAMS, and there was an attempt to create a smaller and more operationally oriented group.

Preplanned or quickly ideated concepts may support collaboration

One core feature of the DLLs was their "doing-first" strategy. In Chapter 6 I suggest that this builds on of the quick setup of an intentionally provisional concept in order to start learning. In Car-Free, LEV-Pool, and CoKitchen, a core concept for the DLL interventions had been ideated in the pre-planning stages, before the projects were funded. This meant that the research groups had more clearly "signed up" to execute those ideas and were better able to include and budget for the required resources. Having a core concept in place also made it easier to get practical support from stakeholders and partners. In contrast,

Mistra SAMS—with its large and diverse research team—jointly approached the complex and open question of what to explore in the labs, which proved to be a question loaded with conflicting research interests, differing research cultures, and ethical issues. The year-long collaborative development processes in Mistra SAMS mostly ended in compromises, and the intervention concepts that were ultimately selected had been present in nascent form much earlier. In contrast, the core interventions in Car-Free and CoKitchen were both ideated in a first meeting, although they built on research developments that the group had been aware of for a long time. In CoKitchen, only five months transpired between the official launch of the project and the start of first physical remodeling of the apartment. Overall, I find that ideas for lab interventions have mainly not emerged through targeted pre-studies or extended ideation processes. Instead, in the more well-functioning DLLs, research and concepts that were known to the group beforehand seems to have been rather quickly turned into concrete concepts for interventions without prior validation. This aligns well with Mintzberg and Westley's (2001) notion of a "doing-first strategy," as was discussed in Section 6.2. The senior design practitioner who had a prominent role in the two first DLLs, likened this to "just jumping straight into the deep end of the pool."

Lack of operative integration of the project team can limit resources

Real-life experimentation is very resource intensive, and in the DLLs, the amount of ground-level work required to stage the interventions in the real world and to support user participation has often been underestimated. Technical problems, such as malfunctional mobile phone apps, has often caused trouble and delays, and managing user participation over longer time periods has proven to be time-consuming (See e.g., Ringensson, 2021). I find that DLLs require flexibility in execution, as well as close coordination of a significant amount of operational work. To operate smoothly, the DLL approach requires a clear mandate for a small group or an individual to lead planning and implementation—something that seems to have worked out better in the smaller and mid-sized labs. In LEV-Pool, user recruitment and providing instructions and user support alone were estimated at six months' full-time equivalent, which was mainly carried out by the design researchers. This, even though in that project, the team was larger and had the support of public and private partners.

In the second phase of Mistra SAMS, there were no clear responsibilities or mandates for planning the lab, and initially no resources available for opera-

tional aspects. The only exception was the Citizen group's explicit responsibility and budgeted resources for recruiting and following a small group of research participants. Consequently, only these participants were recruited. Service providers were not involved at an operational level until a very late stage, and even then, only to a limited extent. In Car-Free, LEV-Pool, and CoKitchen, collaboration with partners provided resources for staging the real-life service concept prototypes. In Mistra SAMS's first phase, the transdisciplinarity came to an end after the lab interventions were finally selected, and staging the labs became solely the design practitioners' responsibility. Aside from the joint decision-making process, I argue that Mistra SAMS' research became "siloed," with very little interaction. The partner organizations in Work Closer were initially only reluctantly and more passively involved, even though they came to be more engaged over time.

Especially in the larger research groups, project management at an operational and hands-on level has been crucial, and in several cases lacking. Bratteteig and Stolterman (1997) argue that the democratic workstyle used in interdisciplinary design projects requires managerial support in the form of balancing or orchestrating creative efforts. On the other hand, they argue that this managerial work must not restrict or control the initiative to force it into technically rational development processes or methods. LEV-Pool had allocated a project manager who worked half-time for the duration of the project, but the final project report still concluded that this was insufficient. The lack of management and integration in Mistra SAMS led to limited resources for staging and managing the living labs, which clearly impeded progress in Work-Closer. Playing-Rules was a very small DLL, and I basically staged it alone after I received the initial go-ahead. However, the lack of additional resources was a significant limitation in analyzing the data and results. In Car-Free, the senior design practitioner and the two design research PhD students who staged and ran the lab had more time, as this was a very small-scale lab, and they also had support from two partner organizations that took care of many practical issues. In CoKitchen, responsibilities were clearly divided among the researchers and partners, which worked well in comparison with other DLLs. Moreover, much of the operational work and day-to-day management of the lab was handled by the KTH Live-In Lab organization that hosted the project.

> *To summarize this section, the larger DLLs, and especially Mistra SAMS met with collaborative and administrative challenges. Both larger and*

> *smaller DLLs are probably best implemented by a small team with clear mandates in order to avoid "design by committee." Interdisciplinary and complementary research collaborations appear to have worked better than transdisciplinarity. When the main intervention concepts were ideated before the research project was funded or launched, many of these problems seem to have been avoided. The intervention concept functioned as a tangible vision which clearly supported alignment, and allowed more support from partner organizations. Finally, physical proximity can help resolve tacit differences, and strong project management can allow for better-integrated project efforts.*

7.2. Design strategies for bridging, orchestration and 'making'

Design for mediating or 'bridging' conflicts and misunderstandings

In product or service development, design projects require teamwork, and designers have been found to serve as *mediators* in multidisciplinary collab-

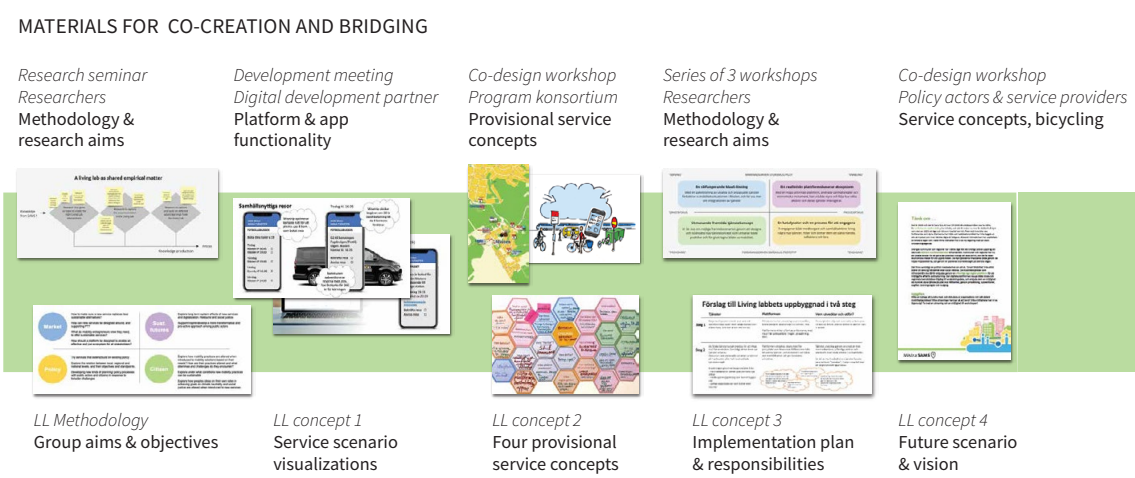
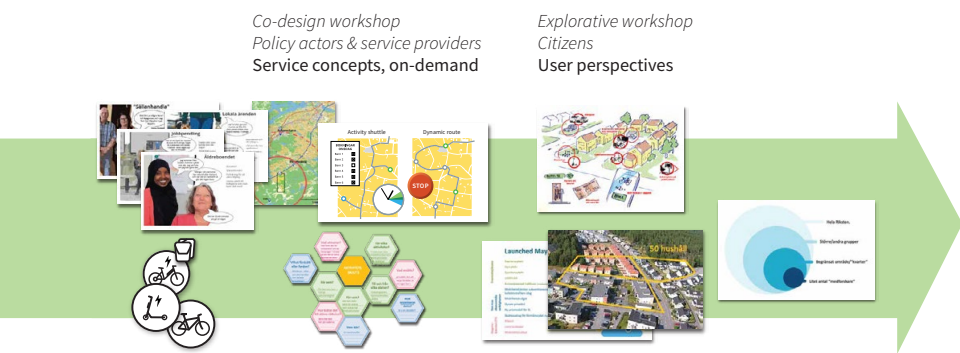


Figure 45. Concept visualizations and other materials I produced for workshops and meetings in Mistra SAMS, and resulting inputs for the living lab planning process. Feb. 2021 to Apr. 2022.

oration (Kelley, 2009). Getting things done in Mistra SAMS required such mediation, by bridging diverse views about the living lab methodology and which interventions should be staged. Importantly, these differences included aims for working with researcher-led, market-driven or community-driven approaches. Altogether, these differences were tacit and resisted disentanglement, and supporting program researchers and partners to coalesce around a shared vision and idea about living labs became a central part of my efforts.

During the decision-making process in Mistra SAMS' second phase, the Living Lab Design group made several attempts to bring about discussion regarding these divergent research views, such as using the four-field matrix workshop materials (see Figure 30, p. 98). The concepts of having a learning vs. a solving aim were originally created as a way to illustrate these internal differences and tensions within Mistra SAMS. Altogether, I designed a series of visualized scenarios and workshop materials to build a shared understanding of the mobility services that the program aimed to explore (See figure 45). This work also includes storytelling, such as in the 2026 vision document (figure 31, p. 101), which came to be widely used in the research program. These bridging attempts also included making illustrations of central concepts to make them easier to grasp and discuss, such as the mixer board (see Figure 42, p. 137), which illustrated the concept of balancing, and the four concentric rings that illustrated potential different types of user engagement in Riksten. In Mistra



SAMS' first phase, the lead design practitioner also made several such illustrations, such as a workshop sketch of a hamburger that helped illustrate the many layers of mobility service provision in MaaS (see Figure 14, p. 78) or the image of the overall design of the research program around the shared Riksten living lab, which became known as "the snail" (see Figure 29, p. 94). Overall, I find that the experiences from the DLLs support Kelley's (2009) claim. I suggest that this drive for mediation and alignment might be connected to the more action- and future-oriented nature of design practice, and also to the design-practitioner-researchers' focus on the user-participants' experience.

Staging a DLL requires orchestration and a sensitivity for experiences

Staging a possible future in the context of people's everyday lives requires a sensibility for making the interventions both "work" and "feel right." In the DLLs, this sensibility has mainly been provided by the design researchers. Efforts such as for creating a "visual identity" for the vehicles in LEV-Pool, the rental service "front-end" in Car-Free, and the future scenario I created for Riksten, are examples of this. In CoKitchen, when the design researchers were purchasing recycling bags for the apartment, they discussed how the recycling system did not cater for international variations in terms of values and awareness. We ultimately designed new, more instructive labels for the bags, as well as matching signs for the recycling rooms on campus (See p. 83). Designers are known for being holistic and able to zoom in and out to consider both the small, practical details and overarching systemic issues (Johansson & Wodilla, 2008; Stolterman, 2008). This capacity to notice details that are lacking and to know what they should be like is likely related to how designers are trained to empathically focus on the user experience.

In CoKitchen, we also asked the architects to make a dedicated space in the kitchen for the recycling bags, and made a concept visualization that convinced Academic Housing to repaint the recycling rooms. Similarly, in more complex labs such as LEV-Pool and CoKitchen, making interventions work and providing the right experience required the orchestration of other people and organizations, both in the project team and externally. Similar to the work of a movie director, orchestrating a DLL includes both visioning, planning, and energizing a team of collaborators. Klaus Krippendorff (2005, 2012) describes how designers both "work out realistic paths, plans to proceed towards desir-

able futures," and "enroll others into their projects" (p. 27). This enrollment requires a clear mandate and a holistic vision that provides the project with a purpose, which also explains the designers' attempts to bridge different views in the project groups.

Krippendorff (2005) further argues that while projects require a purpose or narrative about what needs to be changed, openness to details is what engages stakeholders to see possibilities for their own contributions. In line with this reasoning, in Mistra SAMS' second phase, my intention was not to engage service providers in a market-driven launch in Riksten. Instead, by engaging service providers in the co-design workshops, I sought to let them interact with the future of the 2026 vision document (figure 31, p. 101) by exploring what their own business might look like in a more sustainable future. Similarly, in Work Closer, the partners that let their employees use the NTC had to "act their roles" as employers in interacting with the future prototype, when employee participation conflicted with the companies' current rules or routines. This was similar to the unrealized plans for collaboration with restaurant wholesaler Martin & Servera in the Restaurant Lab, which sought to engage Martin & Servera's operations in delivering their services to the prototype of a future sustainable restaurant. I believe that this type of active engagement with a prototyped sustainable future scenario could be an effective way to provide intense learning experiences for commercial actors—experiences that involve their operations divisions and not just their sustainability office, as might otherwise be the case.

The 2026 future vision helped engage service providers in generative co-creation, even though the resulting mobility concepts were mostly not put to use. If the co-created services would have been staged as interventions in the Riksten lab, the future vision would have functioned as a core intervention, by guiding the design of new services for this projected future society. Internally in the research program, the same vision served to clarify our objectives for the living lab and also functioned as a critique of how public transportation is commissioned to exclude mobility services for individual transportation needs.

Designer as operative 'doer' and maker

It is well known that the volume of operational work involved in design research is often not visible, well-understood, or adequately planned for. For the

design researchers in the DLLs, hands-on operational work was time consuming hard to outsource. We found that the interventions often required special non-standard solutions, and design work was often needed to provide the right experience for the research participants. Outsourcing would have required assistants or contractors to share this designerly sensitivity or competence. It is interesting to contrast this with the architects' design work in CoKitchen. After the technical drawings were finished, the co-living apartment was built by a regular construction contractor. Still, the design researchers in some cases saw that important parts were missing. Noticing that a standard table would be too small to allow the envisioned social interaction, one design researcher took her car after work to purchase a very large second-hand kitchen table mere days before the students moved in.

I have referred to this work as staging and orchestrating. In the same sense as in a theater play, it is the back stage work and story, required to provide the front stage experience. Referring to Bødker (2017) and Eriksen (2012), Botero and Hyysalo (2020) find that even though it is frequently unacknowledged, this practical or "mundane" work of materializing co-design is not unimportant. In participatory design and co-design, practical issues concerning user or stakeholder involvement has been suggested to be "co-constitutive to the processes, results, and further uptake of the collaborative design outcomes" (Hyysalo & Hyysalo, 2018, p. 43) and may have strategic implications (Botero & Hyysalo, 2020).

I argue that designers not only have a disposition for "making" but also often seem to have a combined urge and capacity for taking practical action. When time was running out in Mistra SAMS' first phase, I designed and staged the very small Playing-Rules DLL almost single-handedly, and a few months later, the other design-practitioner in the program staged the NTC in Work Closer, handling all the practical contacts with the architect, the remodeling contractors, and service providers. I suggest that this is in line with what Archer, Baynes, and Roberts (2005) describe as the "executive skills of the doer and maker," such as the designer's sensibility, judgment, and practical capacity for invention (p. 13). Designers are concerned with making, while other researchers are often concerned with what is already there. In Mistra SAMS' second phase, while I engaged service providers in hands-on co-creation, other researchers in the program carried out interviews with the same actors. It was clear that the Living Lab Design group could not stage the lab alone, and we

therefore worked to engage the other research groups in the program, and also made concerted efforts to engage external partners.

> *In summary, when staging a DLL designers rely on their sensitivity for human experiences, their capacity for orchestration and for taking practical action. Orchestration requires a shared vision, which explains why designers place such a strong focus on bridging conflicting views. Visioning is performed by use of stories, visualized ideas and concepts.*

8. The design methodology PhD

In this section, I present a number of reflections and insights from carrying out PhD research with a focus on design methodology. A DLL, as well as Practice-Based Design Research or Research through Design projects in general, mainly aims to produce knowledge in a domain, such as health-care, housing, or mobility. Overall, I have found that carrying out a design methodology PhD using Practice-Based Design Research entails specific challenges which are discussed below.

Challenges of having both methodological and empirical aims

The Research through Design approach has been found to both produce empirical knowledge in a researched field and simultaneously lead to advances in design methodology (Zimmerman et al., 2010). This is likely true for most practice-based PhD projects in the field of design. However, as my PhD research specifically aimed to explore and advance designerly research methods, the fact that the research projects I took part in were founded and set up to produce domain knowledge about mobility services and co-living practices became a challenge. Thus, while conducting RtD projects in collaboration with many other researchers and partner organizations, the additional agenda of performing Research for Design (Frayling, 1993) was mine alone. This meant that the empirics produced and the vast majority of the interviews and observations I conducted as part of the projects have not been related to my own PhD research questions. In this dissertation, the papers I have written about mobility practices and co-living services serve mostly as empirical examples of the type of knowledge or outcomes produced in DLLs.

Design research methods can be applied to many different types of issues and domains, and as a senior design researcher, it may be possible to support a research project by taking on a narrower specialist role that focuses on the

methodologies used from a meta-perspective. As a PhD student, this may be difficult, as student researchers are expected to handle much of the "legwork" of doing interviews and publishing results. I suggest that PhD research in design would be best conducted within a research domain that the candidate plans to continue to work in, to allow them to develop parallel expertise in that domain. Another possible solution to this challenge would be assuming the role of "following" a research project, providing ongoing methodological documentation, reflection, and active support. In Swedish, this role is called följeforskare which could be translated as "companion action researcher." There are, however, few projects that can provide funding for this type of supporting role, and in practice the role is often concerned with top-down research evaluation and management.

Lack of mandate to make methodological choices

Another challenge with PhD-level research on design methodology is that a mandate is needed to influence which methodological choices get tested. In Mistra SAMS' second phase, this lack of power became evident, as the other researchers instead planned to use different research approaches. It is likely that I was allowed to independently design and stage Playing-Rule because that lab was considered a small warm-up project and because time was running out and the other researchers were engaged elsewhere in the program. Even though this project was hampered by limited resources and time, it allowed me to try out methods that provided important input for my dissertation topic.

Since PhD research positions require funding, openings are, in most cases, posted only after a project has been planned. Later on, when a PhD research project begins to take shape, I have found that plans concerning the research methodology were hard to change, since these choices were connected to the roles of specific researchers or parts of the budget. This was especially clear in Mistra SAMS' second phase, as the budgeted number of DLL participants and interviews as well as assigned roles all stayed fixed even though the type of experiment and the design interventions had not been specified in the program plan and were changed several times. As part of the Living Lab Design group, I spent the bulk of my time trying to bridge the diverse research views within the program, suggesting and discussing methodological choices. Still, in the end, the living lab was mainly set up by other researchers, partly as a design intervention, partly as a type of commercial pilot study run by a partner organiza-

tion, and partly building on principles closer to sociology or community-based participatory research.

The 'operative trap' of a design PhD

Another related challenge lies in the real-life and hands-on "making" nature of staging the DLLs. As discussed previously (see Section 7.2), making the future prototypes work and feel realistic in a real-world setting includes tackling a large number of practical and operational issues. This was the case in all the DLLs, but particularly evident in LEV-Pool and Work-Closer. These challenges are, to some extent, a common issue in participatory and action-oriented research, and as Botero and Hyysalo (2020) argue, this "mundane and strategic" work is important for research results and outcomes. At the same time, this practical embeddedness also provides rich empirical data. As discussed in Sections 6.2 and 6.5, many of the DLLs' findings regarding barriers to future developments resulted from the difficult practical staging of the lab interventions. Nevertheless, this operational work means that fellow PhD students who carry out more traditional research based on literature studies, surveys, and interviews enjoy more time for reading and writing.

9. Conclusions

9.1. Research contributions

An approach for designerly living labs

In this dissertation I contribute to the literature on living labs, socio-technical experiments, and real-life experimentation by proposing design-driven, or *designerly* living labs as a complementary approach to the more common notions of market- or policy-driven living labs and pilots. The proposed approach for designerly living labs is encapsulated in the brief description in 4.1 and further detailed in the discussion in Chapter 6, of the key characteristics of the approach and how they relate to the literature.

Specifically, I have described how DLLs can be viewed as exploratory prototyping in the pre-design stage, by staging future sustainability concepts to function as provisional tools for learning. Real-life user participation requires balancing, and I have contributed to our understanding of this need by describing the use of recruitment strategies and short-term challenges to promote more experimentation. I have also outlined how the overall process of staging these design interventions in today's society functions as backcasting set in real-life.

User participation in design-led real-life experimentation

I contribute to several fields of participatory research by proposing and describing a different form of active user participation in real-life experimentation. I find that the role of user participants in DLLs departs from the two dominant groups of participatory approaches and research ideals in this field. First, more mainstream living labs and pilots are mainly concerned with innovation and with the development, testing, and scaling-up of new solutions,

often based on a market or policymaking logic. Here, active user participation mainly takes place either in co-creation workshops or in late stage evaluative concept testing[11]. Second, PD and DSI, as well as several strands of sociology, urban planning, and action research[12], are instead concerned with activating or empowering citizens and grassroot communities in collective approaches that seek to address local matters of concern in the here-and-now. DLLs take a third position, where—in Krippendorff's (2005) words—designers devise paths towards desirable futures and "enroll" others in their projects. DLL user participants neither co-design the concepts nor are activated in the form of insider-led action to change their situation. The DLL participants have instead assumed a very active role in designer-led, open-ended and reflexive experimentation and exploration.

When aiming to solve an issue in the here-and-now, and especially when working with marginalized and vulnerable groups, embeddedness and issues of power and inclusion are important. DLLs have, however, focused primarily on the unsustainable lifestyles of an often disengaged middle-class. The actions of this group are limited by expectations of convenience, as well as current norms and structures, and they are not likely to form an active user community driving insider-led initiatives for sustainability. For the same reason, this group in general is not likely to request or co-create disruptive or challenging sustainable future concepts through a co-design process. Instead, the design researchers have devised more radical interventions that enable or provoke participants to reflect on and experiment with changing their practices. Adding on to the work of Hesselgren (2019) and Finlay (2002), I argue that this research-led and designer-led approach is participatory, in that the user participants are actively engaged in transparent and close collaboration, in the role of reflexive co-researchers. This reflexivity and engagement has, in many cases, grown over time to develop what I suggest calling a co-researcher mentality.

How to manage different research cultures in a DLL

Chapter 7 contributes to the literature on interdisciplinary and transdisci-

11 As noted in 2.3, the term participation general implies more-active user engagement and excludes approaches based on passively logged user data.

12 This also includes a relatively smaller group of living labs and ULLs with a bottom-up grassroot orientation; often concerned with social issues, urban planning, or "placemaking."

plinary research collaboration. First, by describing several reasons for confusion and conflicts in such collaborations. Second, by explaining how truly transdisciplinary collaboration may be challenging in such a context. I suggest that design-led interdisciplinary collaborations may be a more effective approach, and also provide some general recommendations regarding the practical planning and implementation of DLLs. In the second part of this chapter I describe the designerly capacities used for staging and orchestrating the DLLs and how design researchers attempted to bridge differences in order to draw together an operational team around a shared vision.

Most of the DLLs have been design-led, and also had a core idea of what kind of future concepts would be staged and explored before the project started. I argue that this allowed the researchers to focus their efforts on the real-life explorations. In contrast, the transdisciplinary Mistra SAMS research program sought to develop the DLLs jointly among researchers with partly conflicting research cultures and aims; an arduous process that also led to important insights. While the design researchers emphasized "just doing it" and designing provisional concepts more quickly to learn through real-life experimentation, other researchers emphasized building consensus, looking for input from stakeholders, positioning potential interventions in relation to previous research, and to evaluate their effects.

The main takeaway is that different research cultures and understandings are not easy to combine within one project or program. Within Mistra SAMS, four conflicting research positions were identified, each with different priorities regarding how to design the living lab. I suggest that this type of differences are often tacit, value-laden, and difficult to resolve, which may lead to misunderstandings, conflicts, or watered-down compromise decisions. I argue that questions such as whether the project aims to solve or to learn about an issue, and exactly how to involve user participants, are strategic choices that should be answered before a project is started, as they will influence the planning of project methods, resources, and required competences.

The real-life experimentation in a DLL is challenging. Much like managing a start-up company, staging and managing a future prototype requires much time and resources, as well as a tight and very operative team. When everyone involved shares a concrete vision of the future that is to be prototyped, as well as an understanding of the core ideas behind the research approach, the team

can focus on real-life making and exploration. In some DLLs, interdisciplinary research collaboration with clear responsibilities and complementary competences has been found to work well and to provide important perspectives and practical support. Also, the tangible future visions provided in the DLLs have helped engage public and private stakeholders in sustainability research, and to contribute resources.

Challenges of a design methodology PhD

Finally, in chapter eight, I contribute to the design research literature with three insights about the challenges of carrying out a design methodology PhD program in applied and practice-based design research projects. These insights include the difficulties of being alone in pursuing methodological outputs when the overall research project aims to produce domain-related knowledge, as well as the lack of mandate to make methodological choices. The third insight presented relates to the challenge of analyzing and disseminating the diverse knowledge outputs from a DLL, provisionally referred to as the design holism issue.

9.2. Implications for practice

Overall, DLLs offer a designerly approach for learning about sustainable futures through acts of making. Real-life interactions and doings can elicit findings that interviews and surveys will likely miss. The participants' interactions with the DLL interventions have resulted in experimentation and reflection that revealed both unfit norms and structures, and new types of value gained by adopting more sustainable lifestyles. Also, the practical set-up or staging of a future sustainability concept in today's society has repeatedly identified unexpected barriers on several different levels. As a type of backcasting set in real life, both these processes have provided actionable pathways towards sustainable futures. The concept demonstrations and the interactive learning experiences have also helped stakeholders move forward to engage in further developments after the labs ended.

For practitioners and researchers set to embark on real-life experimentation research, Chapter 7 includes many recommendations on how to implement design-driven experimentation and provide conditions for design. However,

given the discussion in Chapter 6 on how DLLs differ from other common real-life experimentation approaches, my main recommendation is to take care to first answer certain strategic questions about the type of results or outcomes that are sought, and what the research context looks like. No research approach is well-suited for all research situations and needs.

- When aiming to solve local or regional social issues, collaborative strategies for activating and supporting citizen communities and initiatives may be a fruitful approach.
- When aiming to innovate, develop and improve product or service concepts, user innovation and user co-creation can provide value, and in late development stages, evaluative testing is likely needed.
- To explore possible sustainable futures or to frame a complex issue in a more open-ended manner, DLLs offer a complement or alternative to other approaches. By "doing first", DLLs can approach issues where stakeholders are not yet ready to invest. By engaging user participants as co-researchers, DLLs can explore more radical sustainability concepts that people are not yet requesting.

User involvement in exploring pathways towards sustainability requires balance. The challenges of changing our daily practices makes it difficult to find users and consumers who are willing to try out new ways, and equally hard to engage user communities in making change. People that do engage actively may still resist relearning and only go so far as to make incremental changes, while passive informants will not provide experimentation and grounded reflections. Well-balanced design-driven experiments can both serve, enable, disrupt, or provoke willing research participants and partners in experimenting with sustainable concepts and lifestyles.

Finally, for market or policy actors planning a DLL, creating favorable conditions for design may require a bit of courage and trust. Designers are known both for creating purposeful future visions and for having the sensitivity and the executive capacity needed to stage them. With a mandate to carry out design research, designers are known for mediating between different research interests and the aims of many stakeholders and research partners. Without a clear mandate for design, research partners may fall back upon better-known modes of research, causing confusion or inhibiting design-driven research altogether.

9.3. Future research

This dissertation was written from a design perspective, and future research is needed that involves researchers from other fields to generate dialog on the suggested methodological differences in sustainability research. Transition experiments always have the dual aims of learning and making change, but this dissertation proposes that the two can be balanced quite differently. Literature overviews and typologies of approaches for real-life experimentation often take for granted an implicit aim to make real change in the here-and-now. When research is framed by this "solving" aim, it risks being limited to incremental changes that can be widely accepted by users today. This limitation can be further exacerbated by the associated evaluative mindset of testing and validating. Thus, learning within experimental governance has commonly focused on successful implementation, with the aim of building transformative capacity at higher infrastructural levels. Other types of research are needed that more clearly include the alternative of exploratory and bounded modes of experimentation, as exemplified by the DLLs. Further discussion regarding what situations and which types of research objectives that different modes of real-life experimentation and user participation are best suited for would be particularly valuable. To further clarify how different approaches may complement each other, designerly modes of real-life experimentation research could be investigated by scholars from other fields, such as Innovation Management and Transition Studies.

To further develop the proposed approach, new designerly living labs are needed to try out the recommendations in this dissertation—especially regarding how research collaborations are planned and organized. There are also opportunities to further develop the DLL approach by including more co-creative and iterative approaches. As has been indicated here, stakeholders can be invited to co-design service concepts that fit within the overall sustainable future vision of a DLL. By providing resources, such stakeholders can then take an active part in staging those concepts as real-life interventions, to be explored by potential "future" users and customers. Another approach would be to invite stakeholders to let their operational organization interact with real-life prototypes of a potential sustainable future. In both cases, DLLs could enable market or policy actors to prepare their operational organization for changes that are likely imminent. Such future prototyping could identify misalignments, trigger innovation, and allow for new business models to be tried out.

9.4. Outro

Radical lifestyle changes are needed to sustain human civilization in the face of climate change. There have been many calls for research that is more action-, future- and learning-oriented—experimental as well as experiential—to enable us to transform the complex socio-technical systems that make up our society. There is a broad drive for real-life experimentation across several fields of research and in society at large. These many emerging approaches can be suited for different types of challenges and might promote change in different ways.

In this dissertation I have investigated a line of design-driven real-life experimentation studies that aimed to catalyze sustainability transitions. I find that designerly living labs offer an important alternative to more widespread notions of pilots and living labs, as well as to approaches such as co-design and design for social innovation. By prototyping provisional sustainable concepts in the context of everyday life, this mainly exploratory approach poses an alternative when the time is not ripe for the more common, late-stage and often temporally compressed research process to evaluate, refine, and scale up innovations. The mainly research-driven designerly living labs are therefore especially well-suited for early research stages when knowledge is limited and when stakeholders and intended users are not yet ready to take the initiative or to change their practices. Exploring sustainability concepts within the context of everyday life and societal structures has proven to identify barriers and pathways for change, and has demonstrated desirable and sustainable futures.

In this dissertation, I have actively explored a designerly research approach in order to understand why it emerged and how it works. I have furthermore experimented with it and made my own contributions to it. Design is unquestionably hard to define. Among many things, it is a distinct human mindset and an abductive approach to development and change. With training comes a stronger future orientation, an aptitude for ambiguities and tensions, a propensity for doing and making, and a rich set of sensitivities, skills, and capacities to these ends. In the DLLs, the very human way of planning and shaping something new, and the designerly art, practices, and traits, have all been present. In the process of writing this dissertation, I started out describing the DLLs as a specific design methodology, but found that they are no less broad and complex than design itself.

References

Akrich, M. 1992. “The De-scription of Technical Objects.” In *Shaping Technology/ Building Society: Studies in Sociotechnical Change*, edited by W. E. Bijker and J. Law, pp. 205–224. Cambridge, MA: MIT Press.

Almirall, E., & Wareham, J., 2011. Living Labs: arbiters of mid-and ground-level innovation. *Technol. Anal. Strat. Manag*. 23 (1), pp. 87-102.

Akrich, M. 1992. The de-scription of technological objects. In *Shaping technology/ building society*, ed. W. E. Bijker and J. Law, 205, 24. Cambridge: MIT Press.

Alvesson, M., & Sköldberg, K. (2017). Reflexive Methodology: new vistas for qualitative research (second edition). Sage. https://doi.org/10.1080/13642531003746857

Ansell, C. K., & Bartenberger, M. (2016). Varieties of experimentalism. *Ecological Economics*, *130*, pp. 64-73.

Archer, B., Baynes, K., & Roberts, P. (2005). A framework for Design and Design Education. *A reader containing papers from 1970s and 80s*.

Argyris, C. (1996). Actionable knowledge: Design causality in the service of consequential theory. *The Journal of applied behavioral science*, *32*(4), pp. 390-406

Arnstein, S.R. (1969) A Ladder of Citizen Participation. J. *Am. Inst. Plan.*, 35, pp. 216–224.

Bergmann, M., Jahn, T., Knobloch, T., Krohn, W., & Pohl, C. (2012). *Methods for transdisciplinary research: a primer for practice*. Campus Verlag.

Bergvall-Kåreborn, B., Ihlström Eriksson, C., Ståhlbröst, A., & Svensson, J. (2009), A milieu for innovation – Defining living labs. *2nd ISPIM Innovation Symposium*, New York, pp. 6-9.

Bernard, H.R. (2011) Research Methods in Anthropology: Qualitative and Quantitative Approaches, *Rowman Altamira*, New York.

Binder, T., & Redström, J. (2006). Exemplary Design Research. *DRS Wonderground*. Conference Proceedings.

Bidwell, D. Thinking through participation in renewable energy decisions. *Nat Energy* **1**, 16051 (2016). https://doi.org/10.1038/nenergy.2016.51

Björgvinsson, E., Ehn, P., & Hillgren, P. A. (2012). Design things and design thinking: Contemporary participatory design challenges. *Design Issues*. 28(3), pp. 101-116. https://doi.org/10.1162/DESI_a_00165

Björklund, T., Maula, H., Soule, S. A., & Maula, J. (2020). Integrating design into organizations: The coevolution of design capabilities. *California Management Review*, *62*(2), pp. 100-124.

Blomberg, J., & Karasti H. (2013). "Ethnography: Positioning Ethnography within Participatory Design." In *Routledge International Handbook of Participatory Design*, edited by J. Simonsen and T. Robertson, pp. 86–116. New York, Routledge.

Boehnert, J., Lockton, D., & Mulder, I. (2018). Designing for transitions. Editorial.

Boks, C., Lilley, D., & Pettersen, I.N. (2017). The Future of Design for Sustainable Behaviour, Revisited. In: Matsumoto, M., Masui, K., Fukushige, S., Kondoh, S. (eds) *Sustainability Through Innovation in Product Life Cycle Design*. EcoProduction. Springer, Singapore. https://doi.org/10.1007/978-981-10-0471-1_46

Botero, A., & Hyysalo, S. (2013). Ageing together: Steps towards evolutionary co-design in everyday practices. *CoDesign*, 9(1), pp. 37-54.

Botero, A., Hyysalo, S., Kohtala, C., & Whalen, J. (2020). Getting participatory design done: From methods and choices to translation work across constituent domains. *International Journal of Design*, *14*(2), pp. 17-34.

Bradbury, H. 2015. Introduction to the Handbook of Action Research. In Bradbury, H. Editor. (2015) *Handbook of Action Research. 3e*. Sage Publications.

Brown, Halina S., & Vergragt, P. J. (2008). Bounded socio-technical experiments as agents of systemic change: The case of a zero-energy residential building. *Technological Forecasting and Social Change*, 75(1), pp. 107–130.

Brown, B., Reeves, S., & Sherwood, S. (2011, May). Into the wild: challenges and opportunities for field trial methods. In *Proceedings of the SIGCHI conference on human factors in computing systems* (pp. 1657-1666).

Brown, T. 2009, Change by Design: How Design Thinking Transforms Organizations and Inspires Innovation, Harper Business, NY.

Brown, A. and P. Dowling (2001) Doing research/reading research: A mode of interrogation for teaching. London: Routledge Falmer.

The British Standards Institution (2008) BS 7000 - Design management systems. https://doi.org/10.3403/BS7000

Buchanan, R. (1992). Wicked Problems in Design Thinking. *Design Issues*. https://doi.org/10.2307/1511637

Buchenau, M., & Suri, J. F. (2000) "Experience Prototyping". *In Proceedings of Designing Interactive Systems (DIS '00) New York, USA*, pp. 424–433. ACM

Bulkeley, H., & Castán Broto, V. (2013). Government by experiment? Global cities and the governing of climate change. Transactions of the Institute of British Geographers, 38(3), pp. 361–375.

Bulkeley, H., Coenen, L., Frantzeskaki, N., Hartmann, C., Kronsell, A., Mai, L., ... & Palgan, Y. V. (2016). Urban living labs: governing urban sustainability transitions. *Current opinion in environmental sustainability*, *22*, pp. 13-17.

Burns, C., Cottam, H., Vanstone, C., & Winhall, J. (2006). *RED paper 02: Transformation design*. London: Design Council.

von Busch, O., & Palmås, K. (2023). The Corruption of Co-Design: Political and Social Conflicts in Participatory Design Thinking. Taylor & Francis.

Bødker, S., Ehn, P., Sjögren, D., & Sundblad, Y. (2000). Co-operative Design—perspectives on 20 years with 'the Scandinavian IT Design Model'. In *proceedings of NordiCHI*, Vol. 2000, pp. 22-24.

Caniglia, G., Luederitz, C., von Wirth, T., Fazey, I., Martin-López, B., Hondrila, K., ... & Lang, D. J. (2021). A pluralistic and integrated approach to action-oriented knowledge for sustainability. *Nature Sustainability*, *4*(2), pp. 93-100.

Caprotti, F., & Cowley, R. (2017). Interrogating urban experiments. *Urban Geography*, *38*(9), pp. 1441-1450.

Carlgren, L., Rauth, I., & Elmquist, M. (2016). Framing design thinking: The concept in idea and enactment. *Creativity and Innovation Management*, 25(1), pp. 38-57.

Cashmore, Stissing Jensen, & Späth 2019. Introduction: The knowledge politics of urban sustainability transitions. p. 9. In Stissing Jensen, Cashmore, and Späth (eds.). *The Politics of Urban Sustainability Transitions: Knowledge, Power and Governance*. Abingdon, Oxon: Routledge.

Ceschin, F. (2013). Sustainable product-service systems: Between strategic design and transition studies. Springer Science & Business Media.

Ceschin, F. (2014). How the design of socio-technical experiments can enable radical changes for sustainability. Doctoral Dissertation

Ceschin, F. (2015). The role of socio-technical experiments in introducing sustainable product-service system innovations. In *The Handbook of Service Innovation*.

Ceschin, F., & Gaziulusoy, I. (2016). Evolution of design for sustainability: From product design to design for system innovations and transitions. *Design studies*, *47*, pp. 118-163.

Chambers, N., Simmons , C., & Wackernagel , M.(2000). Sharing nature's interest: ecological footprints as indicator of sustainability. London: Earthscan

Chesbrough, H. W. (2003). The Era of Open Innovation. *MIT Sloan Management Review*.

Chilvers, J., Pallett, H., & Hargreaves, T. (2018). Ecologies of participation in socio-technical change: The case of energy system transitions. *Energy Research & Social Science*, *42*, pp. 199-210.

Choi, B. C. K., & Pak, A. W. P. (2006). Multidisciplinarity, interdisciplinarity, and transdisciplinarity in health research, services, education and policy. 1. Definitions, objectives, and evidence of effectiveness. *Clinical and Investigative Medicine, 29*, pp. 351-364.

Choi, B. C. K., & Pak, A. W. P. (2007). Multidisciplinarity, interdisciplinarity, and transdisciplinarity in health research, services, education and policy: 2. Promotors, barriers, and strategies of enhancement. *Clinical and Investigative Medicine*, pp. 224-232.

Chow, R. 2013. "The RIP + MIX Method and Reflection on its Prototypes." In *Prototype: Design and Craft in the 21st Century*, Valentine L. (Eds), 155–168. London: Bloomsbury.

Chow, R., & Jonas, W. (2010). Case transfer: A design approach by artifacts and projection. *Design Issues*, 26(4), pp. 9-19.

Coghlan, D. (2004), "Managers as learners and researchers", in Coghlan, D., Dromgoole, T., Joynt, P. and Sorensen, P. (Eds), *Managers Learning in Action: Management Learning, Research and Education*, Routledge, London, pp. 183-9.

Coghlan, D. (2007). Insider action research: opportunities and challenges. *Management Research News*, 30(5), 335-343. https://doi.org/10.1108/01409170710746337

Compagnucci, L., Spigarelli, F., Coelho, J., & Duarte, C. (2021). Living Labs and user engagement for innovation and sustainability. *Journal of Cleaner Production*, 289, 12572

Corsín Jiménez, A. (2014). Introduction. The prototype: More than many and less than one. *Journal of Cultural Economy*, 7(4), pp. 381–398.

Coyne, R. (2005). Wicked problems revisited. *Design Studies*. https://doi.org/10.1016/j.destud.2004.06.005

Damodaran, L. 1996, User involvement in the systems design process – a practical guide for users. Behaviour & Information Technology, 15 (6), pp. 363-377.

Dantec, C. A. L., & DiSalvo, C. (2013). Infrastructuring and the formation of publics in participatory design. *Social Studies of Science*, *43*(2), pp. 241-264.

Dourish, P., & Button G. (1998). On 'Technomethodology': Foundational Relationships between Ethnomethodology and System Design. *Human–Computer Interaction* 13: pp. 395–432.

Dunne, A., & Raby, F. (2013). Speculative everything: design, fiction, and social dreaming. MIT press.

Dutilleul, B., Birrer, F.A. and Mensink, W. (2010), "Unpacking European living labs: analysing innovation's social dimensions", *Central European Journal of Public Policy*, Vol. 4 No. 1, pp. 60-85.

Edwards & Bulkeley 2017 Heterotopia and the urban politics of climate change experimentation. *Environment and Planning D: Society and Space*, 36(2) p. 3.

Ehn, P., & Ullmark, P. (2017) Educating the reflective design researcher, In *Practice-Based Design Research,* Vaughan Ed. Bloomsbury Publishing.

Ehn, P., & Kyng, M. (1987). The collective resource approach to systems design. *Computers and democracy*, 17, 57.

Elliott, J. (1991). Action research for educational change. *Milton Keynes, UK: Open University Press*.

Envall, F. (2021). Experimenting for change?: The politics of accomplishing environmental governance through smart energy pilot projects. Doctoral dissertation. *Linköping University Electronic Press*.

European Environment agency *(2013)*. Environmental pressures from European consumption and production. A study in integrated environmental and economic analysis. Technical Report 2. *European Environment Agency, Copenhagen*.

European Network of Living Labs (2017) U4IoT Living Lab Methodology Handbook. https://issuu.com/enoll/docs/366265932-u4iot-livinglabmethodology-handbook (accessed 20 March 2023)

European Network of Living Labs (2019) ENoLL Application Guidelines - 13th Wave. https://issuu.com/enoll/docs/397044439-enoll-application-guidelines-13th-wave (accessed 19 Feb 2021)

European Network of Living Labs. Activity Report 2020. https://issuu.com/enoll/docs/activity_report_2020 (accessed 20 Nov 2022)

Evans, J. (2016). Trials and tribulations: Problematizing the city through/as urban experimentation. *Geography Compass*, 10(10), pp. 429-443.

Evans, J., & Karvonen, A. (2010). Living laboratories for sustainability: exploring the politics and epistemology of urban transition. In *Cities and low carbon transitions* (pp. 142-157). Routledge.

Evans, J., Karvonen, A., & Raven, R. (Eds.). (2016). The experimental city. Routledge.

European Commission. (2006). The Helsinki Manifesto. European Commission. http://elivinglab.org/files/Helsinki_Manifesto_201106.pdf (accessed 11 Feb 2021)

Fazey, I., Schäpke, N., Caniglia, G., Hodgson, A., Kendrick, I., Lyon, C., ... & Saha, P. (2020). Transforming knowledge systems for life on Earth: Visions of future systems and how to get there. Energy Research & Social Science, 70, 101724.

Felt, U., & Fochler, M. (2010). Machineries for making publics: Inscribing and de-scribing publics in public engagement. *Minerva*, 48(3), pp. 219–238.

Finlay, L. (2002). Negotiating the swamp: the opportunity and challenge of reflexivity in research practice. *Qualitative research*, 2(2), pp. 209-230.

Frayling, C. (1993). Research in Art and Design (Royal College of Art Research Papers, Vol 1, No 1, 1993/4). *Royal College of Art Research Papers*.

Følstad, A. (2008). Living Labs for innovation and development of information and communication technology: a literature review. *Electronic Journal of Virtual Organisations,* 10, pp. 99-131.

Gaver, B., Dunne, T., & Pacenti, E. (1999) Cultural Probes. *Interactions*, January and February 1999.

Geels, F. W. (2002). Technological transitions as evolutionary reconfiguration processes: a multi-level perspective and a case-study. *Research policy*, *31*(8-9), pp. 1257-1274.

Geels, F.W. (2011), The multi-level perspective on sustainability transitions: Responses to seven criticisms. *Environmental Innovation and Societal Transitions*, 1(1), pp. 24–40. http://dx.doi.org/10.1016/j.eist.2011.02.002

Geels, F. W., & Schot, J. (2007). Typology of sociotechnical transition pathways. *Research Policy*. https://doi.org/10.1016/j.respol.2007.01.003

Gray, C., & Malins, J. (2016). Interpreting the map: methods of evaluation and analysis. In *Visualizing Research*, pp. 129-158. Routledge.

Hacker, K. (2013). Community-based participatory research. SAGE Publications, Inc. https://www.doi.org/10.4135/9781452244181

Hakkarainen, L (2017) Caring for Technology Evolving Living Lab Collaboration. *Aalto University publication series Doctoral Dissertations*. 103/2017. http://urn.fi/URN:ISBN:978-952-60-7459-7

Hakkarainen, L., & Hyysalo, S. (2016). The evolution of intermediary activities: Broadening the concept of facilitation in living labs. *Technology Innovation Management Review*, 6(1).

Hartswood, Mark, Rob Procter, Roger Slack, Alex Voß, Monika Büscher, Mark Rouncefield, & Phillip Rouchy. 2002. Co-realisation: Towards a principled synthesis of ethnomethodology and participatory design. *Scandinavian Journal of Information Systems*, Vol. 14 (2), pp. 9-30.

Hartwick, J., & Barki, H. (1994). Explaining the role of user participation in information system use. *Management science*, *40*(4), pp. 440-465.

Hasselqvist H., & Hesselgren M. (2017) Bridging citizen and stakeholder perspectives of sustainable mobility through practice-oriented design, *Sustainability: Science, Practice and Policy*, 15:1, 1-14, DOI: 10.1080/15487733.2018.1533781

Henderson, A., & Kyng, M. (1991) There's no place like home: Continuing design in use. in *Design at work: Cooperative Design of Computer systems*. Greenbaum and M. Kyng Eds., Lawrence Erlbaum Associates, Hillsdale.

Henning, M. (2006). The Rise and Fall of CORBA: There's a lot we can learn from CORBA's mistakes. Queue, 4(5), pp. 28-34.

Hermans, G. (2015). *Opening up design: Engaging the layperson in the design of everyday products*. Doctoral dissertation, Umeå University.

Herr, K. & Anderson, G. L., "The Action Research Dissertation : A Guide for Students and Faculty" (2015). *All Books and Media by Montclair State University*.

Herth A., Verburg R., & Blok K. (2022) Enabling innovations for the energy transition: The university campus as a living lab. *Proceedings of CINet 2022*, Pisa

Hesselgren M. (2019) Designing for sustainability practices: RE-DO Design Doings, Strategies and Postures. (PhD dissertation, KTH Royal Institute of Technology.

Hesselgren M., Hasselqvist H., & Sopjani L. (2017) Design strategies for exploring and bridging: Intersections of everyday life and change-making decisions for sustainability. *Design Management Academy*, Hong Kong, 2017.

Hesselgren, M., & Hasselqvist, H. (2016) Giving car-free life a try: Designing seeds for changed practices. In: *Proceedings of DRS 2016, Design Research Society 50th Anniversary Conference*. Brighton, UK, 27–30 June 2016.

Hillgren P-A., Seravalli A., & Emilson A. (2011) Prototyping and infrastructuring in design for social innovation, *CoDesign*, 7:3-4, 169-183. DOI: 10.1080/15710882.2011.630474

Hodgson, Fred, Bailey, & Hall (eds.) 2019. The Projectification of the Public Sector. *New York*: Routledge.

Hollands (2015). Critical Interventions into the Corporate Smart City. *Cambridge Journal of Regions, Economy and Society*.

Hossain, M., Leminen, S., & Westerlund, M. (2019). A systematic review of living lab literature. *Journal of cleaner production*, 213, pp. 976-988.

Hyysalo, V., & Hyysalo, S. (2018). The mundane and strategic work in collaborative design. *Design issues*, *34*(3), pp. 42-58.

Ilstedt S., & Enlund T. (2012) Centrum för design och hållbar utveckling. KTH.

Ilstedt, S., & Sjöman, M. (2022) The value of being close: Social and ecological sustainability in co-living for students, in Lockton, D., Lenzi, S., Hekkert, P., Oak, A., Sádaba, J., Lloyd, P. (Eds.), DRS2022.

Ilstedt, S., & Wangel, J. (2013). Designing sustainable futures. *Nordes*, 1(5).

Intille, S.S., Larson, K., Tapia, E.M., Beaudin, J.S., Kaushik, P., Nawyn, J., & Rockinson, R. (2006), "Using a live-in laboratory for ubiquitous computing research", *Pervasive Computing*, Springer, Berlin, Heidelberg, pp. 349-365.

Irwin, T., Tonkinwise, C., & Kossoff, G. (2022). Transition design: An educational framework for advancing the study and design of sustainable transitions. Cuadernos del Centro de Estudios en Diseño y Comunicación. Ensayos, (105), pp. 31-72.

ISO, ISO 13407 Human-centered design processes for Interaction Systems, ISO, Geneva, Switzerland, 1999

ISO, ISO 9241-210 Human-centered design processes for Interaction Systems, ISO, Geneva, Switzerland, 2010

Jackson, T. D. (2008). The challenge of sustainable lifestyles. In *State of the World 2008. Innovations for a Sustainable Economy*, No. 4, pp. 45-60, Routledge

Jahn, S., Newig, J., Lang, D. J., Kahle, J., & Bergmann, M. (2022). Demarcating transdisciplinary research in sustainability science – Five clusters of research modes based on evidence from 59 research projects. *Sustainable Development*, 30 (2), pp. 343– 357. https://doi.org/10.1002/sd.2278

Johansson, U., Sköldberg, K., & Svengren, L. (2003, April). Industrial design as a balancing artistry: Some reflections upon industrial designer's competence. In *Proceedings of Fifth European Academy of Design Conference: Techne Design Wisdom, Barcelona*

Jose, R., & Ramakrishna, S. (2021). Comprehensiveness in the Research on Sustainability. *Materials Circular Economy*, *3*, 1-2.

Kankainen, A., Vaajakallio, K., Kantola, V., & Mattelmäki, T. (2012). Storytelling Group-a co-design method for service design. *Behaviour and Information Technology*. https://doi.org/10.1080/0144929X.2011.563794

Karasti, H. (2014). Infrastructuring in participatory design. In Proceedings of the 13th Participatory Design Conference: Research Papers-Volume 1 (pp. 141-150).

Kelley, D. (2002) Human Centered Design (YouTube). TED. Online 26.09.2014, http://www.ted.com/talks/david_kelley_on_human_centered_design (retrieved 22 nov 2022)

Keyson, D., Guerra-Santin, O., & Lockton, D. (2017a). Living Labs. *Design and Assessment of Sustainable Living. Netherlands: Springer*. https://doi.org/10.1007/978-3-319-33527-8_12

Keyson, D. V., Morrison, G. M., Baedeker, C., & Liedtke, C. (2017b). Living labs to accelerate innovation. In: *Living Labs*: Design and assessment of sustainable living*; D. V. Keyson, G. M. Morrison, C. Baedeker, & C. Liedtke (Eds.*) pp. 55-61, *Netherlands: Springer*.

Kiesler, S., & Cummings, J. N. (2002). What do we know about proximity and distance in work groups? A legacy of research. *Distributed work*, *1*, pp. 57-80

Kimbell, L., & Bailey, J. (2017). Prototyping and the new spirit of policy-making. CoDesign, 13(3), pp. 214-226.

Koen, P. A., Ajamian, G. M., Boyce, S., Clamen, A., Fisher, E., Fountoulakis, S., ... & Seibert, R. (2002). Fuzzy front end: effective methods, tools, and techniques. *The PDMA toolbook 1 for new product development*.

Kommonen, K. H., & Botero, A. (2013). Are the users driving, and how open is open? Experiences from Living Lab and user driven innovation projects. *The Journal of Community Informatics*, 9(3).

Koskinen, I., Zimmerman, J., Binder, T., Redstrom, J., & Wensveen, S. (2012). Design Research Through Practice. In *Design Research Through Practice*. https://doi.org/10.1016/C2010-0-65896-2

Kotter, J. P. (1996). Leading change. Boston, MA: *Harvard Business School Press*.

Kraff, H. (2020). A critical exploration of agonistic participatory design. *The Design Journal*, *23*(1), pp. 31-48.

Krippendorff, K. (2005). The semantic turn: A new foundation for design. crc Press.

Krippendorff, K. (2012). Design research, an oxymoron?. In *Design research now* (pp. 67-80). Birkhäuser.

Kuijer, L. (2014). Implications of Social Practice Theory for Sustainable Design. (Doctoral dissertation)

Kujala, S. (2003). User involvement: a review of the benefits and challenges. *Behaviour & information technology*, *22*(1), 1-16.

Laakso, S. (2017). Giving up cars: The impact of a mobility experiment on carbon emissions and everyday routines. Journal of Cleaner Production, 169, pp. 135-142.

Latour B. (2004). 'From "matters of facts" to "states of affairs". Which protocol for the new collective experiments?' in Henning Schmidgen (Ed.), *Experimental Cultures*.

Laurent, B., & Pontille, D. (2018). Towards a study of city experiments. In *Creating smart cities* (pp. 90-103). Routledge.

Leminen, S. (2015). Q&A. What are living labs? *Technology Innovation Management Review*, *5*(9), pp. 29–35.

Leminen, S., Westerlund, M., & Nyström, A. G. (2014). On becoming creative consumers–user roles in living labs networks. *International Journal of Technology Marketing*, 9(1), 33-52.

Leminen, S., Nyström, A. G., & Westerlund, M. (2015). A typology of creative consumers in living labs. *Journal of Engineering and Technology Management*, *37*, pp. 6-20. https://doi.org/10.1016/j.jengtecman.2015.08.008

Lettl, C. (2007). User involvement competence for radical innovation. *Journal of engineering and technology management*, *24*(1-2), pp. 53-75.

Liedtke, C., Baedeker, C., Hasselkuß, M., Rohn, H., & Grinewitschus, V. (2015). User-integrated innovation in Sustainable LivingLabs: An experimental infrastructure for researching and developing sustainable product service systems. *Journal of Cleaner Production*. https://doi.org/10.1016/j.jclepro.2014.04.070

Lilley, D. (2009) Design for sustainable behaviour: strategies and perceptions. *Design Studies,* 30, pp. 704-720.

Linse, C. (2017). Ambiguity at the heart of design work : Sensing and negotiating ambiguity in knowledge-creation work (PhD dissertation, KTH Royal Institute of Technology).

Lockton, D., Harrison, D., & Stanton, N. (2008) Making the user more efficient: design for sustainable behavior. *International Journal of Sustainable Engineering*, 1, pp. 3-8.

Lockton, D., & Candy, S. (2019). A vocabulary for visions in designing for transitions. *Cuadernos del Centro de Estudios en Diseño y Comunicación. Ensayos*, (73), pp. 27-49.

Loorbach, D. (2007). Transition management. New mode of governance for sustainable development. Utrecht: International Books.

Loorbach, D., Frantzeskaki, N., & Avelino, F. (2017). Sustainability transitions research: transforming science and practice for societal change. Annual Review of Environment and Resources, 42(1), pp. 599-626.

Manzini, E. (2014). Making things happen: Social innovation and design. *Design issues*, *30*(1), pp. 57-66.

Manzini, E. (2015). Design, when everybody designs: An introduction to design for social innovation. *MIT press*.

Manzini, E., & Rizzo, F. (2011). Small projects/large changes: Participatory design as an open participated process. *CoDesign*, *7*(3-4), pp. 199-215.

Marvin, S., Bulkeley, H., Mai, L., McCormick, K., & Palgan, Y. V. (2018). Urban living labs: Experimenting with city futures. In *Urban Living Labs: Experimenting with City Futures*. https://doi.org/10.4324/9781315230641

Mastelic, J., Sahakian, M., & Bonazzi, R. (2015). How to keep a living lab alive?. *info*.

Mattelmäki, T. (2008). Probing for co-exploring. *CoDesign*, *4*(1), pp. 65–78. https://doi.org/10.1080/15710880701875027

McCrory, G., Schäpke, N., Holmén, J., & Holmberg, J. (2020). Sustainability-oriented labs in real-world contexts: An exploratory review. *Journal of Cleaner Production*, *277*, 123202.

Meagher, L., Lyall, C., & Nutley, S. (2008). Flows of knowledge, expertise and influence: a method for assessing policy and practice impacts from social science research. *Research Evaluation*, *17*(3), pp. 163-173.

Mintzberg H., & Westley F. (2001) Decision Making: It's Not What You Think. *MIT Sloan Management Review*; Spring 2001; 42, 3.

Mogensen, P. H. (1992). Towards a Prototyping Approach in Systems Development. *DAIMI Report Series*, *21*(412). https://doi.org/10.7146/dpb.v21i412.6725

Mulder, I., & Stappers, P. J. (2009, June). Co-creating in practice: results and challenges. In *2009 IEEE International Technology Management Conference* (pp. 1-8). IEEE.

Nevens, F., Frantzeskaki, N., Gorissen, L., & Loorbach, D. (2013). Urban Transition Labs: co-creating transformative action for sustainable cities. *Journal of Cleaner Production*, 50, pp. 111-122.

Nilstad Pettersen I. (2015) Towards practice-oriented design for sustainability: the compatibility with selected design fields, *International Journal of Sustainable Engineering*, 8:3, 206-218, DOI: 10.1080/19397038.2014.1001468

Norman. D. A. (2005). Human-centered design considered harmful. interactions 12/4, 14–19. https://doi.org/10.1145/1070960.1070976

Oudshoorn, N. & Pinch, T. J. (2003). Introduction: How Users and Non-Users Matter. In *How Users Matter - The Co-Construction of Users and Technology*. Cambridge: MIT Press.

Owen, C. (2007). Design thinking: Notes on its nature and use. *Design Research Quarterly*, *2*(1), pp. 16-27.

Pallot, M., Trousse, B., Senach, B., & Scapin, D. (2010, August). Living lab research landscape: From user centred design and user experience towards user cocreation. In First European Summer School "Living Labs".

Pihkala, S., & H. Karasti. 2016. "Reflexive Engagement–Enacting Reflexivity in Design and for 'Participation in Plural." In *Proceedings of the 14th Participatory Design Conference*, pp. 21–30.

Prahalad, C. K., & Ramaswamy, V. (2004). Co-creation experiences: The next practice in value creation. *Journal of interactive marketing*, *18*(3), pp. 5-14.

Purao, S., Bagby, J., & Umapathy, K. (2008, July). Standardizing Web Services: Overcoming'Design by Committee'. In *2008 IEEE Congress on Services-Part I* (pp. 223-230). IEEE.

Radón, A., Sjöman, M., & Svengren Holm, L. (2013) Entreprenörskap och kreativitet i framgångsrika svenska modeföretag, In: Emma Senström och Lars Strannegård (ed.), *Kreativt kapital: Om ledning och organisation i kulturella och kreativa näringar,* pp. 117-155. Stockholm: 8tto - Volante

Randall, R. (2009). Loss and climate change: The cost of parallel narratives. *Ecopsychology*, *1* (3), pp. 118-129.

Reason P., & Bradbery H., 2008. The SAGE Handbook of Action Research 3e. Sage publications.

Reckwitz A. (2002). Toward a theory of social practices: a development in culturalist theorizing. Eur. J. of Social Theory 5(2). https://doi.org/10.1177/1368431022225432.

Ringenson T. (2021) Mobilising digitalisation to serve environmental goals. (PhD dissertation, KTH Royal Institute of Technology)

Rip, A., & Kemp, R. (1998). Technological change. Human choice and climate change, 2(2), pp. 327-399.

Rittel, H. W. J., & Webber, M. M. (1973). Dilemmas in a general theory of planning. *Policy Sciences*. https://doi.org/10.1007/BF01405730

Robinson, J. (2004). Squaring the circle? Some thoughts on the idea of sustainable development. *Ecological economics*, *48*(4), pp. 369-384

Romero Herrera, N. (2016). In-situ and mixed-design interventions. In: *Living Labs*: Design and assessment of sustainable living*; D. V. Keyson, G. M. Morrison, C. Baedeker, & C. Liedtke (Eds.*) *Netherlands. Springer*

Romero Herrera, N. (2017). The emergence of living lab methods. In: *Living Labs*: Design and assessment of sustainable living*; D. V. Keyson, G. M. Morrison, C. Baedeker, & C. Liedtke (Eds.*) 9-22, *Netherlands. Springer*

Rotmans, J., & Loorbach, D. (2010). Towards a better understanding of transitions and their governance—A systemic and reflexive approach. In: J. Grin, J. Rotmans & J. Schot (Eds.), Transitions to sustainable development—new directions in the study of longterm transformative change (pp. 105–220). *New York. Routledge.*

Russell, J. Y. (2010). A philosophical framework for an open and critical transdisciplinary inquiry. In *Tackling Wicked Problems: Through the Transdisciplinary Imagination*. https://doi.org/10.4324/9781849776530

Ryghaug, M., Skjølsvold, T. M., & Heidenreich, S. (2018). Creating energy citizenship through material participation. *Social studies of science*, *48*(2), pp. 283-303.

Sabel, C. F., & Zeitlin, J. (2011). Experimental Governance. In Levi-Faur, D. (Ed.). (2012). *The Oxford handbook of governance*. Oxford University Press.

Sagaris, L., & Berrios, E., & Tiznado Aitken, I. (2020). Using PAR to frame sustainable transport and social justice on policy agendas. A pilot experience in two contrasting Chilean cities. Journal of Transport Geography. 83. 102654. 10.1016/j.jtrangeo.2020.102654.

Salminen, J., Konsti-Laakso, S., Pallot, M., Trousse, B., & Senach, B. (2011). Evaluating user involvement within living labs through the use of a domain landscape. In *17th International Conference on Concurrent Enterprising* (pp. 1-10). IEEE.

Sanders, E. B. N. (2006). Scaffolds for building everyday creativity. *Design for effective communications: Creating Contexts for Clarity and Meaning*, 65-77.

Sanders, E. B. N., & Stappers, P. J. (2008). Co-creation and the new landscapes of design. *CoDesign*. https://doi.org/10.1080/15710880701875068

Sanders, E. B. N., & Stappers, P. J. (2012). Convivial toolbox: Generative research for the front end of design. Bis.

Sanders, E. B. N., & Stappers, P. J. (2014). Probes, toolkits and prototypes: Three approaches to making in codesigning. *CoDesign*. https://doi.org/10.1080/15710882.2014.888183

Sanders, E.B.N., & William, C.T. (2001) Harnessing People's Creativity: Ideation and Expression through Visual Communication, In Langford, J. and McDonagh-Philp, D. (Eds.) *Focus Groups: Supporting Effective Product Development*, Taylor and Francis

Schreier, M. (2014). Qualitative content analysis. In The SAGE Handbook of Qualitative Data Analysis (pp. 170–183). *SAGE Publications Ltd. 10.4135/9781446282243*.

Schulman, S., 2010. Design thinking is not enough (as referenced in Hillgren et al., 2011)

Schuurman, D. (2015). Bridging the gap between Open and User Innovation? Exploring the value of Living Labs as a means to structure User Contribution and manage Distributed Innovation. *Doctoral dissertation, Ghent University*

Schuurman, D., De Marez, L., & Ballon, P. (2015). Living Labs: a systematic literature review. Open Living Lab Days 2015

Schön D.A. (1985). The design studio: An exploration of its traditions and potential. London: Royal Institute of British Architects.

Schön, D.A. (1995), "Knowing in action: the new scholarship requires a new epistemology", *Change*, November/December, pp. 27-34.

Scoones, Newell, & Leach 2015. The politics of green transformations. In Scoones, Leach, and Newell (eds.) *The politics of green transformations*. pp. 1-24.

Scott, K., Bakker, C., & Quist, J. (2012). Designing change by living change. *Design Studies*. https://doi.org/10.1016/j.destud.2011.08.002

Sengers, F., Wieczorek, A. J., & Raven, R. (2019). Experimenting for sustainability transitions: A systematic literature review. *Technological Forecasting and Social Change*, *145*, 153-164.

Shapin, S., & Schaffer, S. (1985). Leviathan and the Air-Pump: Hobbes, Boyle, and the Experimental Life. Princeton, NJ: *Princeton University Press*.

Schot, J.W., & Geels, F.W., 2008. Strategic niche management and sustainable innovation journeys: theory, findings, research agenda and policy. *Tech. Anal. Strat. Manag*. 20 (5), pp. 537–554.

Shove, E. (2003). Converging conventions of comfort, cleanliness and convenience. *Journal of Consumer policy*, *26*, pp. 395-418.

Shove, E., & Watson, M. (2006). POPD Manifesto, Durham University.

Shove, E., Watson, M., Hand, M., & Ingram, J. (2007). The Design of Everyday Life. In *The Design of Everyday Life*. https://doi.org/10.5040/9781474293679

Shove, E., Watson, M., Ingram, J., & Hand, M. (2008). The design of everyday life. Oxford: Berg.

Sjöman, M. (2022) Designerly Living Labs: Real-life experimentation in early research stages. *Continuous Innovation Network. Proceedings of CINet 2022*.

Sjöman, M., & Hesselgren, M. (2020) Designerly Living Labs: Early-stage exploration of future sustainable concepts, in Boess, S., Cheung, M. and Cain, R. (Eds.), *Synergy - DRS International Conference 2020*.

Sjöman, M., & Hesselgren, M. (2022) Designerly Living Labs: Design-driven experimentation. In: Oldbury, K., Isaksson, K. & Marsden, G. (2022) Experimentation for sustainable transport: Risks, strengths, and governance implications. *Linnefors förlag, Boxholm*.

Sjöman, M., Ringenson, T., & Kramers, A. (2020) Exploring everyday mobility in a living lab based on economic interventions. *European Transport Research Review*.

Smith, A., Stirling, A., & Berkhout, F. (2005). The governance of sustainable socio-technical transitions. *Research policy*, *34*(10), pp. 1491-1510.

Sopjani, L., Hesselgren, M., Janhager Stier, J., & Ritzén, S. (2016). Aligning private and public domains for sustainable disruptive innovation. In 17th International CINet Conference.

Sopjani, L., Janhager Stier, J., & Ritzén, S. (2016) User involvement in disruptive innovation – A study on users of a light electric vehicle sharing system. In: *Nuran ACUR (ed.), Proceedings of 23rd Innovation and Product Development Management Conference* Glasgow

Sopjani, L., Stier, J. J., Ritzén, S., Hesselgren, M., & Georén, P. (2019). Involving users and user roles in the transition to sustainable mobility systems: The case of light electric vehicle sharing in Sweden. *Transportation Research Part D: Transport and Environment*, *71*, pp. 207-221.

Spaargaren, G. (1997). The ecological modernisation of production and consumption: Essays in environmental sociology. Wageningen, The Netherlands: Landbouw. Universitiet Wageningen.

Spaargaren, G. (2003). Sustainable Consumption: A Theoretical and Environmental Policy Perspective. *Society & Natural Resources: An International Journal*, 16 (October 2002), pp. 687–701.

Spaargaren, G. (2011). Theories of practices: Agency, technology, and culture: Exploring the relevance of practice theories for the governance of sustainable consumption practices in the new world-order. *Global Environmental Change*, *21*(3), pp. 813-822.

Steen, K., & Van Bueren, E. (2017). The defining characteristics of urban living labs. *Technology Innovation Management Review*, 7(7).

Steen, M., Kuijt-Evers, L. & Klok, J. (2007) Early user involvement in research and design projects – A review of methods and practices. EGOS Colloqium. Vienna.

Stolterman, E. (2008). The Nature of Design Practice and Implications for Interaction Design Research. *International Journal of Design*, 2(1), pp. 55–65.

Strengers, Y. (2014). Smart energy in everyday life: are you designing for resource man? *Interactions*, *21*(4), pp. 24-31.

Ståhlbröst, A. (2012). A set of key principles to assess the impact of Living Labs. *International Journal of Product Development,* 17(1), pp. 60-75

Swann, C. (2002). Action research and the practice of design. *Design issues*, *18*(1), pp. 49-61.

Tironi, M. (2020). Prototyping public friction: Exploring the political effects of design testing in urban space. *The British journal of sociology*, *71*(3), pp. 503-519.

Tripp, D. (2005). Action research: a methodological introduction. *Educação e Pesquisa*, 31, pp. 443-466.

Tukker, A., Haag, E., Eder, P., Vercalsteren, A., Tischner, T. W. U., Charter, M., ... & van der Vlugt, M. (2000). Eco-design: European state of the art. *Seville, Spain: Institute for Prospective Technological Studies*.

Zuber-Skerritt, O. (2018). An educational framework for participatory action learning and action research (PALAR). *Educational Action Research*, *26*(4), pp. 513-532.

United Nations (2015). Transforming Our World: the 2030 Agenda for Sustainable Development. *General Assembly 70 Session*.

United Nations Environment Programme (2016). Fostering and Communicating Sustainable Lifestyles: Principles and emerging practices. https://wedocs.unep.org/bitstream/handle/20.500.11822/17016/fostering_Communicating_Sust_Lifestyles.pdf (retrieved 20 Jan 2023)

Vaajakallio, K., & Mattelmäki, T. (2014). Design games in codesign: As a tool, a mindset and a structure. *CoDesign*. https://doi.org/10.1080/15710882.2014.881886

Vaddadi, B., Ringenson, T., Sjöman, M., Hesselgren, M., & Kramers, A. (2022). Do they work? Exploring possible potentials of neighbourhood Telecommuting centres in supporting sustainable travel. *Travel Behaviour and Society*, *29*, pp. 34-41.

Van der Bijl-Brouwer, M., & Dorst, K. (2017). Advancing the strategic impact of human-centred design. *Design Studies*, 53, pp. 1-23.

Vaughan, L. (2017). Designer/Practitioner/Researcher. In *Practice-based design research* (No. 2, pp. 9-17). Bloomsbury Publishing.

Verganti, R. (2009). Design-driven innovation: Changing the rules of competition by radically innovating what things mean. *Cambridge, MA: MIT Press*.

Vergragt, P. J., & Brown, H. S. (2007). Sustainable mobility: from technological innovation to societal learning. Journal of Cleaner Production, 15(11-12), pp. 1104-1115.

Von Hippel, E. (1986). Lead users: a source of novel product concepts. Management Science 32, pp. 791–805

Von Wirth, T., Fuenfschilling, L., Frantzeskaki, N., & Coenen, L. (2019). Impacts of urban living labs on sustainability transitions: Mechanisms and strategies for systemic change through experimentation. *European Planning Studies*, *27*(2), pp. 229-257.

Voss, A., Hartswood, M., Procter, R., Slack, R., Büscher, M., & Rouncefield, M. (2009). *Introduction: Configuring User-Designer Relations: Interdisciplinary Perspectives*. https://doi.org/10.1007/978-1-84628-925-5_1

Voytenko, Y., McCormick, K., Evans, J., & Schliwa, G. (2016). Urban living labs for sustainability and low carbon cities in Europe: Towards a research agenda. *Journal of cleaner production*, *123*, pp. 45-54.

Warde, A. (2005). Consumption and theories of practice. *Journal of Consumer Culture*. https://doi.org/10.1177/1469540505053090

Weiland, S., Bleicher, A., Polzin, C., Rauschmayer, F., & Rode, J. (2017). The nature of experiments for sustainability transformations: A search for common ground. *Journal of Cleaner Production*, 169, pp. 30–38.

Welch, D., & Warde, A. (2015). Theories of practice and sustainable consumption. In *Handbook of research on sustainable consumption*. Edward Elgar Publishing.

White, M. & Langenheim, N. (2021) 'A ladder-truss of citizen participation: re-imagining Arnstein's ladder to bridge between the community and sustainable urban design outcomes', *J. Design Research*, Vol. 19, Nos. 1/2/3, pp. 155–183.

Woolrych, A., Hornbæk, K., Frøkjær, E., & Cockton, G. (2011). Ingredients and meals rather than recipes: A proposal for research that does not treat usability evaluation methods as indivisible wholes. *International Journal of Human-Computer Interaction*, *27*(10), 940-970.

Wrigley, C., Nusem, E., & Straker, K. (2020). Implementing design thinking: Understanding organizational conditions. *California Management Review*, *62*(2), pp. 125-143.

Zimmerman, J., Forlizzi, J., & Evenson, S. 2007. "Research through Design as a Method for Interaction Design Research in HCI." *Proceedings of CHI 2007, New York: ACM Press, pp.* 493–502. doi:10.1145/1240624.1240704.

Zimmerman, J., & Stolterman, E. 2014. An Analysis and Critique of Research through Design: towards a formalization of a research approach. DIS 2010

World Wildlife Fund (2018) One Planet Plate.